Binary outcomes and maximum likelihood

From coefficients to quantities we care about

Sebastian Ramirez-Ruiz
European University Institute (EUI)
November 13, 2026

Where we are: into the modeling block

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BLOCK 1 · DONE ✓

Inference and the practice of research

W1 · The logic of statistical inference

W2 · Regression anatomy

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Causal thinking and measurement

W3 · Estimands, DAGs, specification

W4 · Measurement I: concepts to numbers

W5 · Measurement II: latent variables

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W7 · Nested data: fixed and random effects

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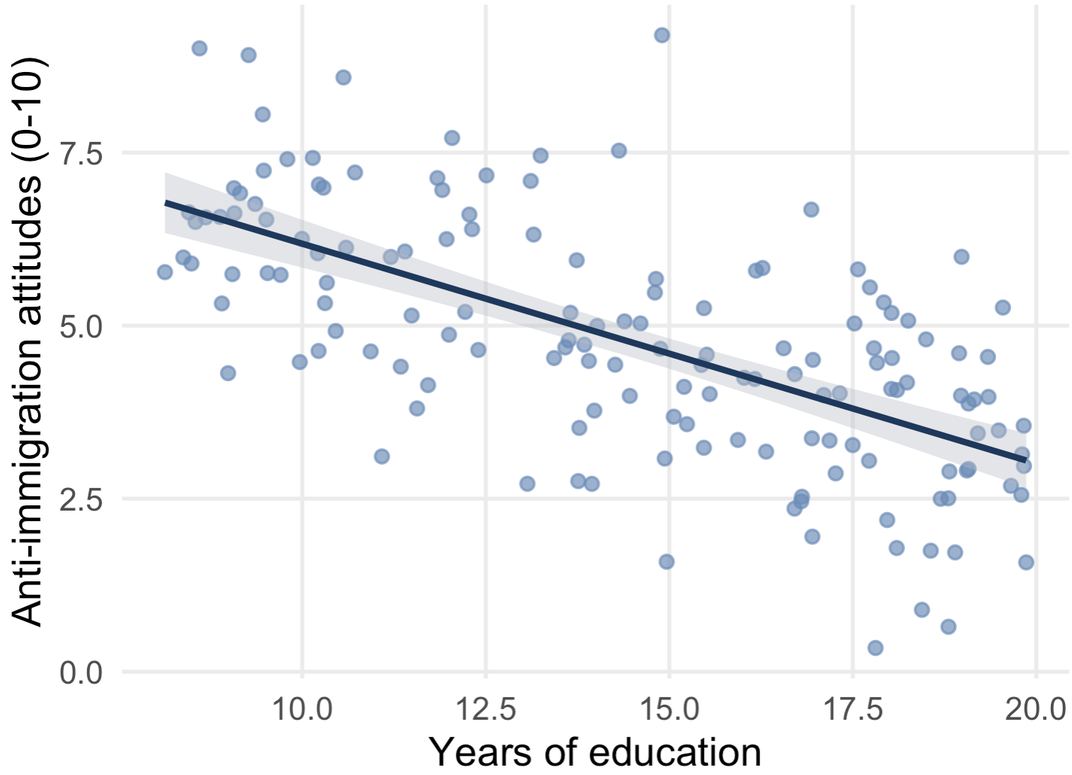
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- Today, both threads come back: the estimand decides **which quantity we report**, and a measured 0/1 outcome is still a measure.

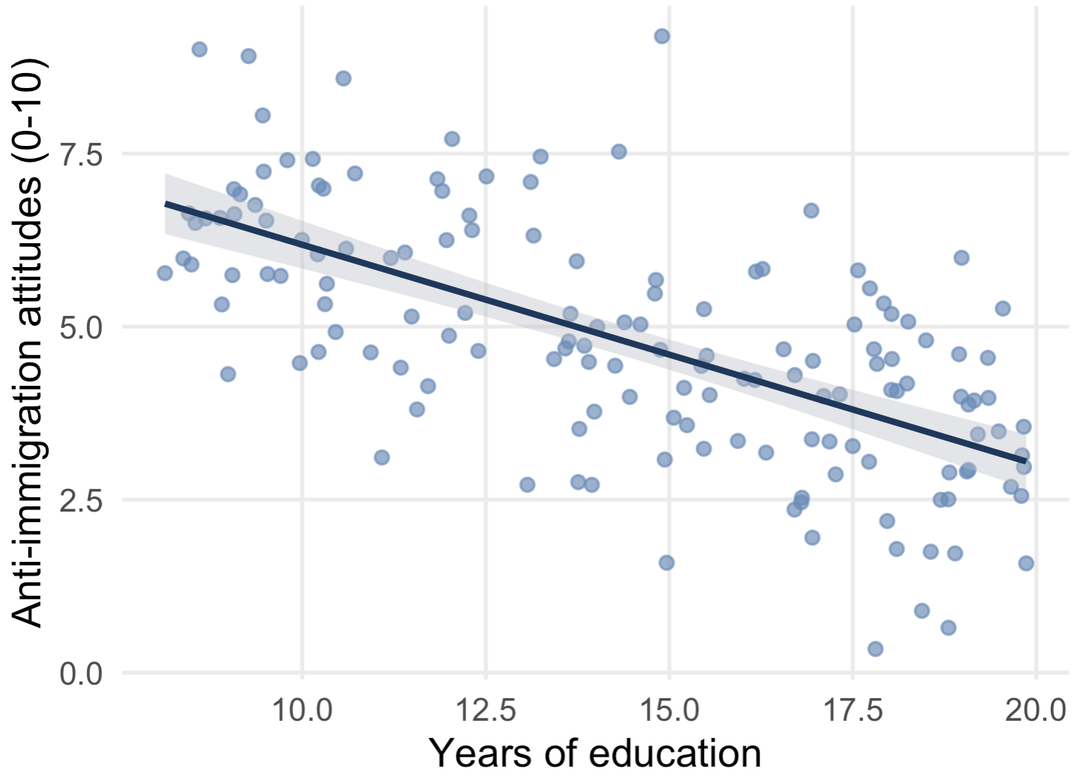
OLS is our workhorse



Simulated. One line, one slope: about -0.32 points per year of education.

OLS is our workhorse

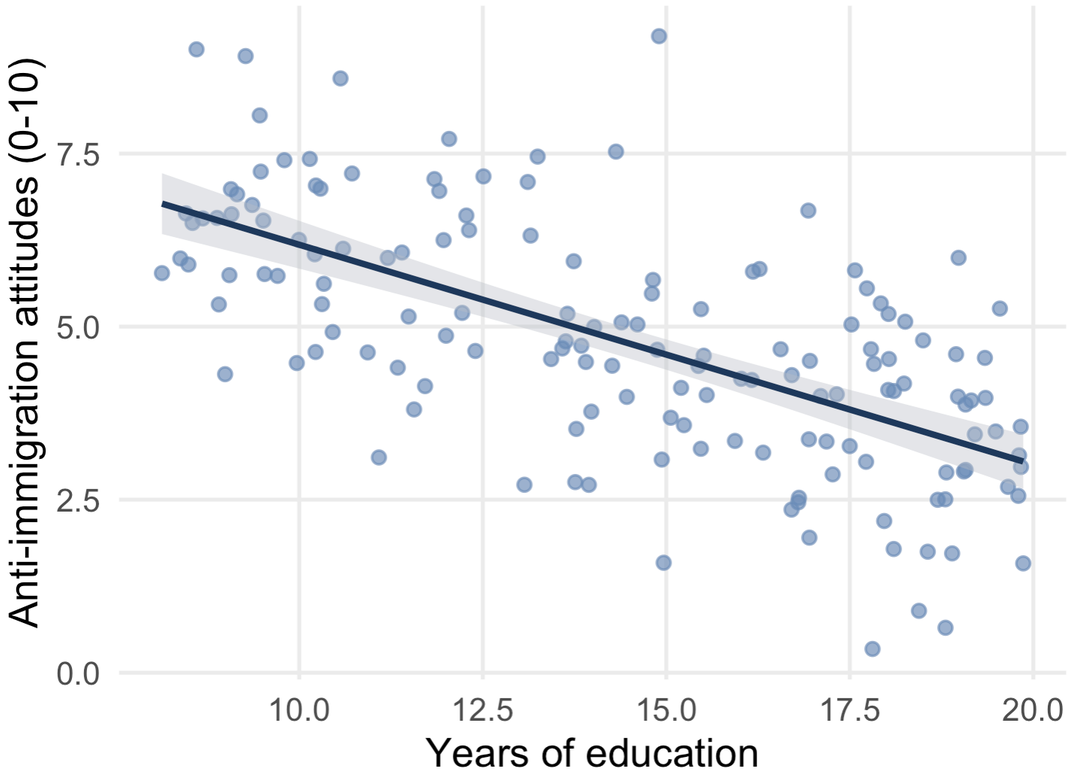
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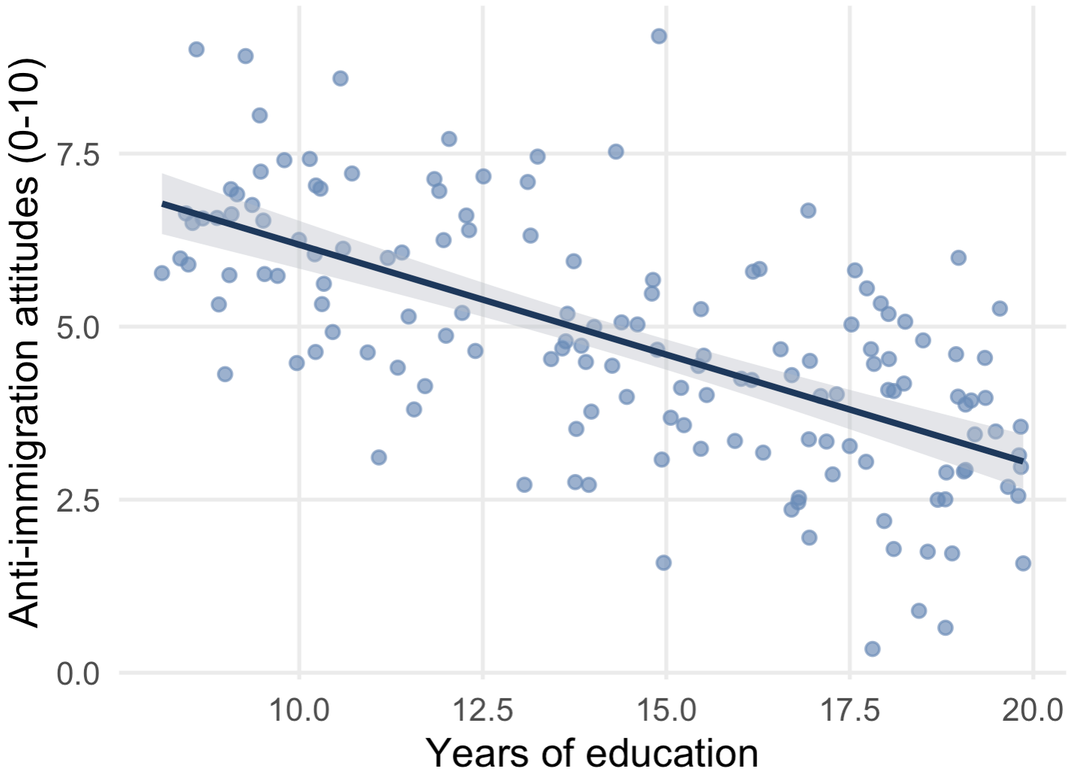
- Most of what you have fit so far is a linear regression, $Y_i = \alpha + \beta X_i + \varepsilon_i$, estimated by ordinary least squares.
- It is everywhere for good reasons. It is **simple**: one line of code, one formula. It is **interpretable**: β is the change in the average outcome for a one-unit change in X. And it is **forgiving**: even when the truth is not a straight line, it gives the best straight-line summary.



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- For the most part, it does fine. Today is about the outcomes where it starts to stress.



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Many outcomes we care about are “switches”

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💼

Employed or not

Labor-market sociology, welfare states

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- We code them 0/1.The average of a 0/1 variable is a share, so a model for its average is a **model for a probability**.
- Can our workhorse pull that? **Mostly**. Today: when OLS is enough, what to use when it isn’t, and what to report either way.

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OLS on a switch (the linear probability model), when it strains, and its nonlinear alternatives: logit and probit

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Practice

Short exercises along the way, ending with a binary outcome from your PhD; likelihoods by hand in Lab 6

Binary outcomes

The linear probability model

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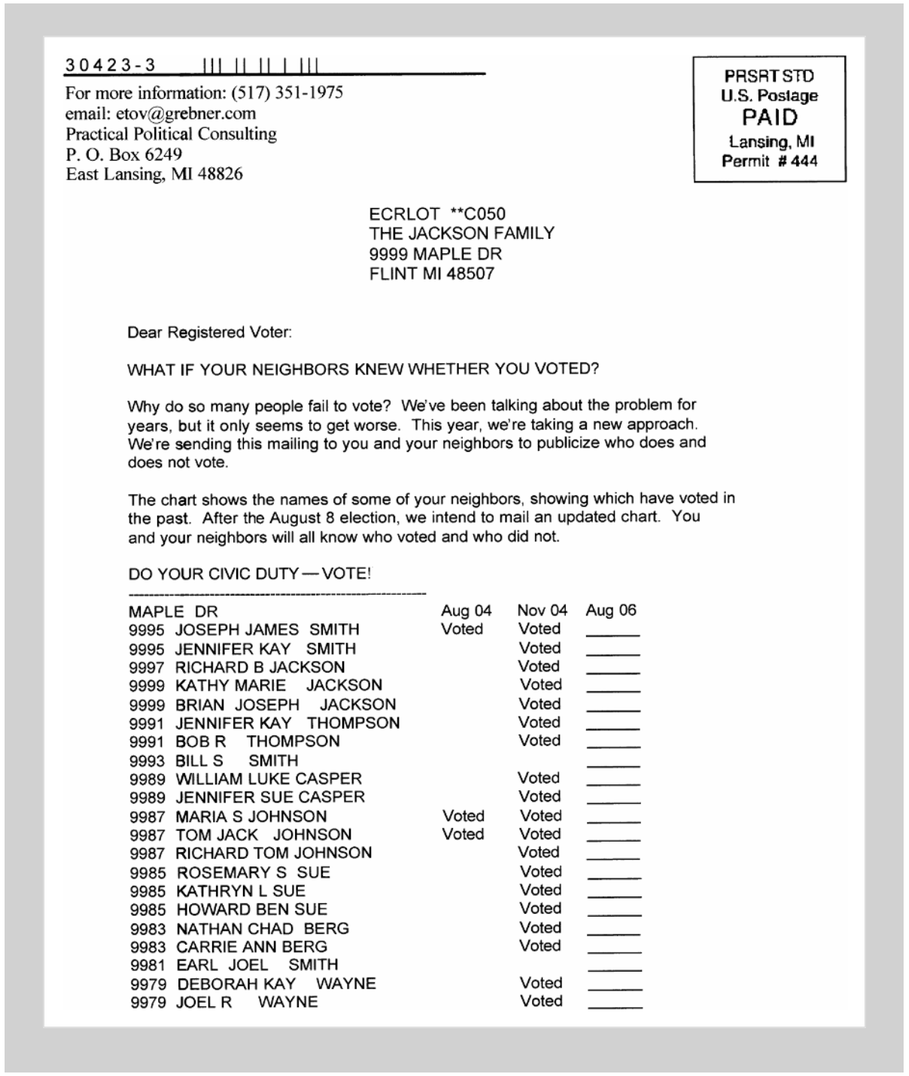
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- So OLS on a 0/1 outcome models a probability: the **linear probability model** (LPM). Each coefficient is a change in **percentage points**.
- With only a randomized binary treatment on the right-hand side, the LPM coefficient **is** the difference in proportions. No functional-form assumption needed.

A letter that tells your neighbors whether you voted



30423-3

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ECRLOT **C050
THE JACKSON FAMILY
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FLINT MI 48507

Dear Registered Voter:

WHAT IF YOUR NEIGHBORS KNEW WHETHER YOU VOTED?

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse. This year, we're taking a new approach. We're sending this mailing to you and your neighbors to publicize who does and does not vote.

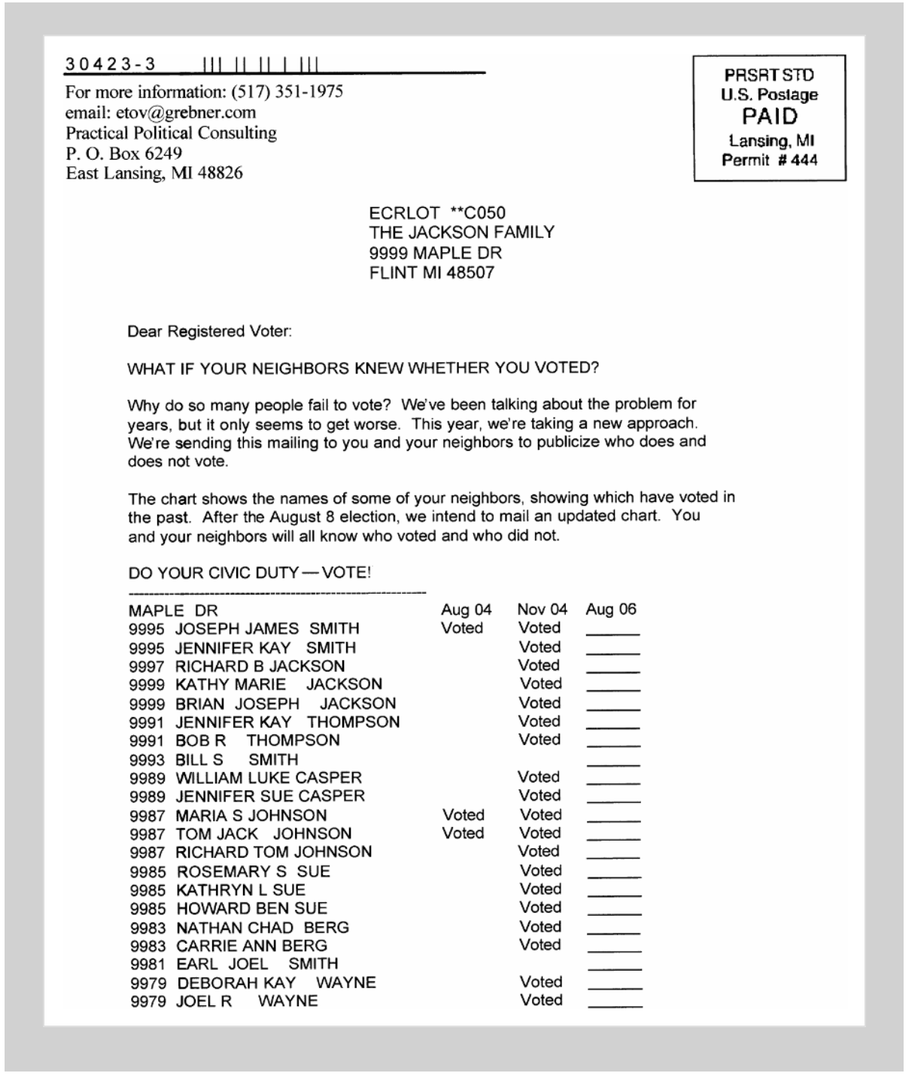
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DO YOUR CIVIC DUTY — VOTE!

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9999 KATHY MARIE JACKSON		Voted	_____
9999 BRIAN JOSEPH JACKSON		Voted	_____
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9991 BOB R THOMPSON		Voted	_____
9993 BILL S SMITH			_____
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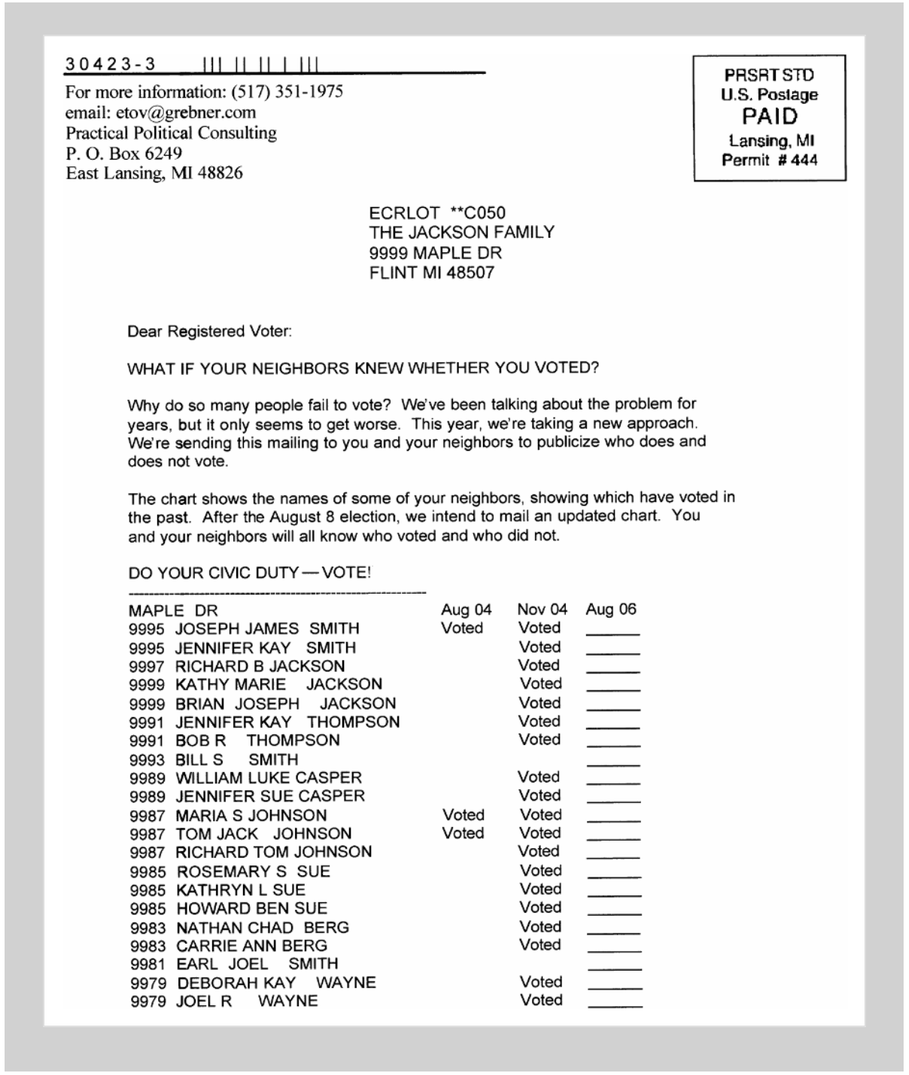
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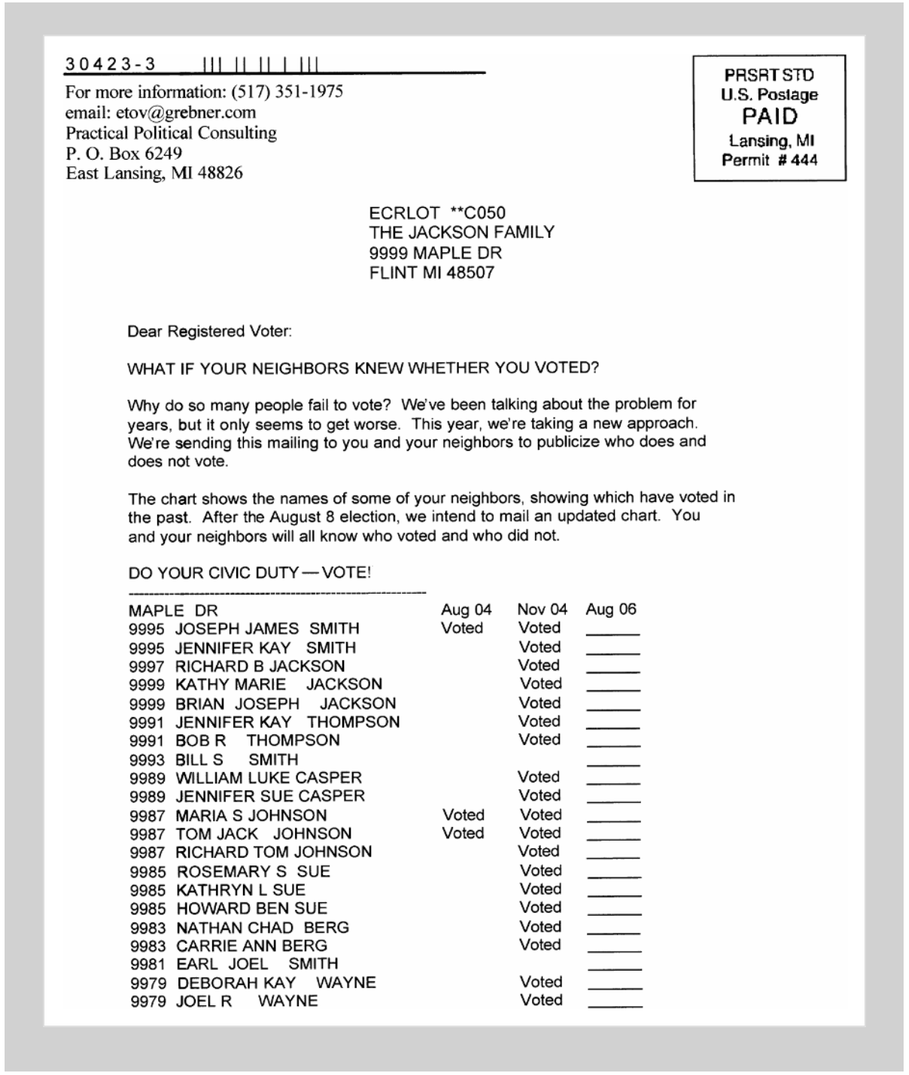
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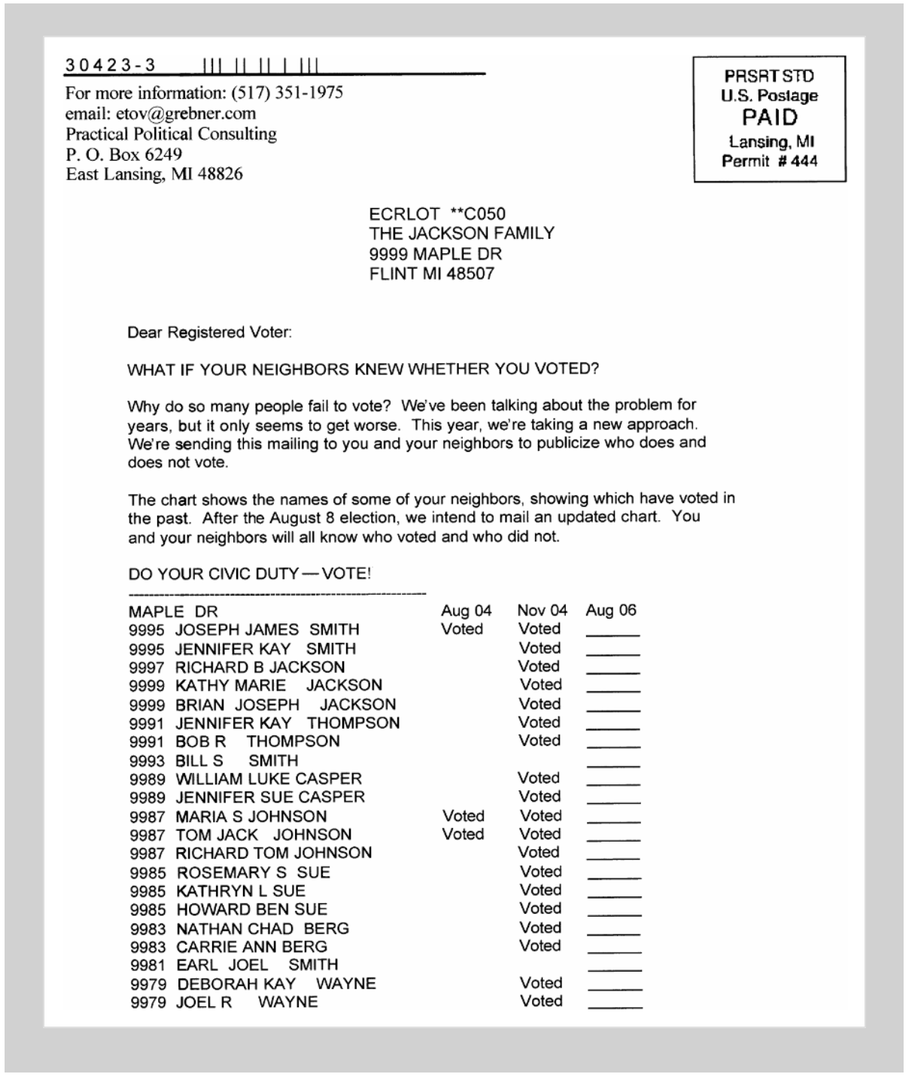
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- The milder letters: a civic-duty reminder; "you are being studied" (the Hawthorne letter, Week 4); your own household's record only (Self).

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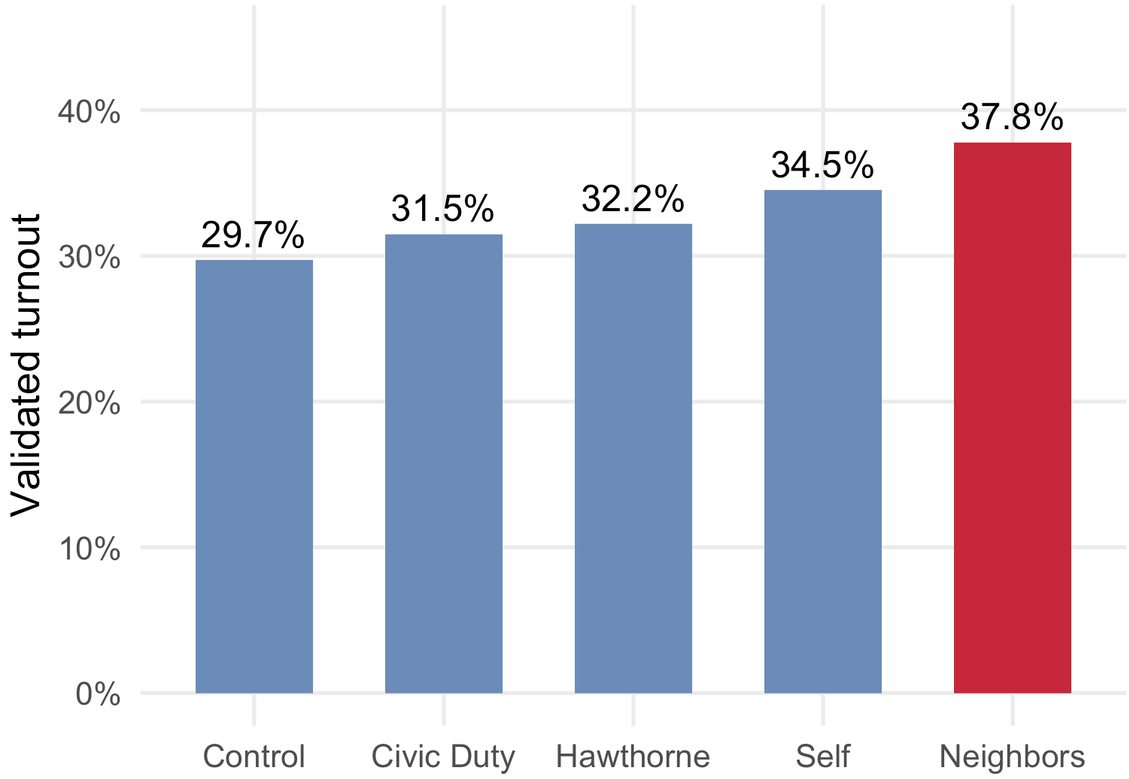
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9979 JOEL R WAYNE		Voted	_____

The "Neighbors" mailing. Gerber, Green & Larimer (2008), Appendix A.

- Michigan, August 2006 primary. 180,002 households are randomly sent one of four letters, or nothing (Gerber, Green & Larimer 2008).
- The strongest dose of social pressure: your household's voting record **and your neighbors'**, with a promise to "mail an updated chart" after the election.
- The milder letters: a civic-duty reminder; "you are being studied" (the Hawthorne letter, Week 4); your own household's record only (Self).
- The outcome: turnout **validated** from voter records, not self-reported.

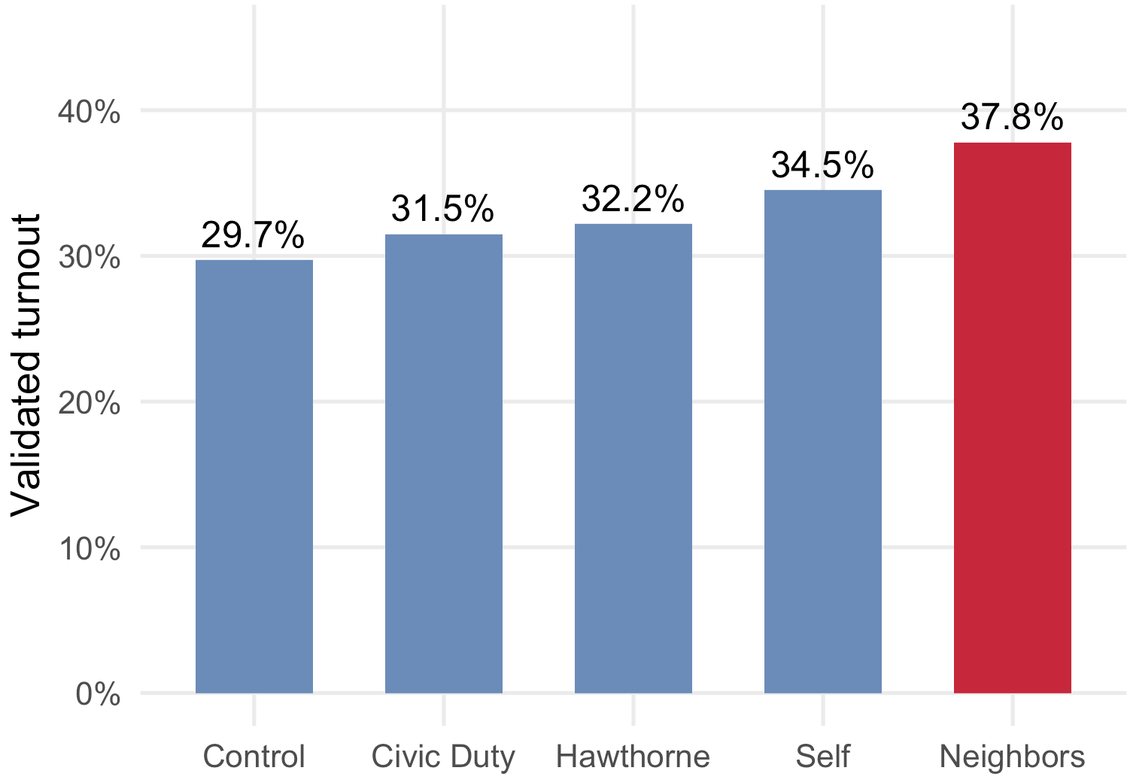
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Validated turnout by letter. Gerber, Green & Larimer (2008), Table 2.

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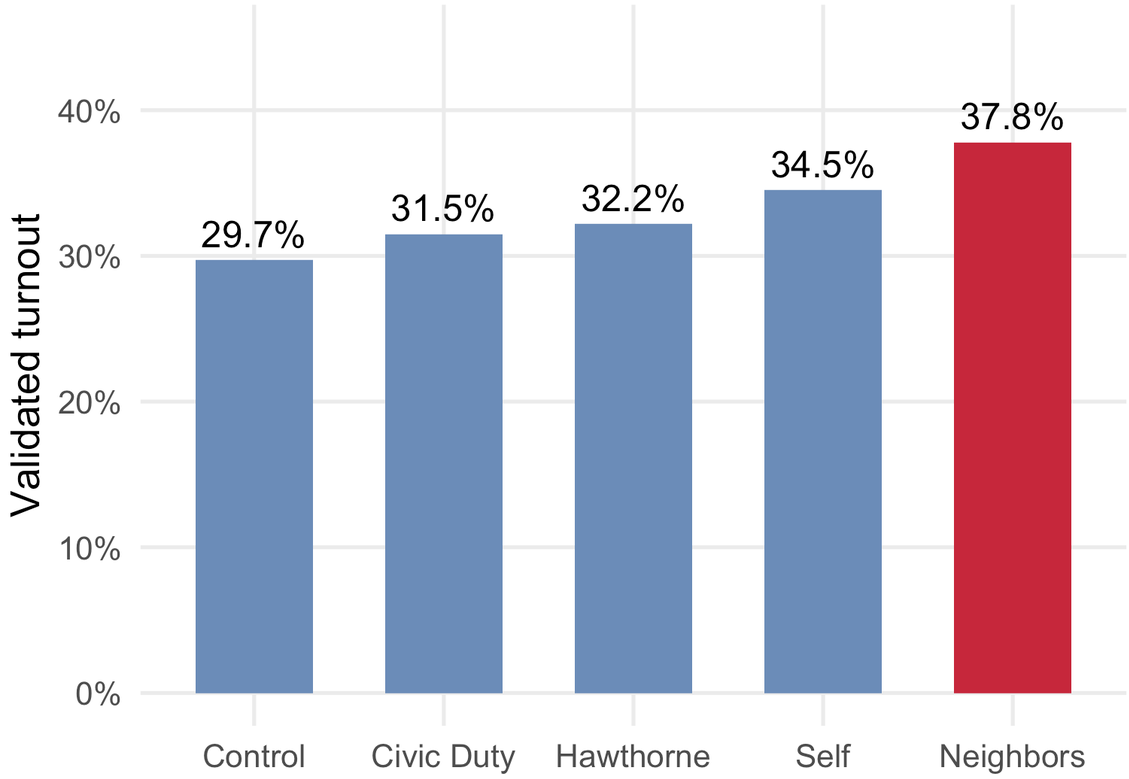
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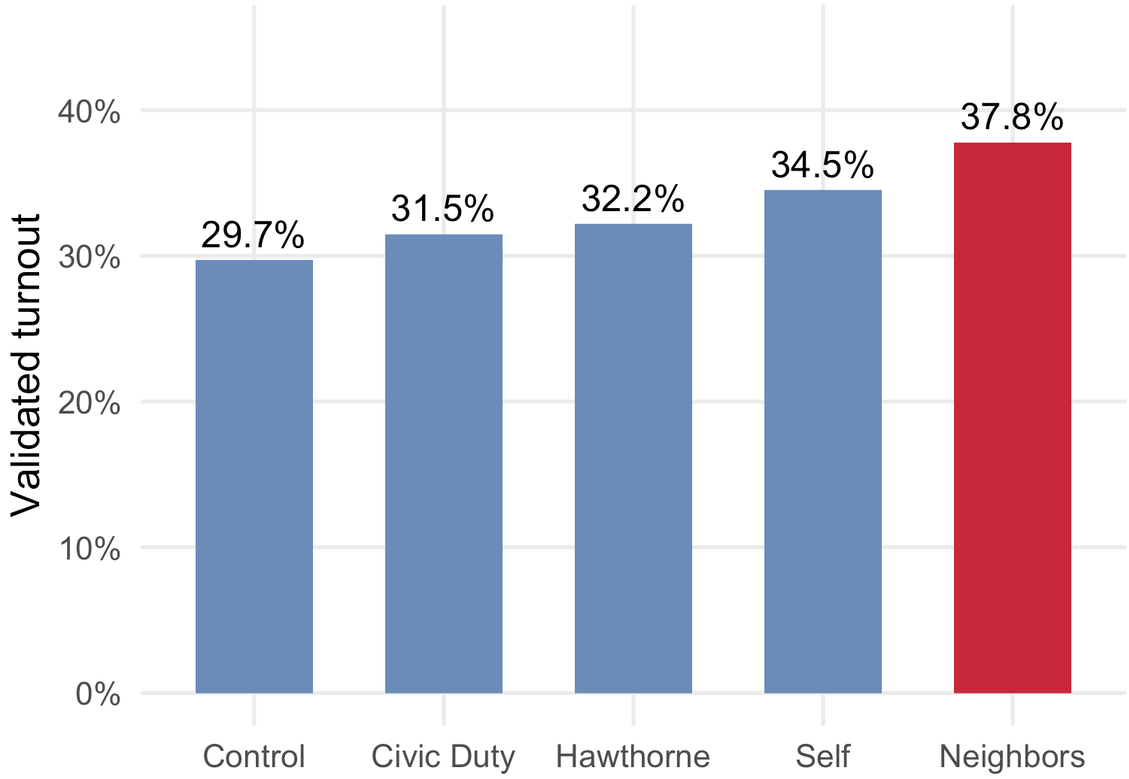
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- Each letter adds a bit more social pressure, and turnout climbs with it.
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- A randomized treatment in five groups and a 0/1 outcome: the easiest case for OLS on a switch.



Validated turnout by letter. Gerber, Green & Larimer (2008), Table 2.

How Gerber, Green & Larimer reported it

OLS, turnout (0/1)	(a)	(b)	(c)
Civic Duty	.018	.018	.018
Hawthorne	.026	.026	.025
Self	.049	.049	.048
Neighbors	**.081**	**.082**	**.081**
Block fixed effects	no	yes	yes
Past turnout (5 elections)	no	no	yes

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- Hold that thought: a **logit** coefficient would move (later today).

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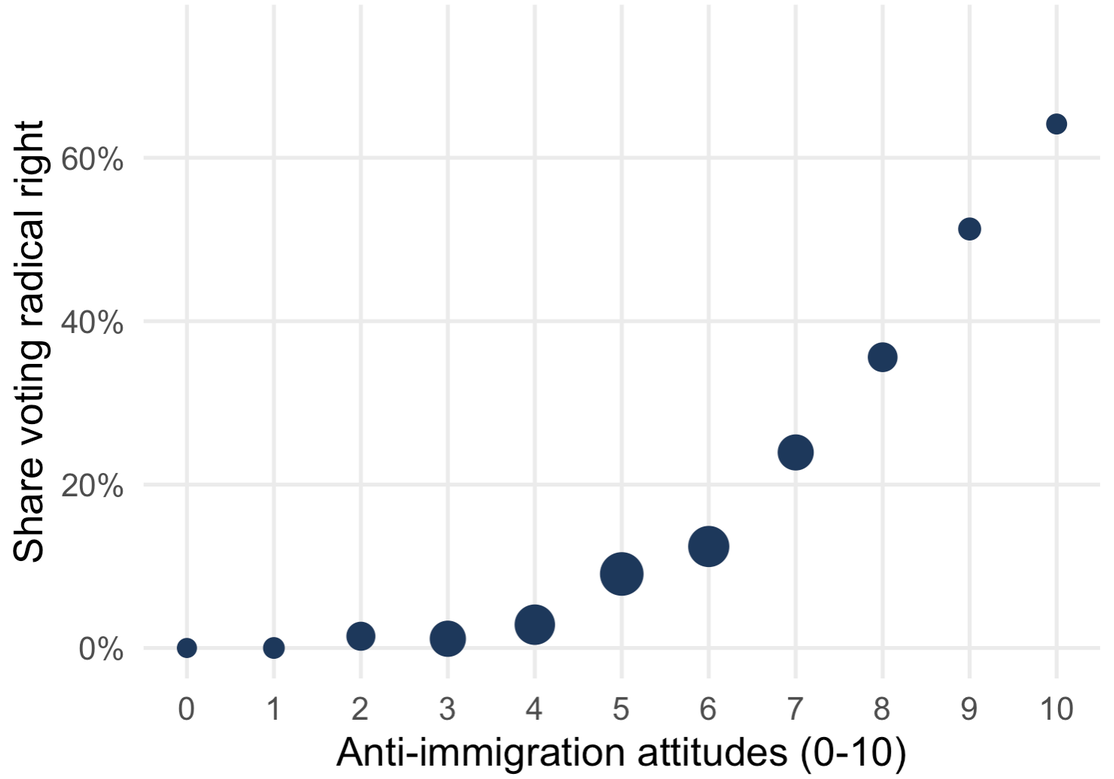
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> Same outcome type, a much harder job for the straight line. We will use a simulated survey of radical-right voting as our test case for the rest of the session.

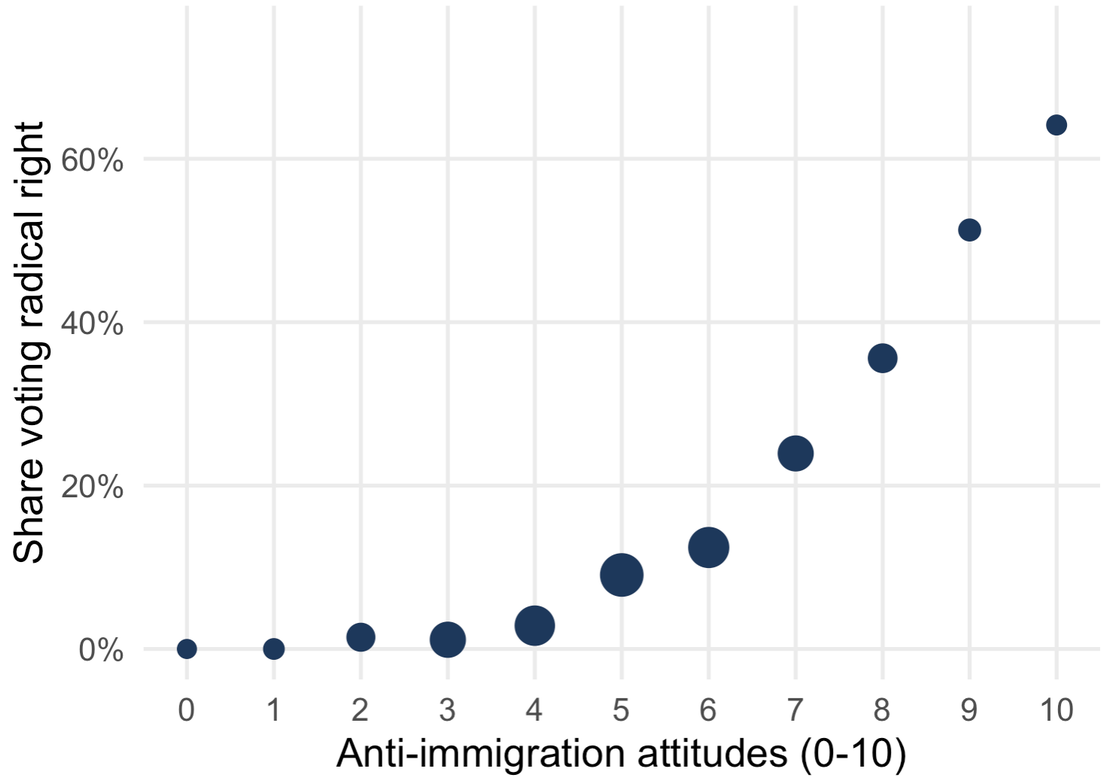
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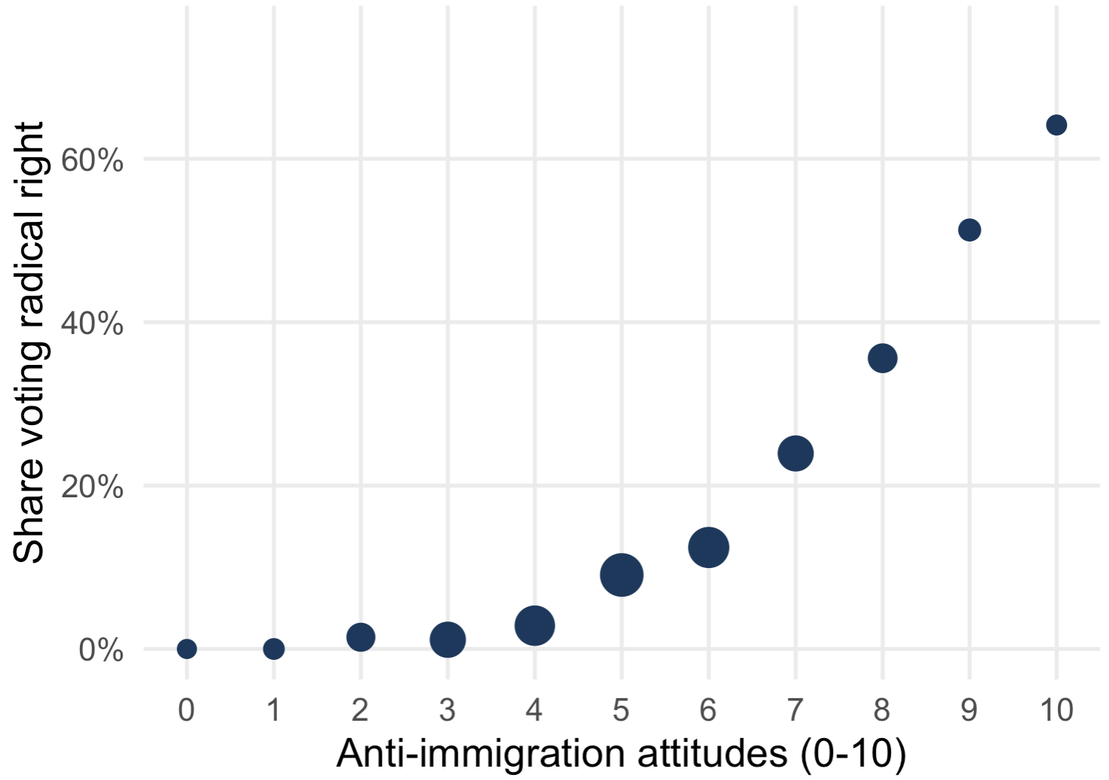
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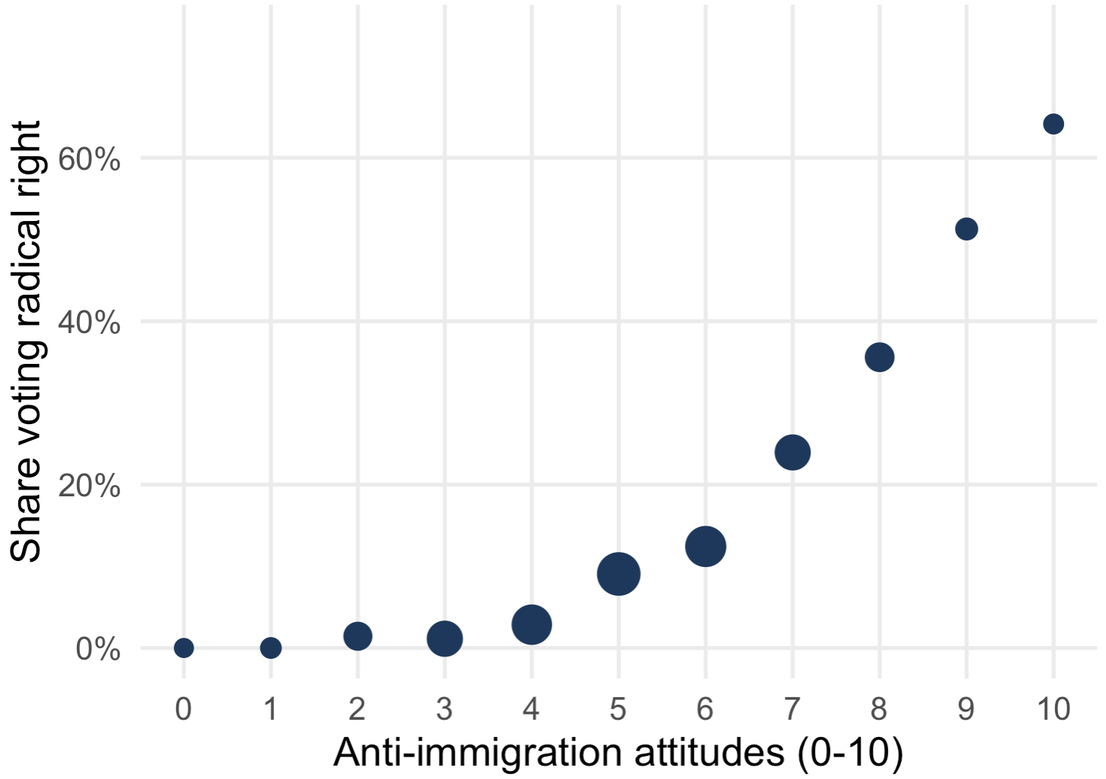
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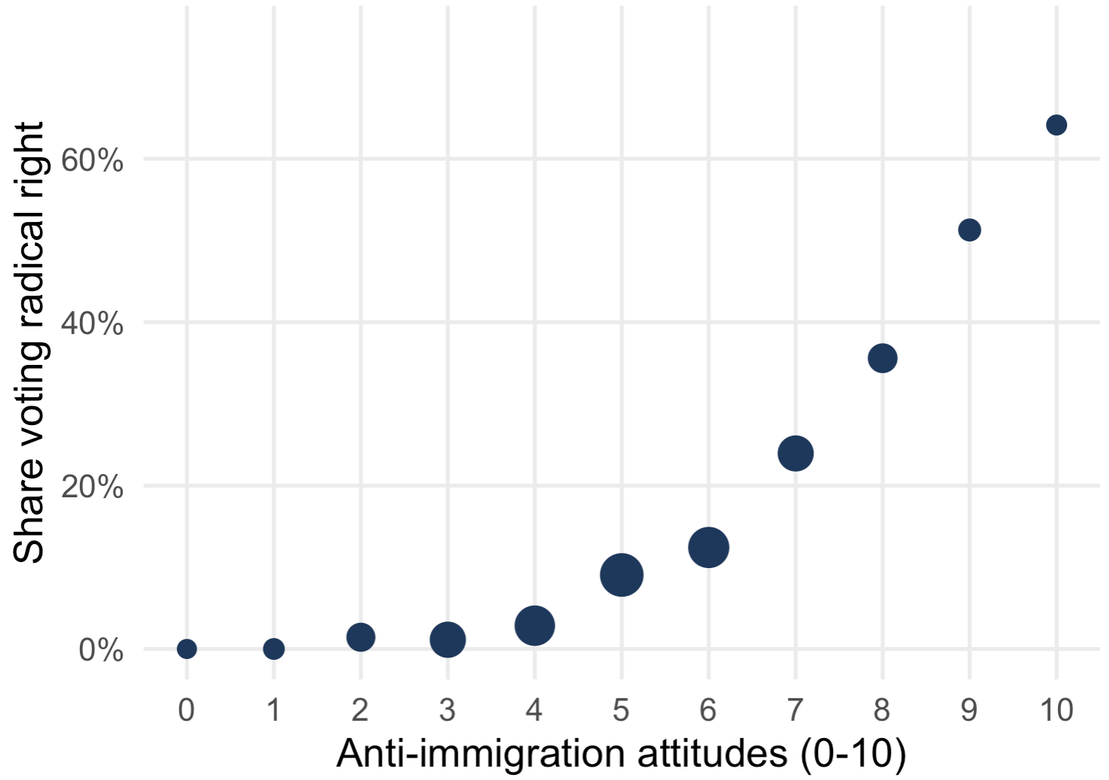
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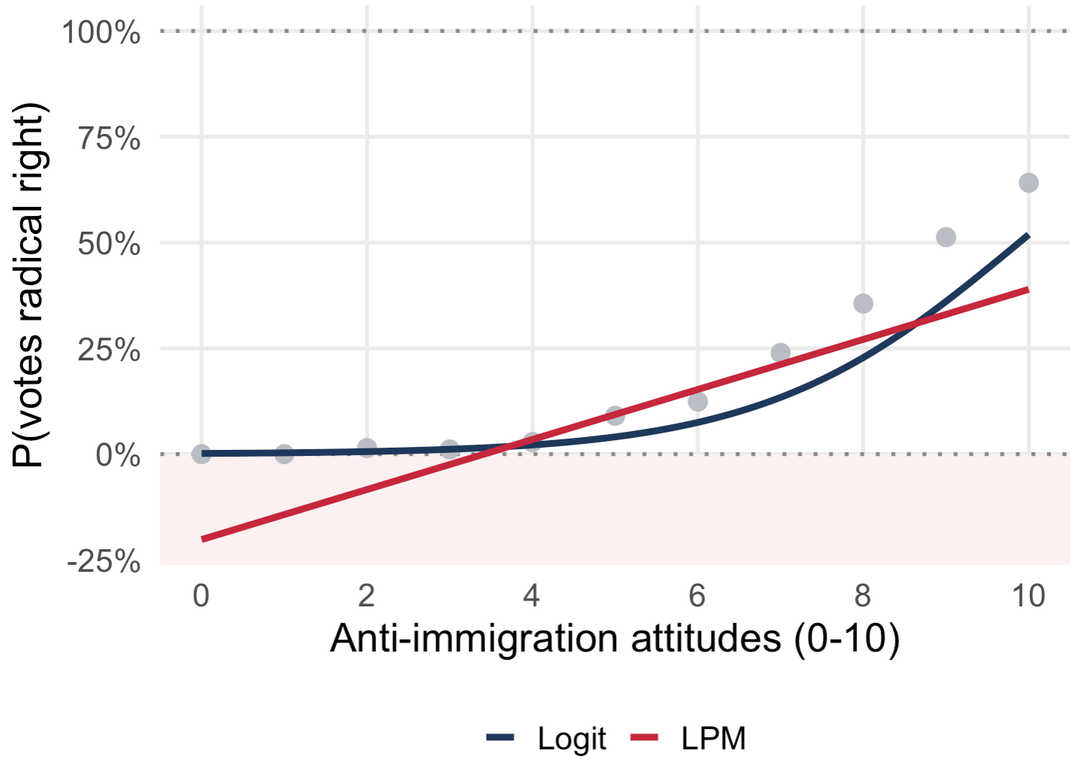
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- Our running example for LPM, logit, and probit.



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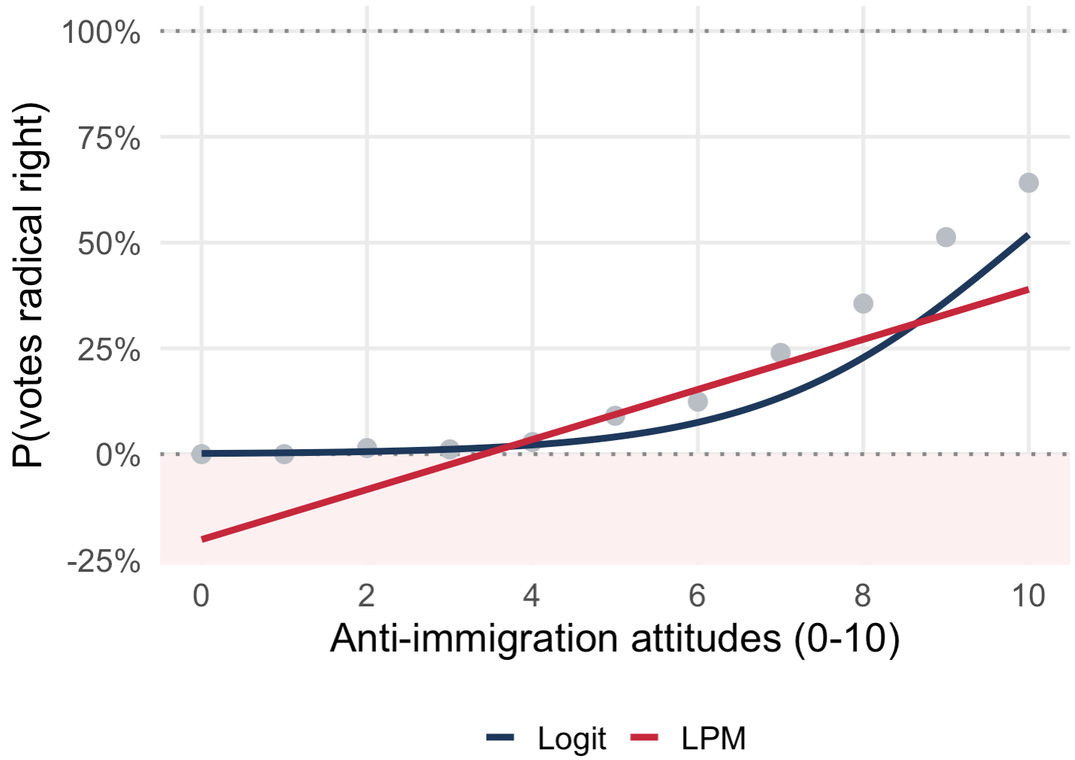
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Predictions for a 45-year-old urban woman with a degree; gray dots: observed shares (all voters). Shaded: probabilities that cannot exist.

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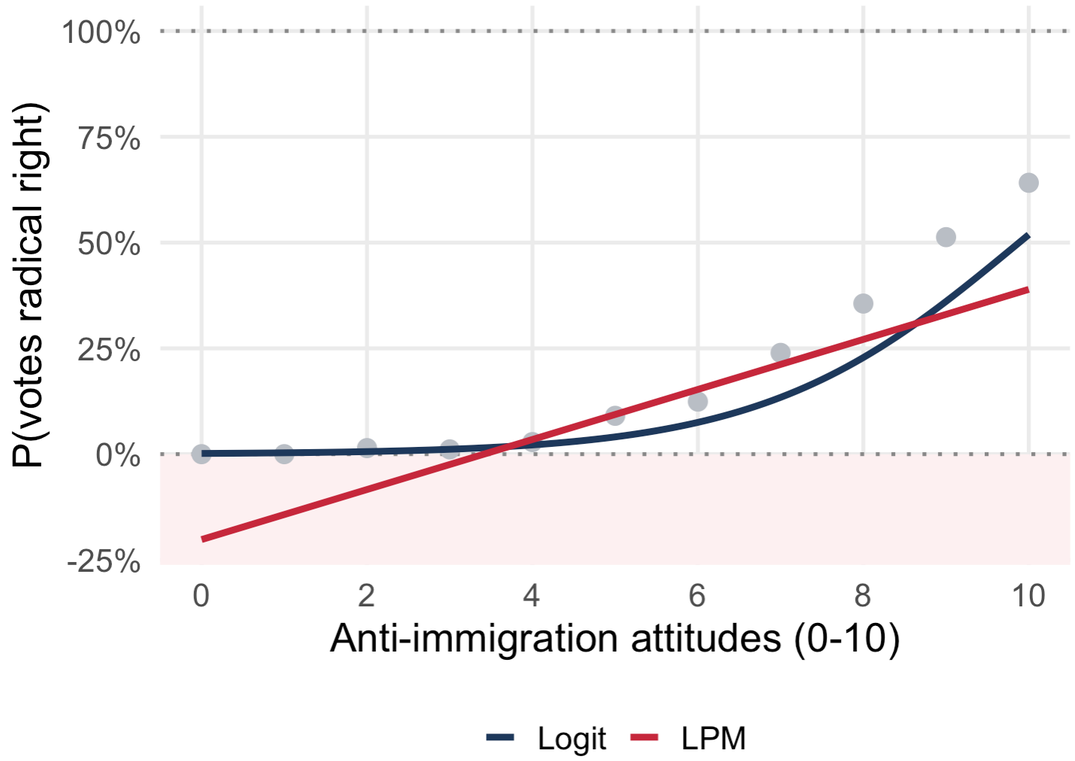
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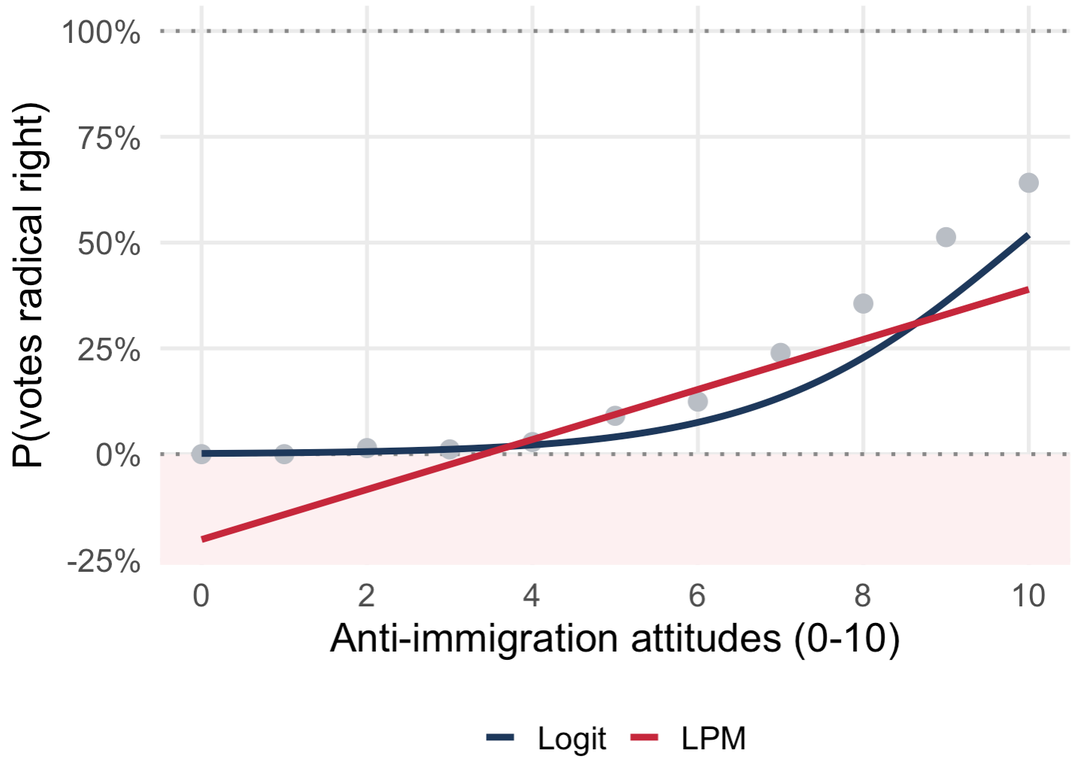
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- The straight line goes **below 0** for voters with liberal views on immigration: impossible probabilities for 17% of the simulated respondents.
- And it misses the curve: too high in the middle, too low among those most opposed to immigration, where support rises fastest.



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The LPM debate

VS

The LPM debate

In its defense

- ✔ **Speaks in percentage points.** No translation needed.
- ✔ **Exact when saturated.** Only categories and all their interactions: it reproduces each cell's share, as in the letter experiment.
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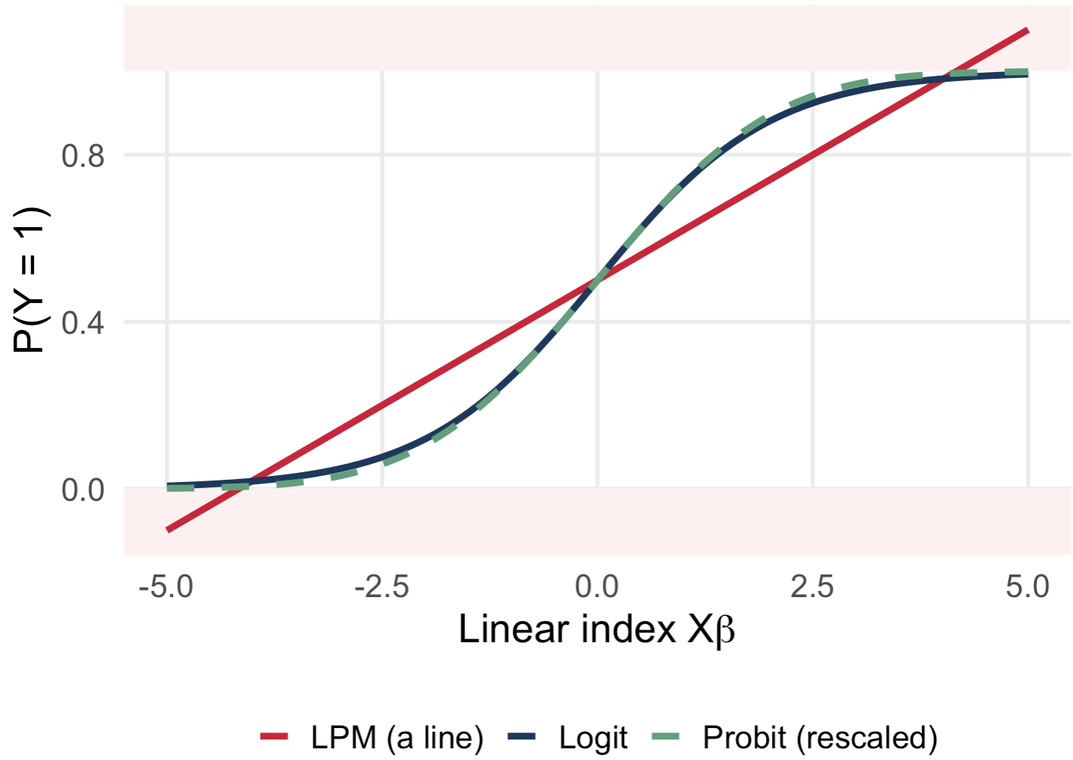
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Verdict: your estimand decides. An average effect of a binary treatment? The LPM is usually fine, and economists and experimentalists often default to it. Probabilities for specific profiles, or outcomes near 0 or 1? Not a bad idea to try nonlinear models.

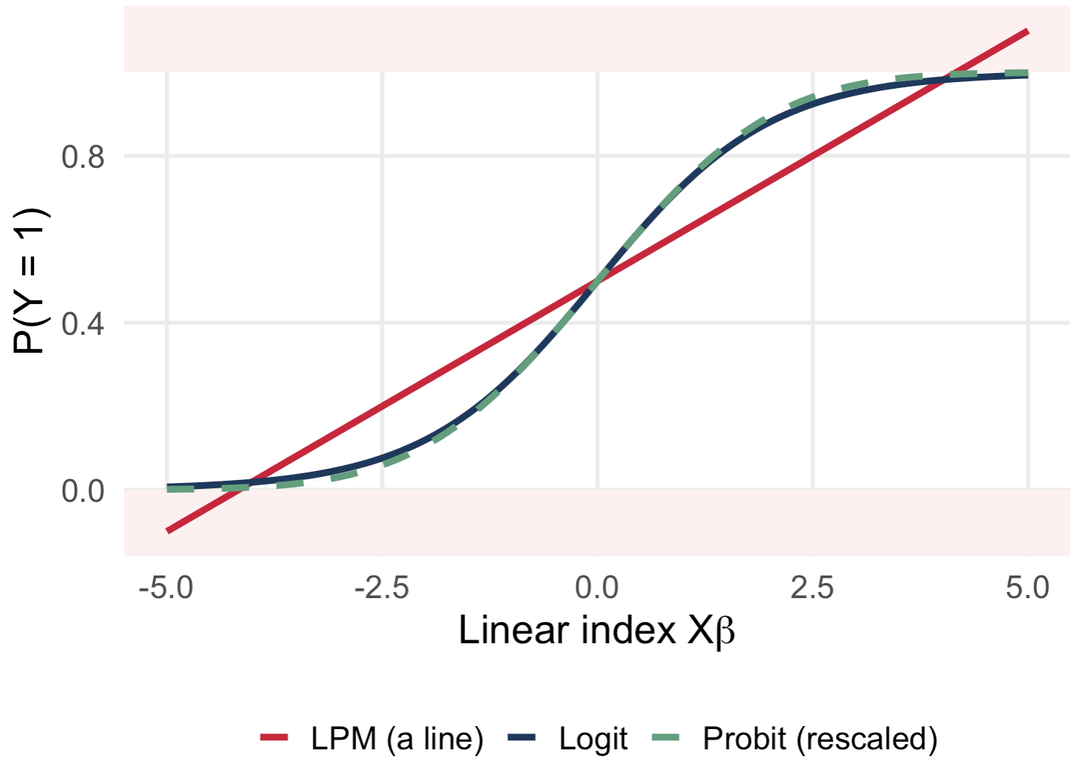
Bend the line: link functions



The line leaves [0, 1] (shaded); the logistic and normal curves bend to stay inside. Probit rescaled to show how similar they are.

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- Keep the linear index $X_i\beta$, but **link** it to the probability through a function g that maps (0, 1) onto all numbers: $g\big(P(Y_i{=}1)\big) = X_i\beta$.

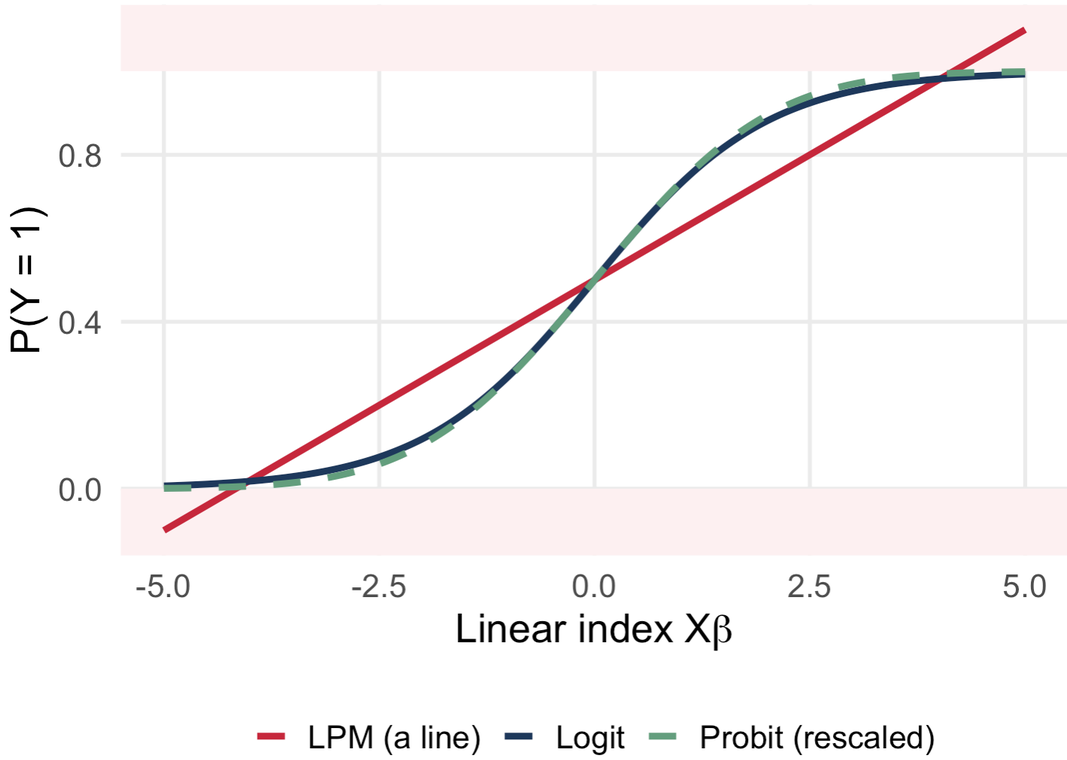


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LPM	P (identity)	$X_i\beta$
Logit	$\log \frac{P}{1-P}$ (log-odds)	$\Lambda(X_i\beta) = \frac{1}{1+e^{-X_i\beta}}$
Probit	$\Phi^{-1}(P)$	$\Phi(X_i\beta)$



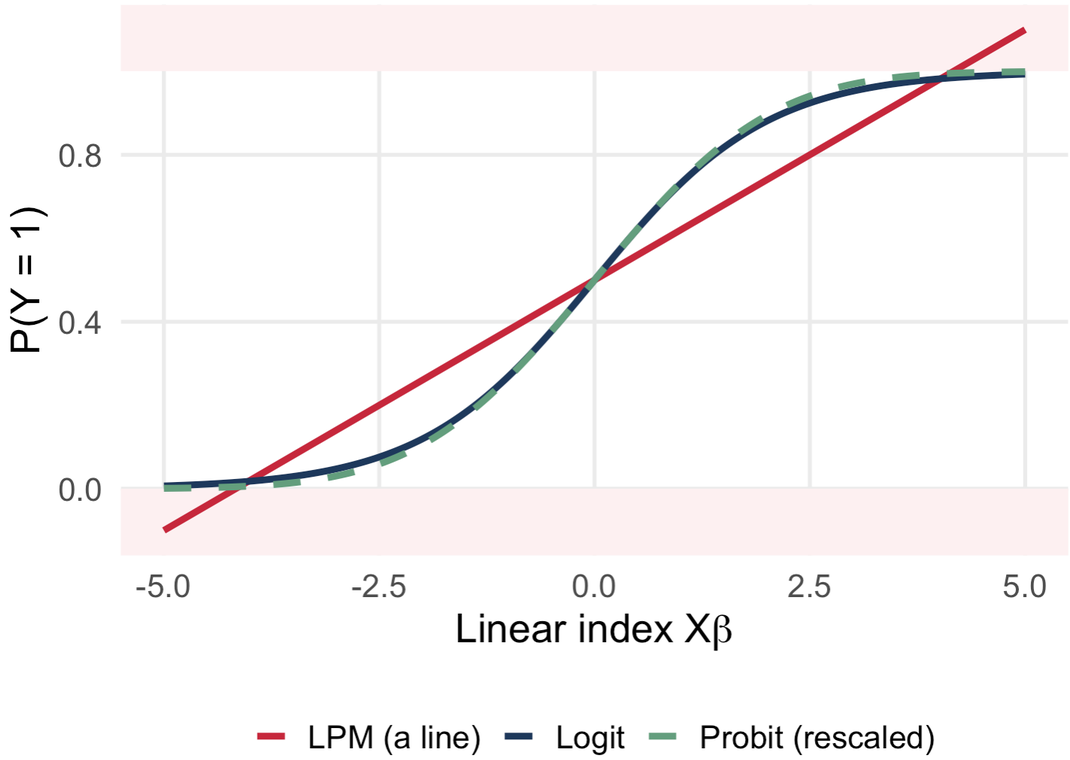
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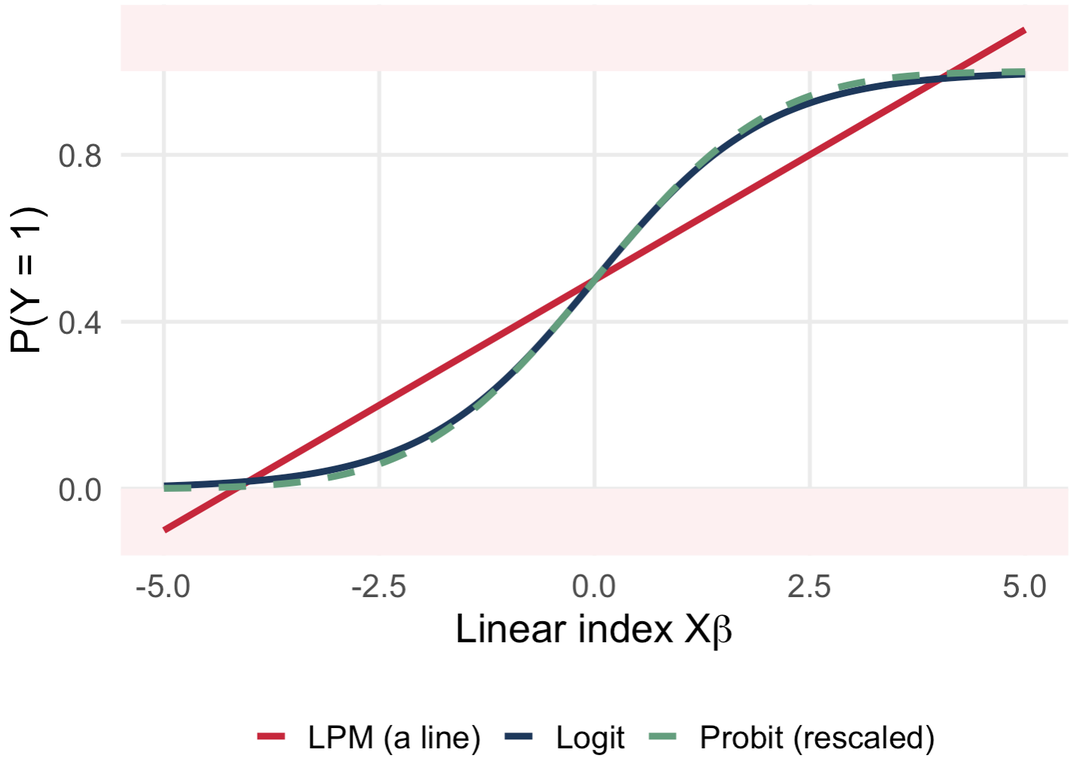
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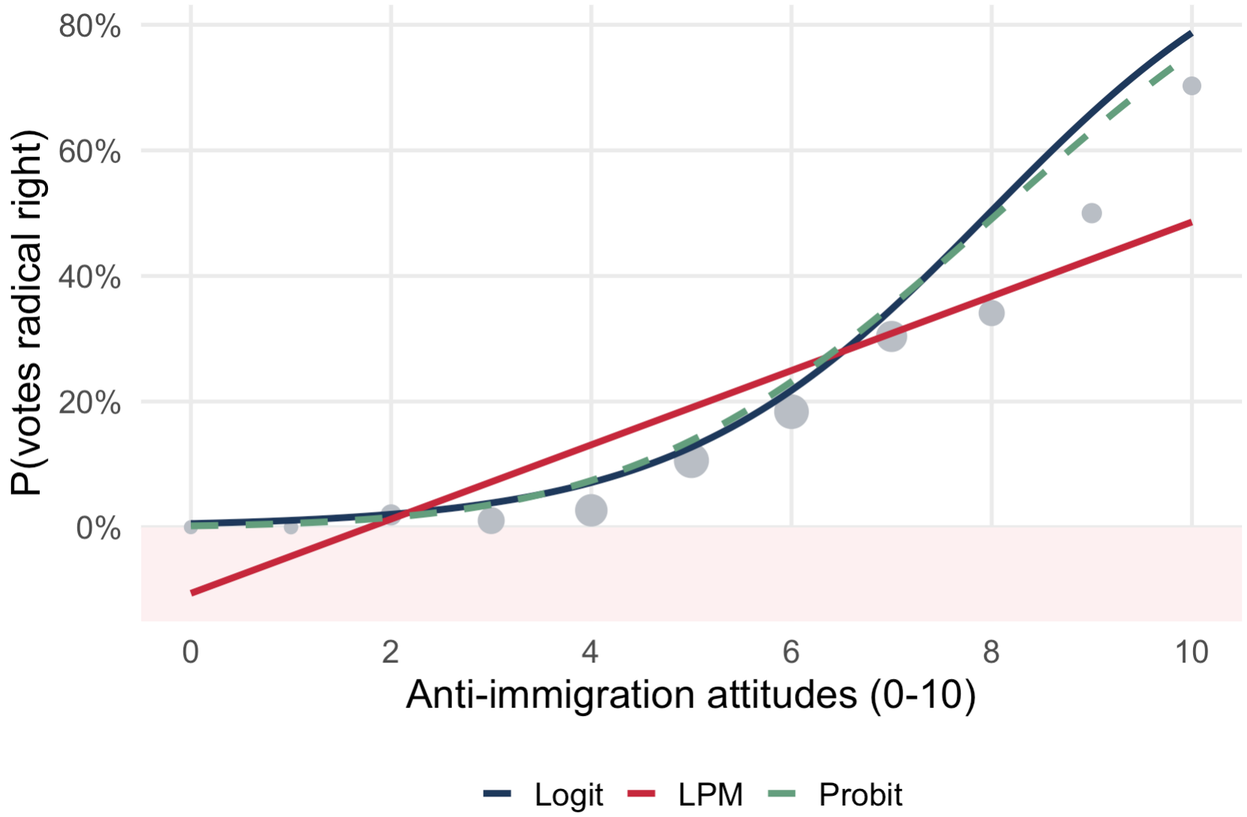
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- The same family covers counts (Poisson, a log link). And last week's 2PL curve was a logit with a latent predictor.



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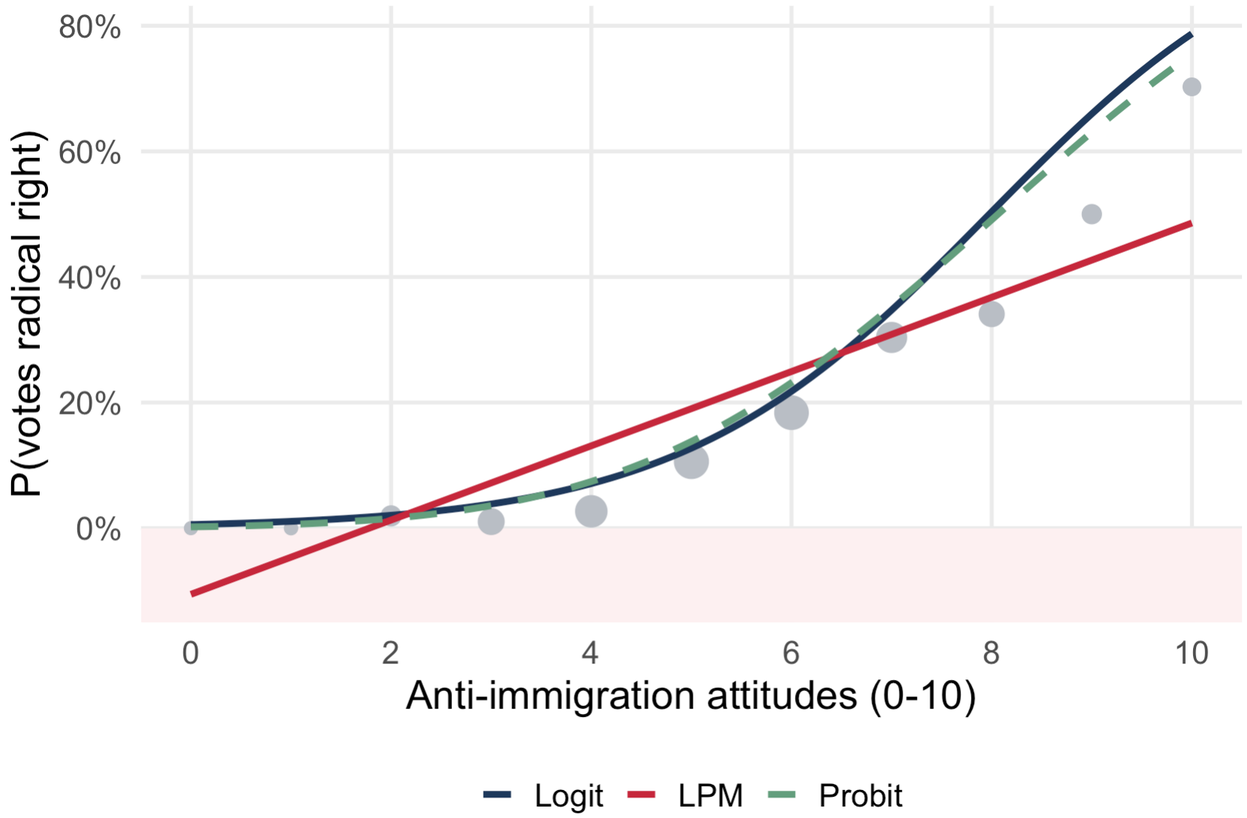
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Gray dots: observed shares among men without a degree (dot size: respondents). Shaded: probabilities that cannot exist.

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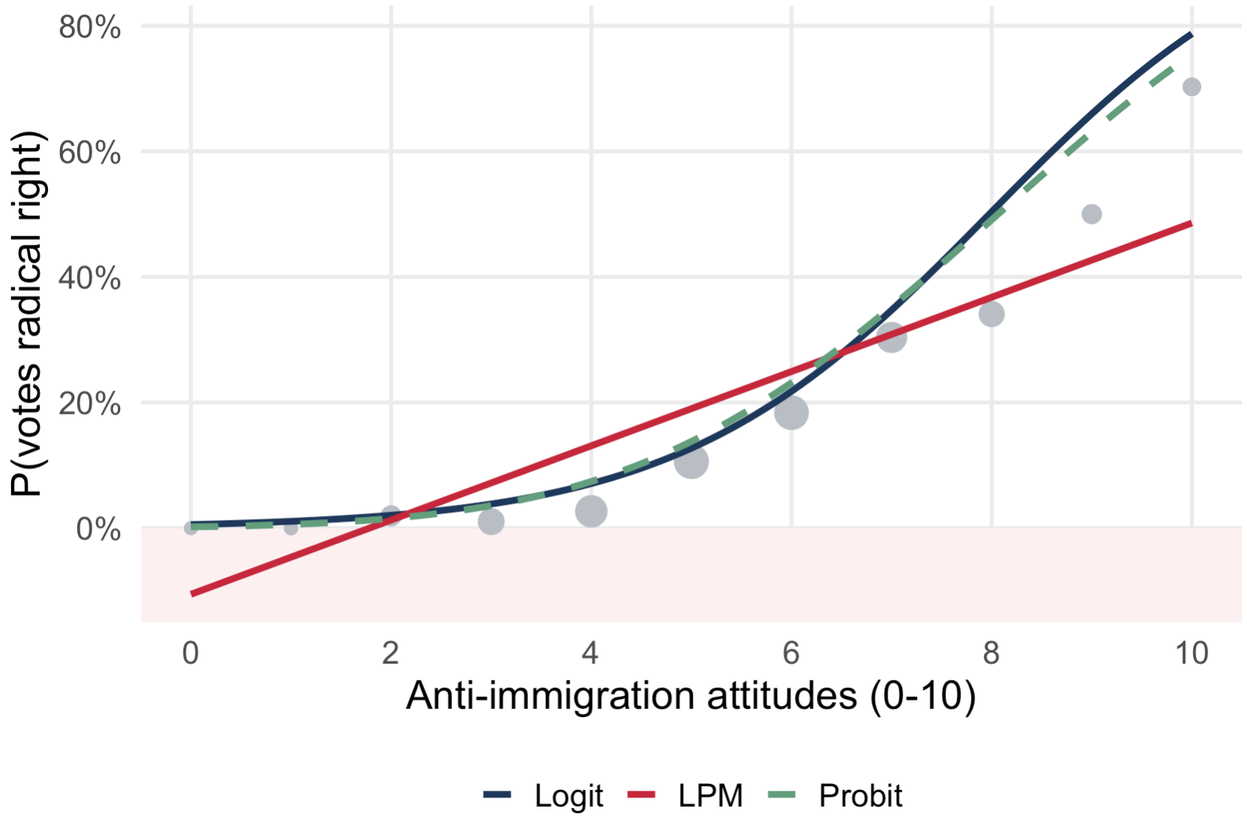
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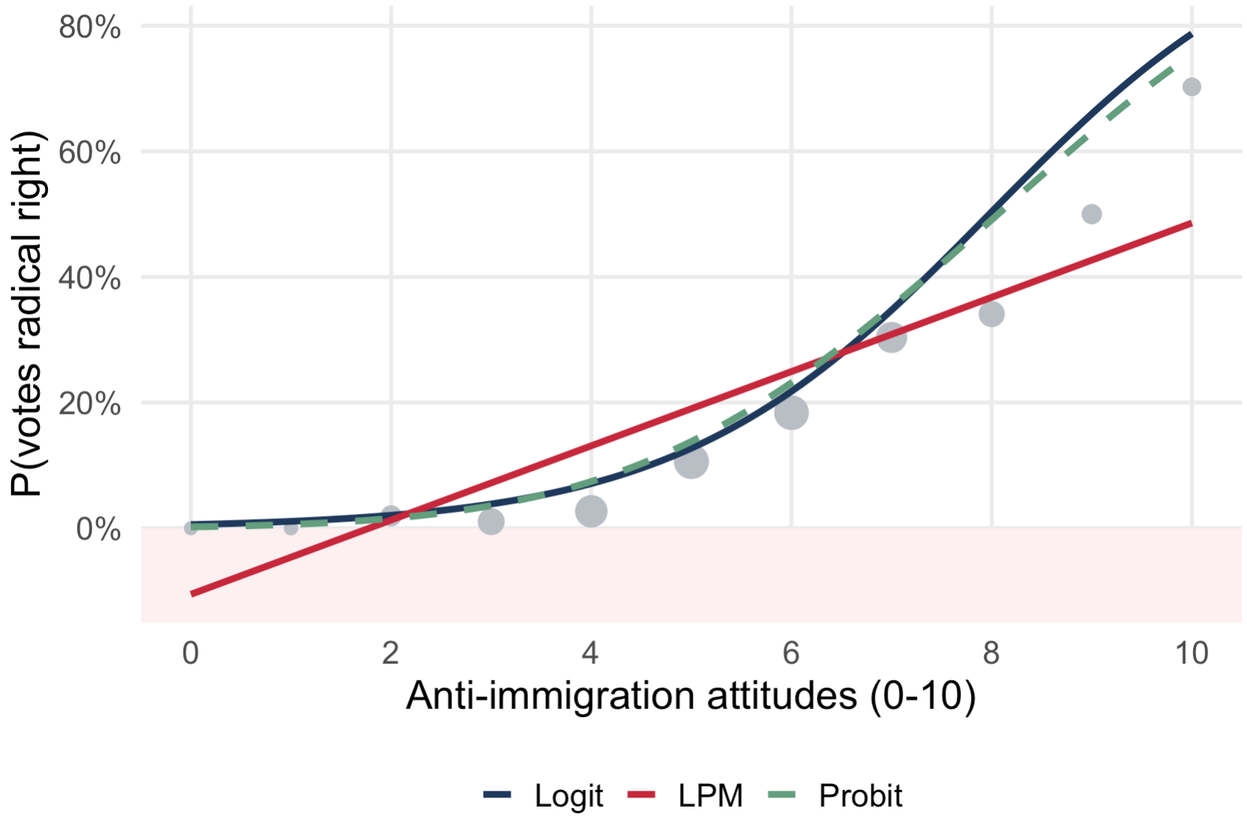
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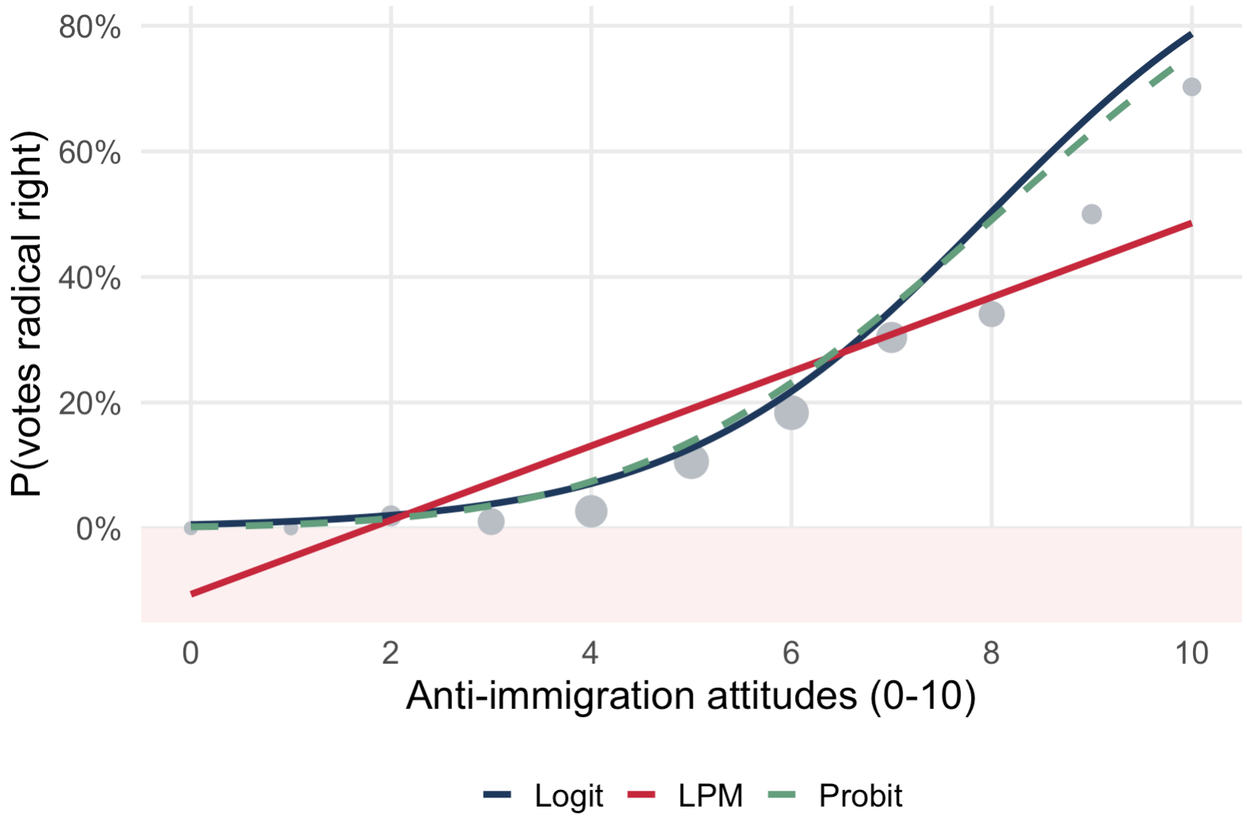
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- Where most voters are, all three agree. They part ways near 0 and 1.



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Reading the output: three models, three languages

	LPM	Logit	Probit
Anti-immigration attitudes (per point)	0.059 (0.003)	0.647 (0.035)	0.355 (0.019)
Male	0.036 (0.011)	0.439 (0.122)	0.251 (0.067)
University degree	-0.025 (0.012)	-0.375 (0.148)	-0.192 (0.078)

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The log-odds rise by 0.44: the **odds** are multiplied by $e^{0.44} = 1.55$. Not to be confused with “men are 55% more likely”.

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Male	0.036 (0.011)	0.439 (0.122)	0.251 (0.067)
University degree	-0.025 (0.012)	-0.375 (0.148)	-0.192 (0.078)

LPM: male

Men are **3.6 percentage points** more likely to vote radical right, holding the rest fixed. Readable as it stands.

Logit: male

The log-odds rise by 0.44: the **odds** are multiplied by $e^{0.44} = 1.55$. Not to be confused with "men are 55% more likely".

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An unobserved index rises by 0.25 standard deviations of its error. No direct reading on any scale your readers use.

Reading the output: three models, three languages

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Same data, three numbers for "male", three meanings. To compare them, translate each into the **same quantity**: the average gap in the probability is 3.6, 4.0, and 4.1 points. Making that translation is what we do after the break.

LPM, logit, probit at a glance

LPM

Curve: a straight line, $X\beta$

A coefficient is: a change in the probability (pp)

Stays in [0, 1]: ✘

Effect varies by person: ✘ (by assumption)

Strength: readable; easy with fixed effects

Often in: economics, political science, experiments

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Curve: logistic, $\Lambda(X\beta)$

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Logit or probit? The curves differ only slightly in the tails, so the choice rarely changes predicted probabilities or average effects. Use the one your field reads, and **never** compare a logit coefficient with a probit one.

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Next: what is a logit coefficient?

What is a logit coefficient?

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- Rearranging the logit: $\log \frac{P(Y_i=1)}{1-P(Y_i=1)} = X_i\beta$. A logit coefficient is a change in the **log-odds**; e^{β} is an **odds ratio**.

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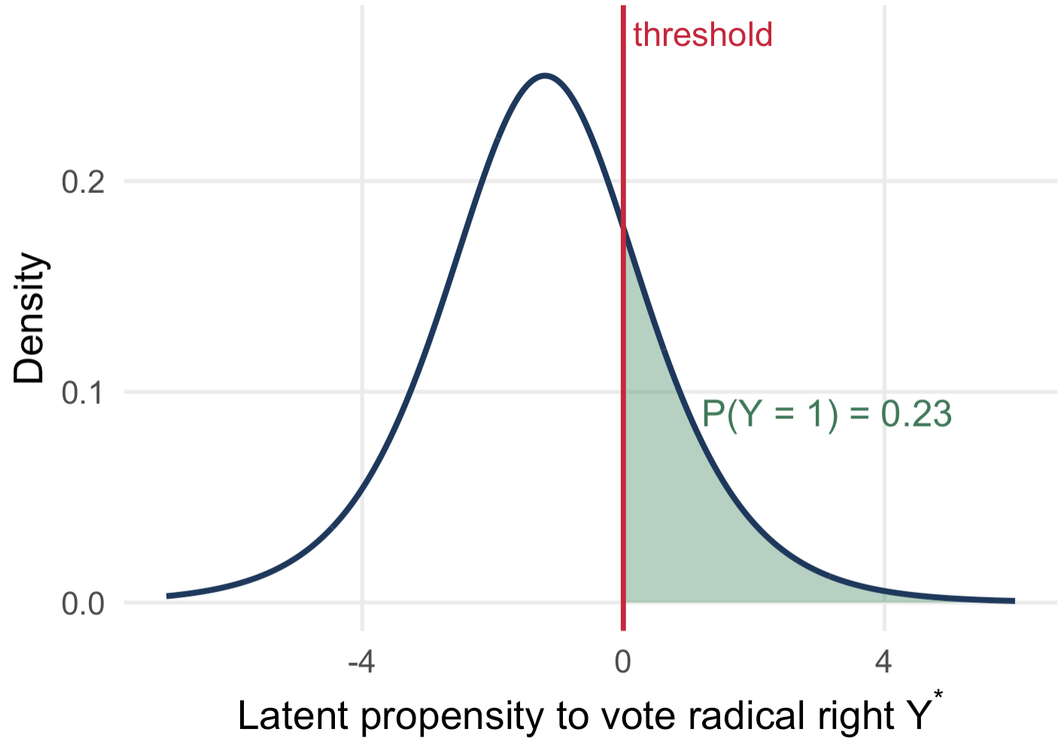
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The same odds ratio, **1.44**, at four baselines

Turnout without the letter	With the letter	How much more likely?	Change
5%	7.0%	**+41%**	+2.0 pp
30%	38.1%	**+27%**	+8.1 pp
70%	77.0%	**+10%**	+7.0 pp
95%	96.5%	**+2%**	+1.5 pp

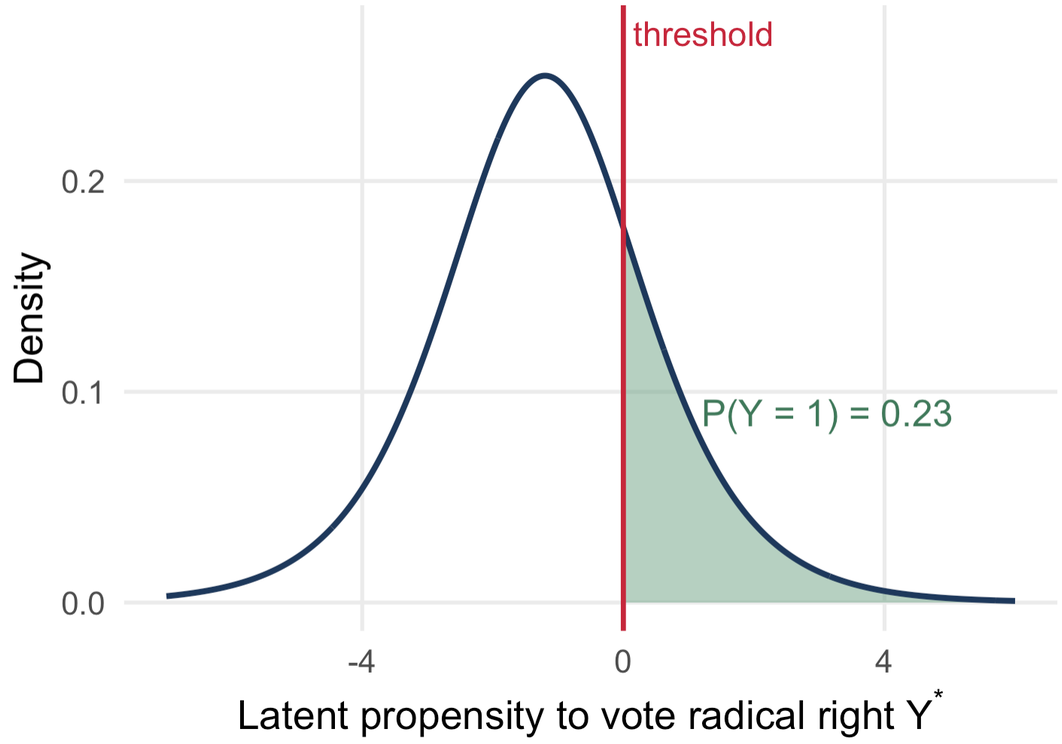
Red: what readers want to know, and not what the odds ratio says. Shaded: the letters.

Another way to read the same curve



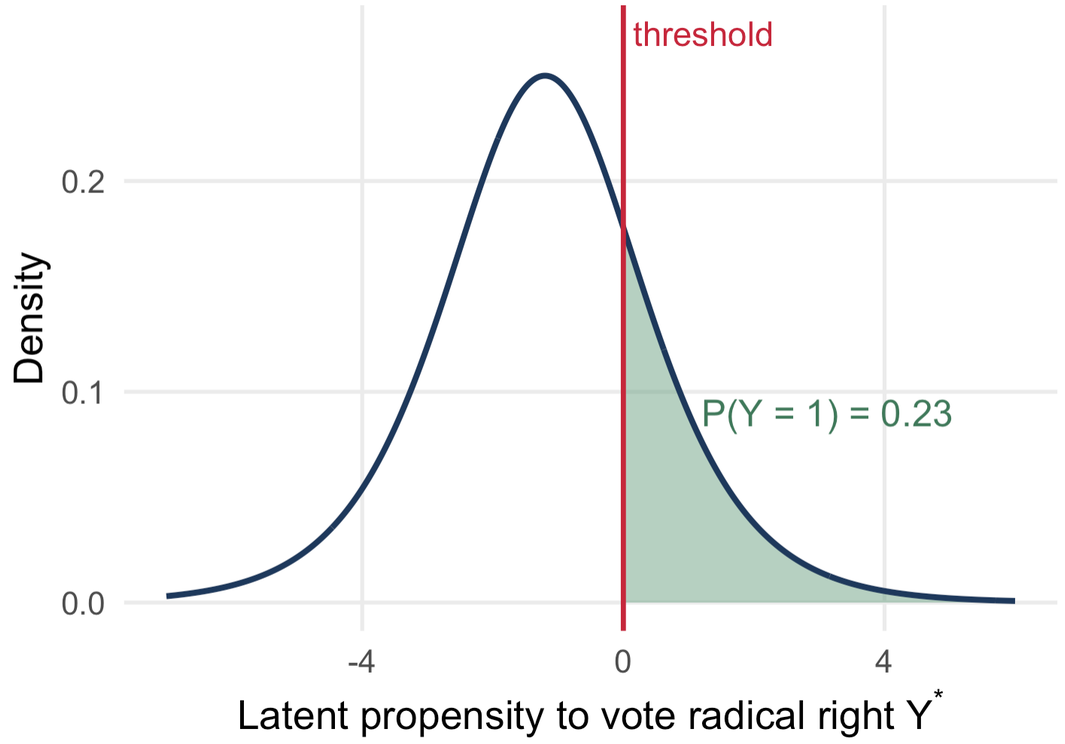
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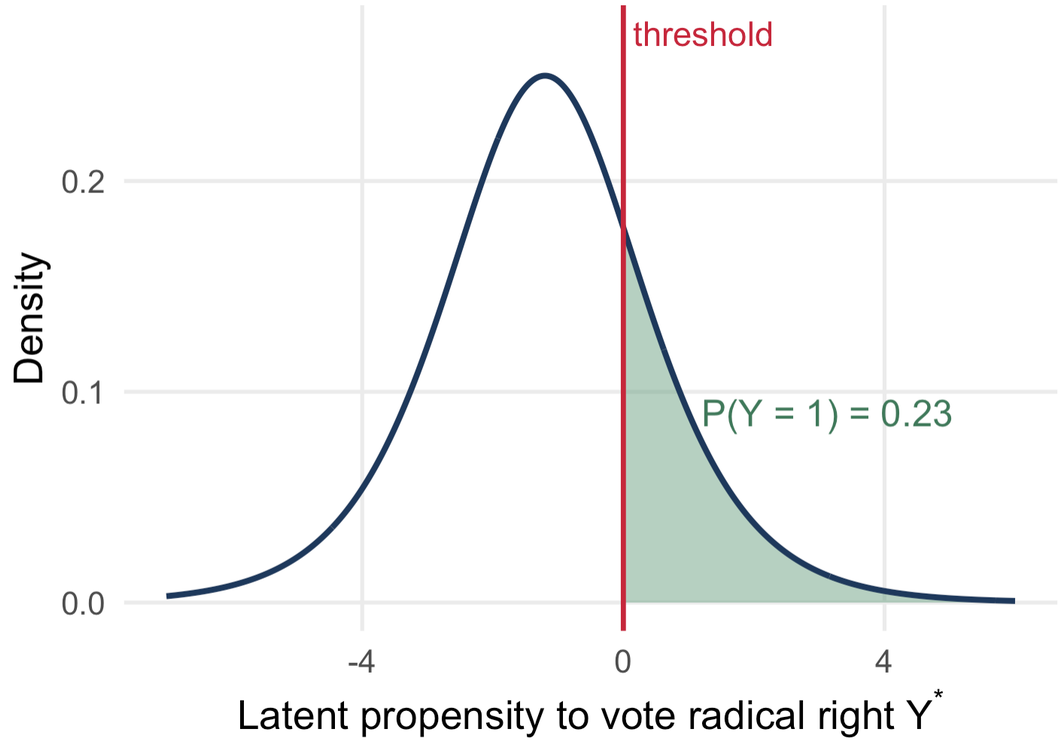
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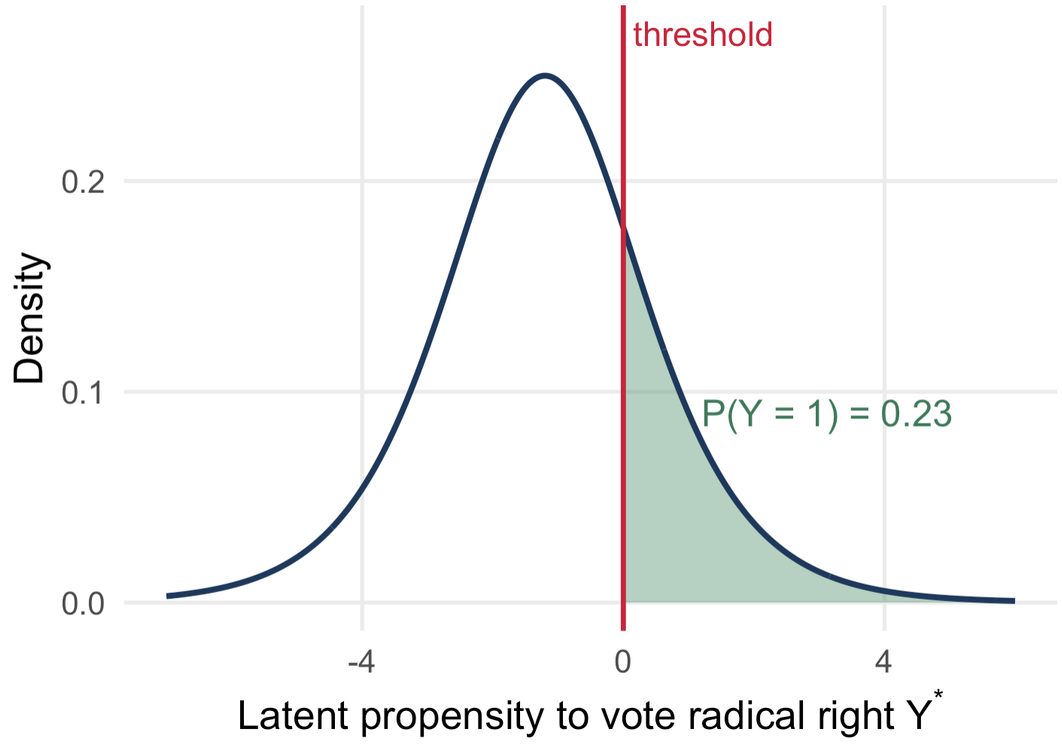
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- Political scientists know this logic from **rational-choice models of voting**: act if the utility of the option beats the alternative (the calculus of voting, Riker & Ordeshook 1968).
- Same model, a different **reading** of what the outcome is. That reading decides what a logit coefficient means.



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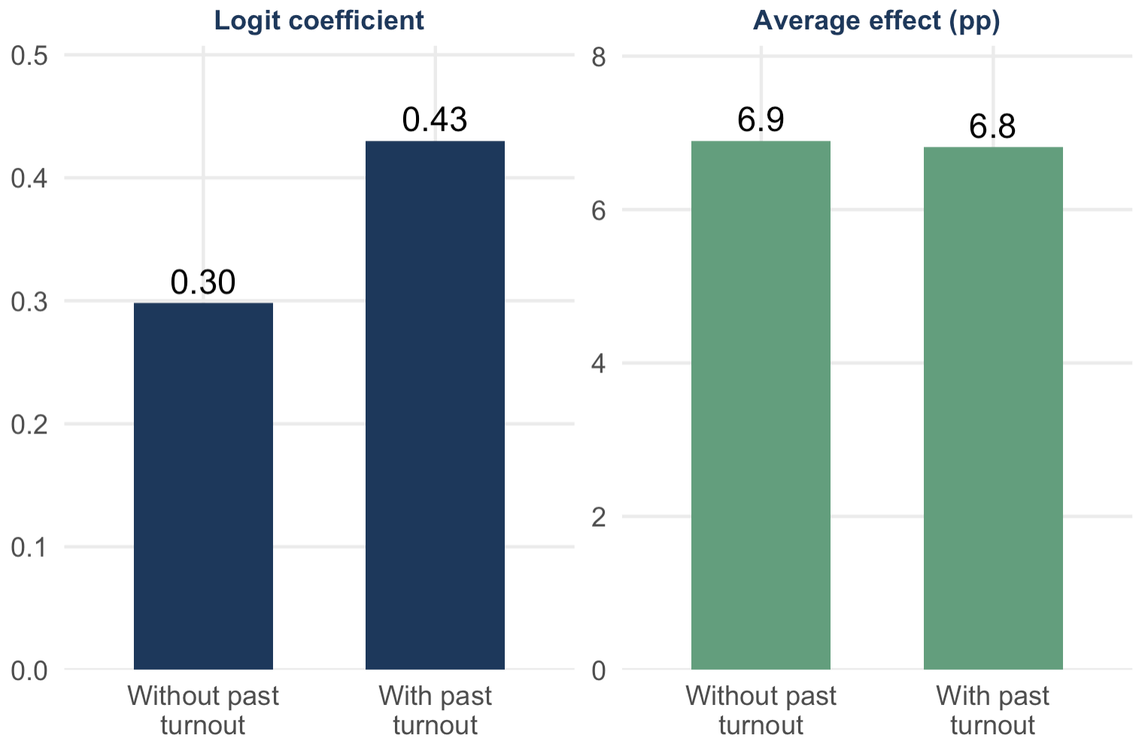
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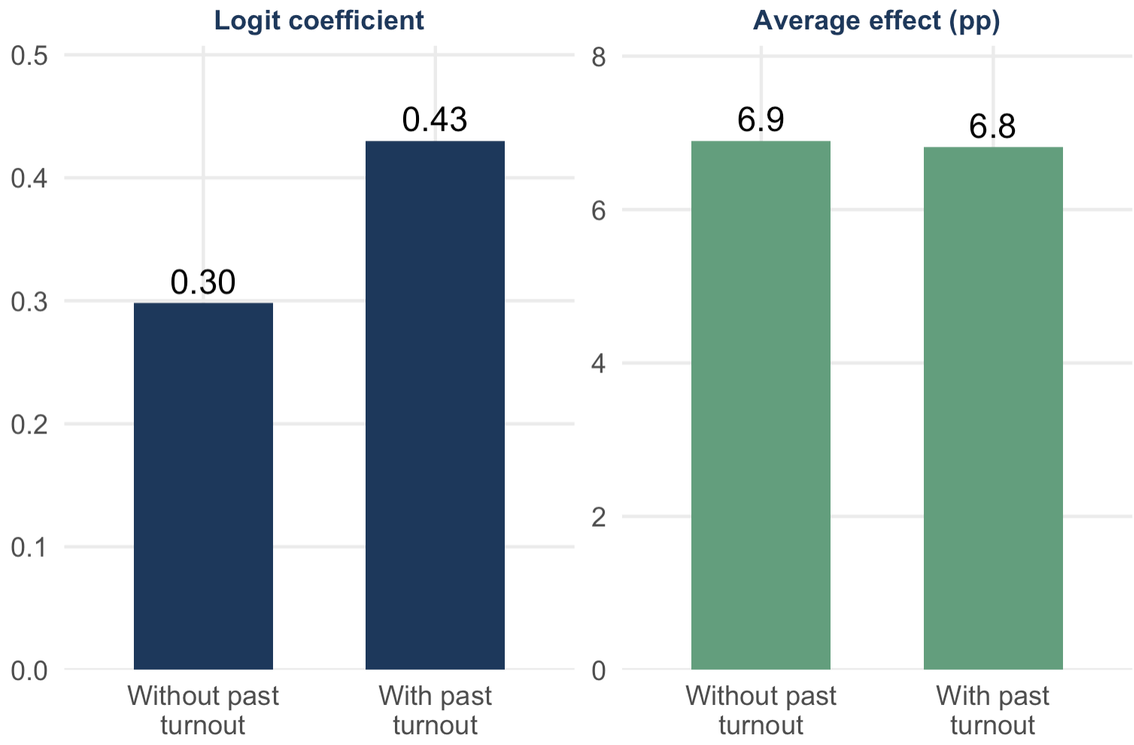
- Anything that changes the **unexplained variation** changes **every** coefficient, even if nothing about the true effect changes.

Add a covariate that has nothing to do with the treatment



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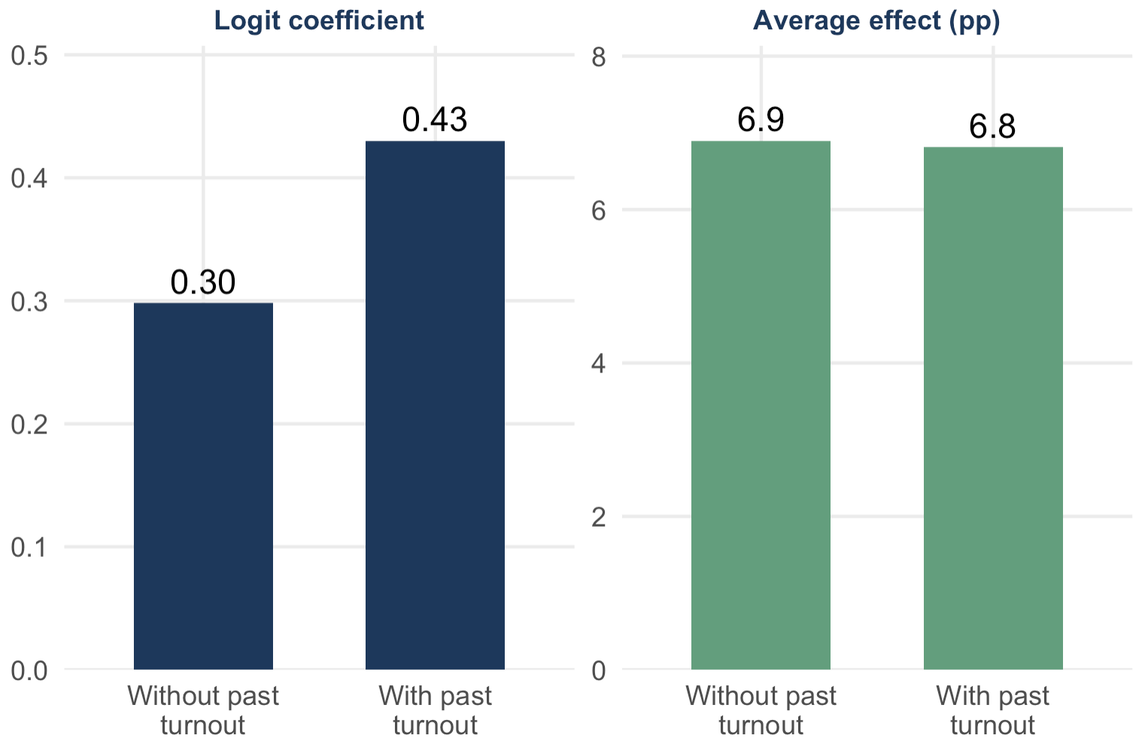
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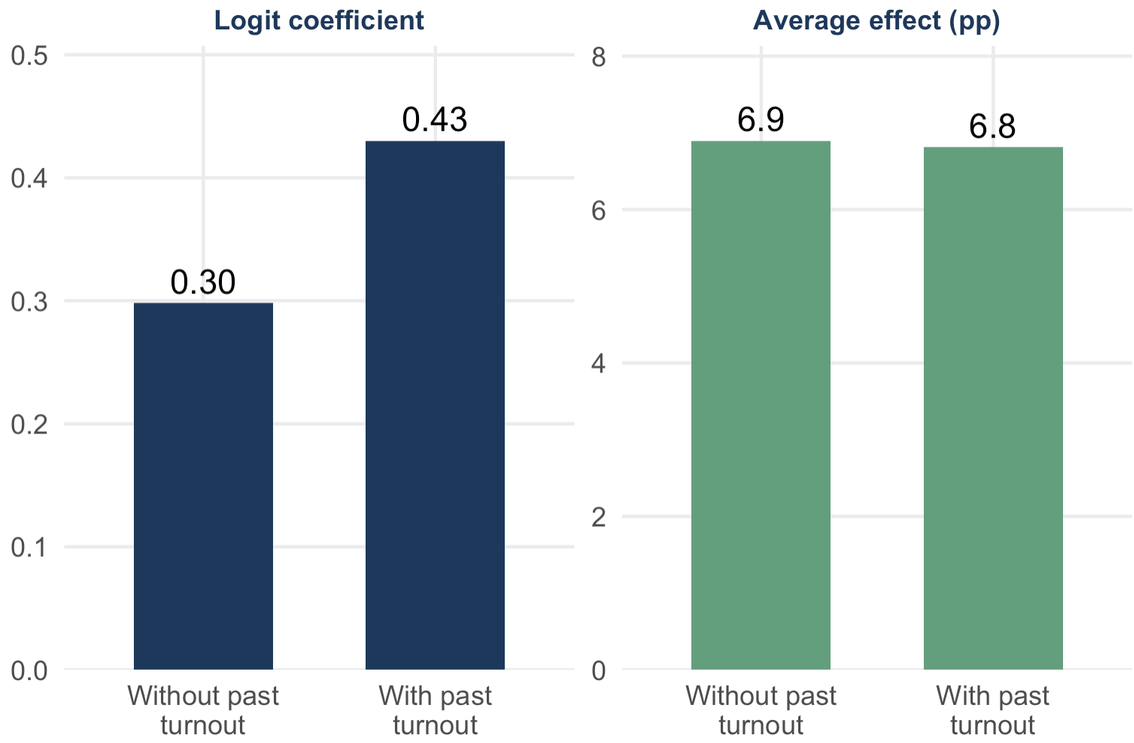
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- The logit coefficient on the letter jumps from 0.30 to 0.43. No confounding: **non-collapsibility**.
- The average effect on the probability barely moves: 6.9 vs 6.8 pp.



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	P(enters), no degree	P(enters), degree	Difference	Odds ratio
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- Why: pooling averages probabilities, and an average of probabilities is not an average of odds. Same story as the letter: add or drop a covariate that predicts the outcome, and the odds ratio moves.

Mood (2010) and Buis (2017): the takeaways

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THE OUTCOME IS A DECISION ON A LATENT PROPENSITY

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Where they agree: don't read mediation off a change in logit coefficients (use a decomposition such as KHB: Karlson, Holm & Breen 2012); say what your 0/1 outcome **means**; and show probability-scale quantities either way.

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A latent propensity

You have a theory of individual decisions. The calculus of voting: vote if the expected benefit, minus the cost, plus civic duty, is positive (Riker & Ordeshook 1968). Coefficients are effects on that utility: compare them across models and groups only with care.

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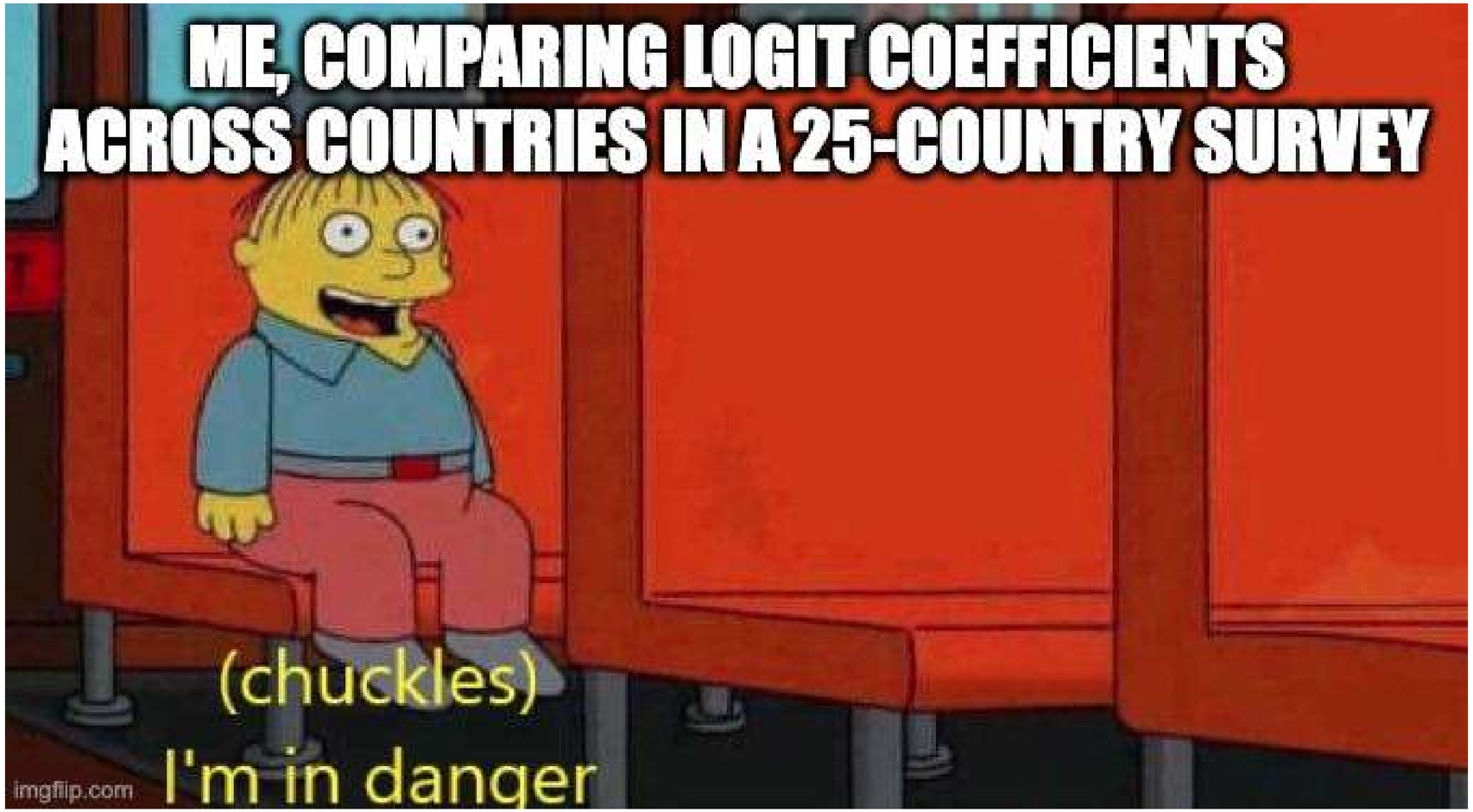
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- Gerber et al. do both: OLS for the effect of each letter, logits for “underlying voting propensities” (footnote 10). Either way, readers act on the **probability scale**.

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After the break: quantities of interest

Break

10 minutes

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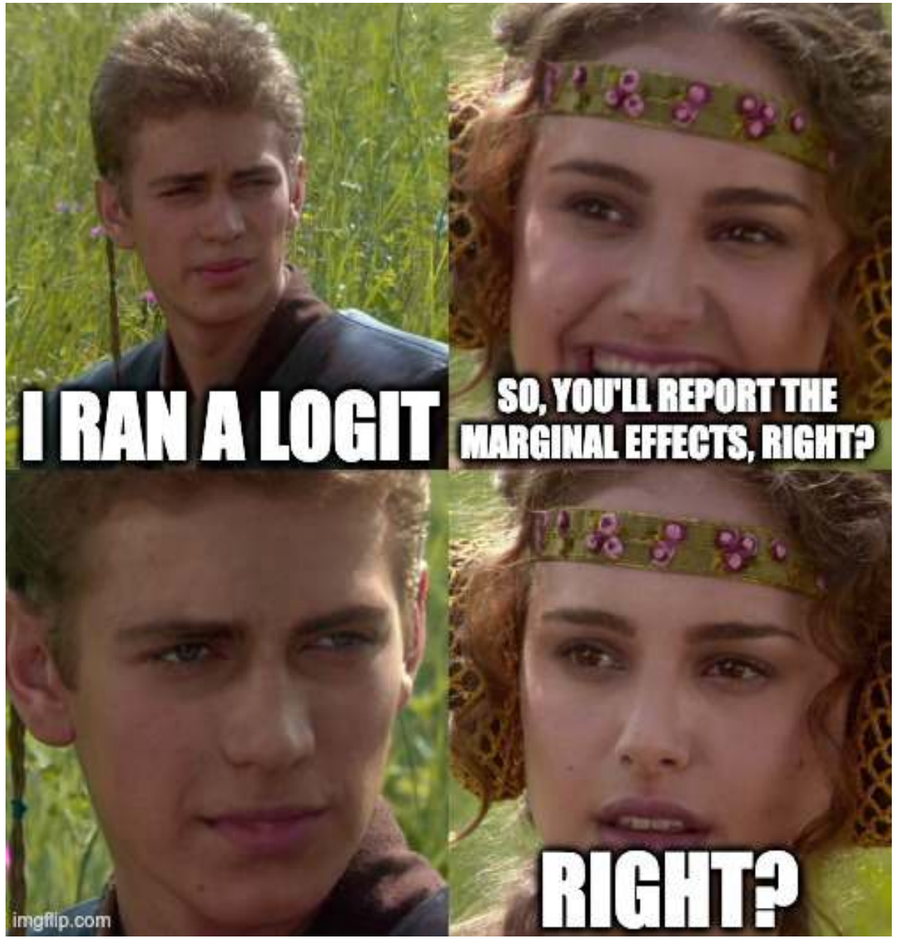
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Difference in probability	Ratio of probabilities	Odds ratio	Logit coefficient
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= their OLS coefficient, .081	"27% more likely to vote"	multiplies the odds	change in the log-odds

Which one goes in your abstract? Week 3: $\hat{\beta}$ is not an estimand. **Today, we see another example of this.**

One result, four numbers



Your turn ✏️ (~6 min, solo, calculators OK)

The **Self** letter showed only your own household's voting record. Turnout: **34.5%**, vs **29.7%** in the control group.

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A press release says "The letter made people 25% more likely to vote." Right or wrong? Which number would you report?

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29.7% · 37.8%

→

ODDS, P / (1 – P)
0.42 · 0.61

→

ODDS RATIO
0.61 / 0.42 = 1.44

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- Interactions make it worse: in a logit, the coefficient on the product term no longer tells you the size of an interaction, **or even whether there is one** (Mize 2019; McCabe et al. 2022). We will see it in the audit.

Why logit coefficients are hard to read

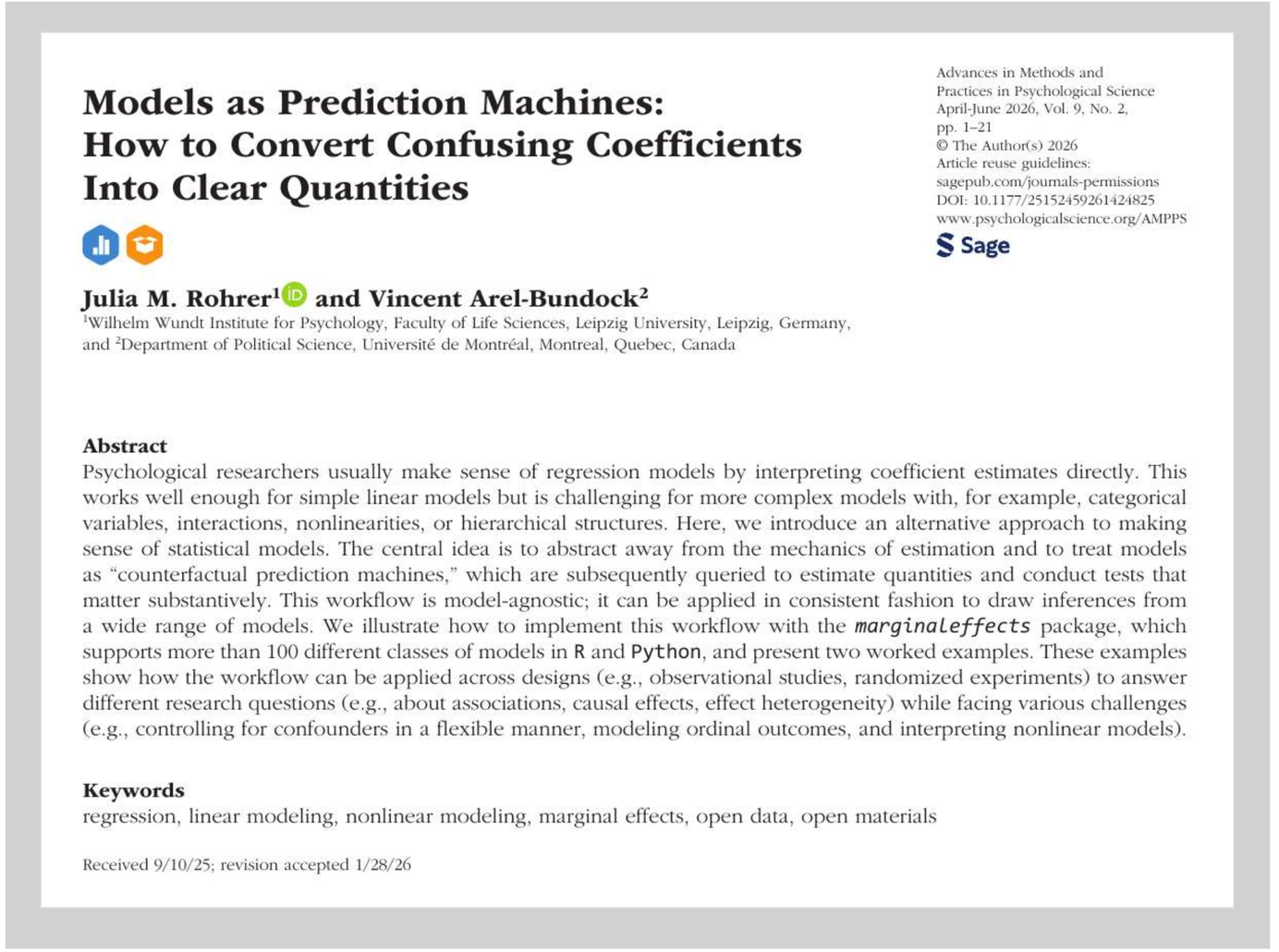
- A logit coefficient is a log odds ratio: the log of a **ratio of ratios of probabilities**. Unpacked, for the Neighbors letter:

PROBABILITIES	→	ODDS, P / (1 − P)	→	ODDS RATIO	→	LOGIT COEFFICIENT
29.7% · 37.8%		0.42 · 0.61		0.61 / 0.42 = 1.44		log 1.44 = 0.36

- People already struggle with simple probabilities. **A ratio of ratios is (certainly) harder still**, for readers and for researchers.
- Interactions make it worse: in a logit, the coefficient on the product term no longer tells you the size of an interaction, **or even whether there is one** (Mize 2019; McCabe et al. 2022). We will see it in the audit.

> **Next week**, multilevel models add random effects and variance components: more numbers that only make sense in combination. The remedy is the same, so we build it today.

I invite you to read



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Models as Prediction Machines: How to Convert Confusing Coefficients Into Clear Quantities

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Abstract
Psychological researchers usually make sense of regression models by interpreting coefficient estimates directly. This works well enough for simple linear models but is challenging for more complex models with, for example, categorical variables, interactions, nonlinearities, or hierarchical structures. Here, we introduce an alternative approach to making sense of statistical models. The central idea is to abstract away from the mechanics of estimation and to treat models as "counterfactual prediction machines," which are subsequently queried to estimate quantities and conduct tests that matter substantively. This workflow is model-agnostic; it can be applied in consistent fashion to draw inferences from a wide range of models. We illustrate how to implement this workflow with the `marginaleffects` package, which supports more than 100 different classes of models in `R` and `Python`, and present two worked examples. These examples show how the workflow can be applied across designs (e.g., observational studies, randomized experiments) to answer different research questions (e.g., about associations, causal effects, effect heterogeneity) while facing various challenges (e.g., controlling for confounders in a flexible manner, modeling ordinal outcomes, and interpreting nonlinear models).

Keywords
regression, linear modeling, nonlinear modeling, marginal effects, open data, open materials

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Rohrer & Arel-Bundock (2026), *Advances in Methods and Practices in Psychological Science* 9(2): 1–21.

Models as prediction machines

> *"...interpreting model coefficients directly is a difficult and error-prone business. But what would be an alternative? How else can one interpret statistical models? A good starting point is the realization that often, model coefficients are not the logical end of a model; they are* ***merely the statistical machinery*** *that allows the model to achieve its goal. That goal is straightforward: to predict the outcome that one should expect for an individual or unit given some characteristics and to quantify the uncertainty of that prediction. The core of our alternative approach thus involves thinking of the model as a* ***prediction machine*** *and transforming the quantities it produces to answer substantive research questions."*

Rohrer & Arel-Bundock (2026, p. 3), emphasis added

The prediction-machine workflow

Adapted from Rohrer & Arel-Bundock (2026), Fig. 1. Italics: our running example.

The prediction-machine workflow

1 · Build a model

Select a model that matches the research question and the field's requirements.
Fit it in your preferred framework: frequentist, Bayesian, machine learning.

A logit of radical-right voting on attitudes, sex, degree, rural residence, and age.

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2 · Compute estimands

Target quantities: predictions, comparisons, slopes.
Decisions: for whom (one unit, or an average) and on which scale.

The gender gap in the probability of a radical-right vote, averaged over all voters.

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3 · Test them

Uncertainty for each quantity.
Tests: null hypotheses (not only zero) and equivalence tests.

Is the gap different from zero? Is it larger among voters without a degree?

Adapted from Rohrer & Arel-Bundock (2026), Fig. 1. Italics: our running example.

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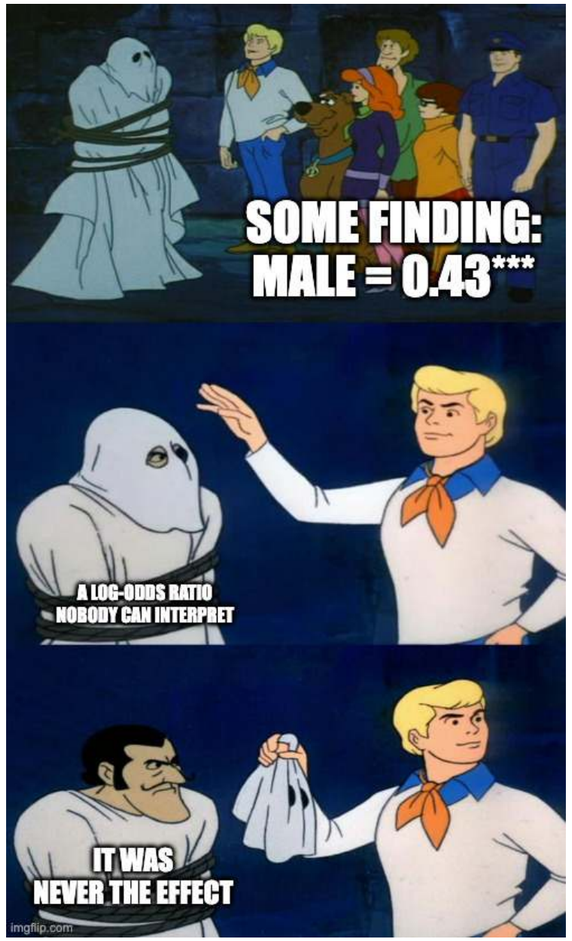
Uncertainty for each quantity.
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Is the gap different from zero? Is it larger among voters without a degree?

The standard workflow jumps from step 1 to reading and testing the coefficients themselves. Steps 2 and 3 are where the two part ways.

Adapted from Rohrer & Arel-Bundock (2026), Fig. 1. Italics: our running example.

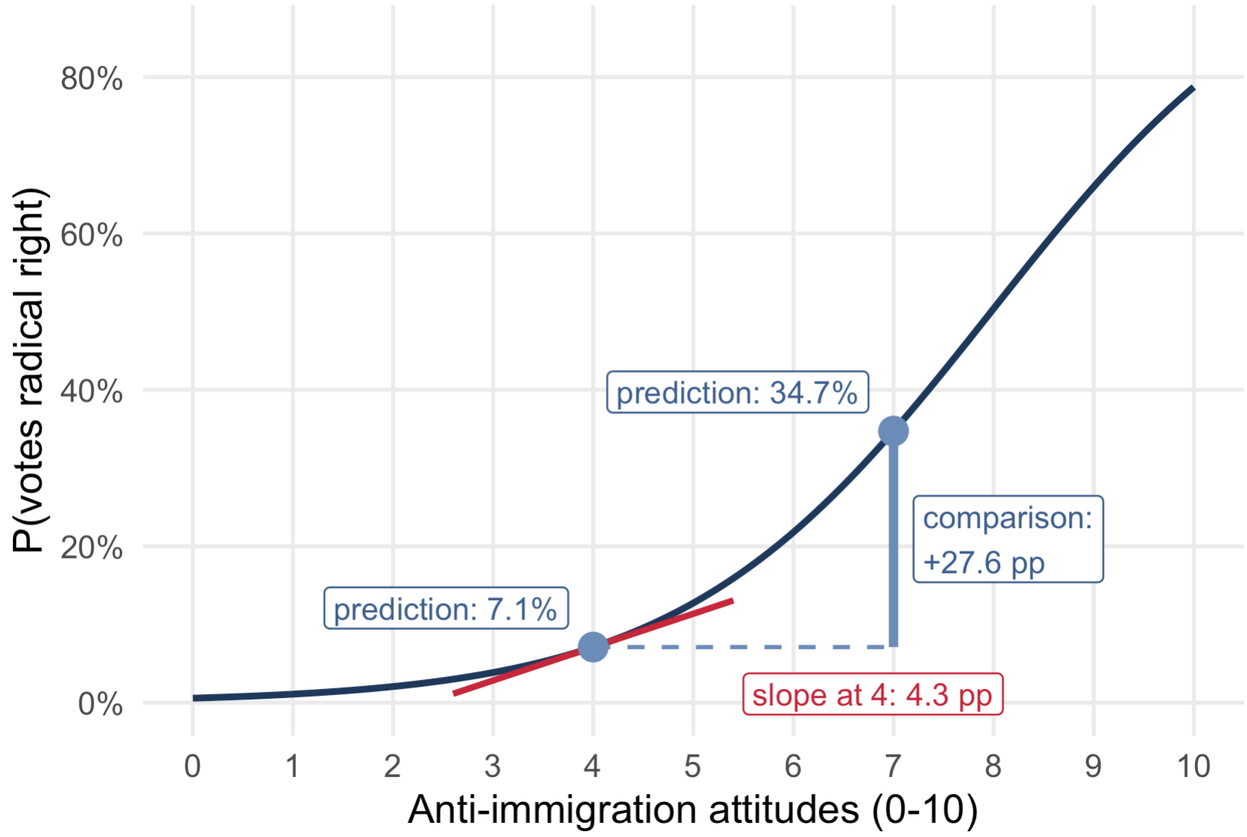
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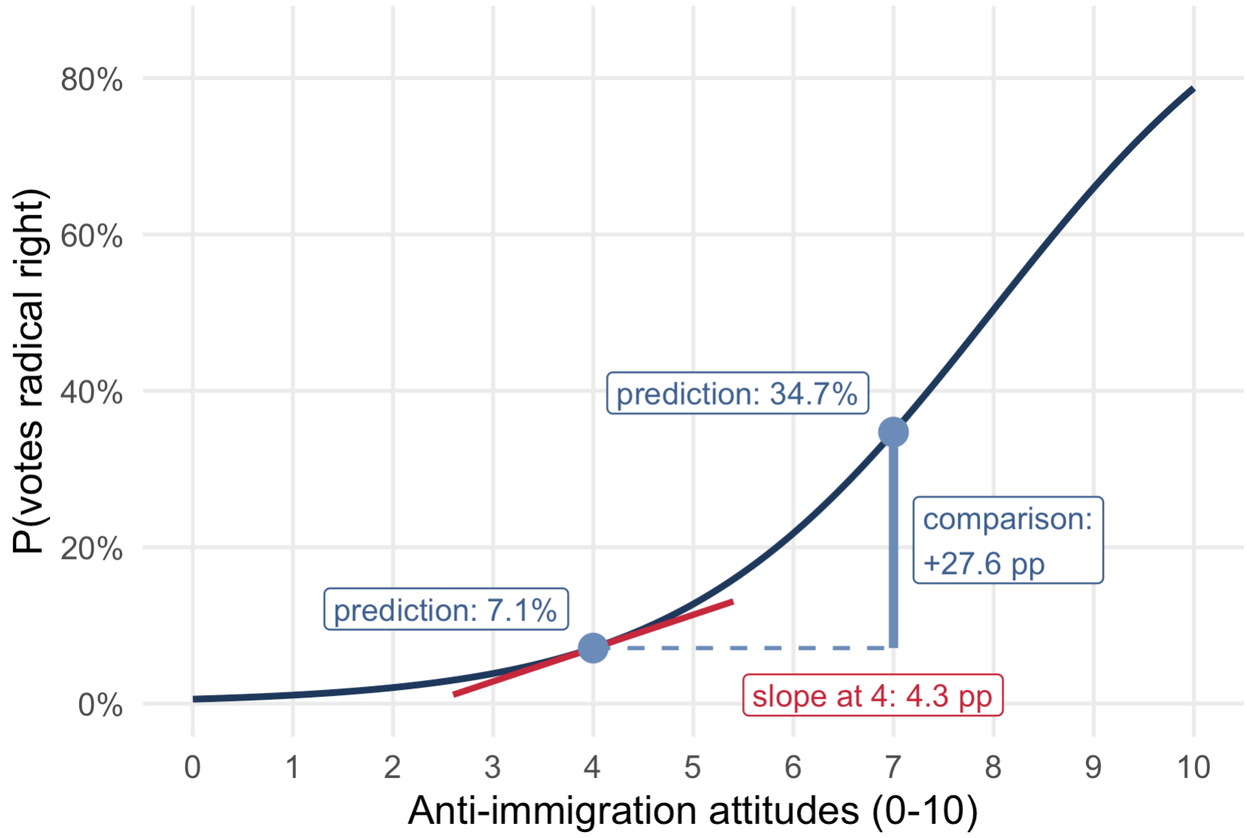
Three questions for the machine



Our radical-right logit, after Rohrer & Arel-Bundock (2026), Fig. 2 (life satisfaction and income in the German SOEP).

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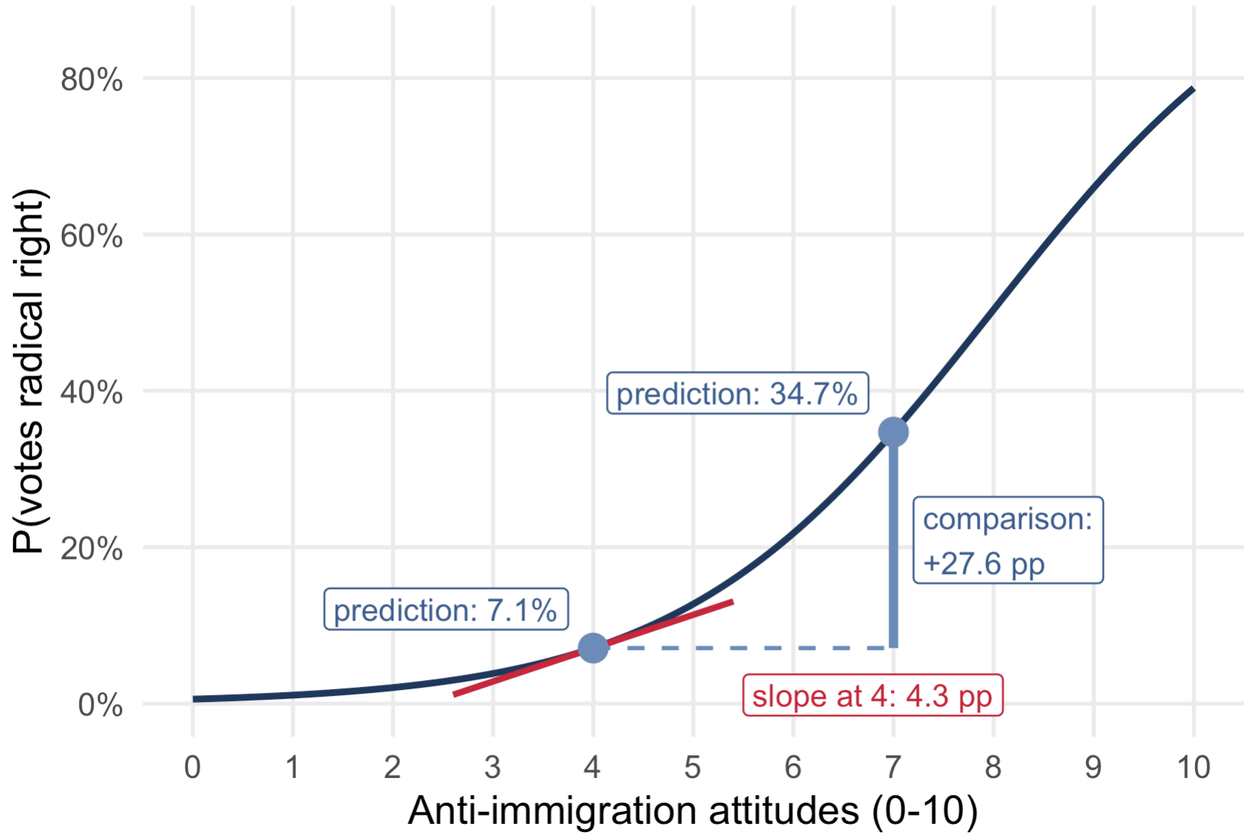
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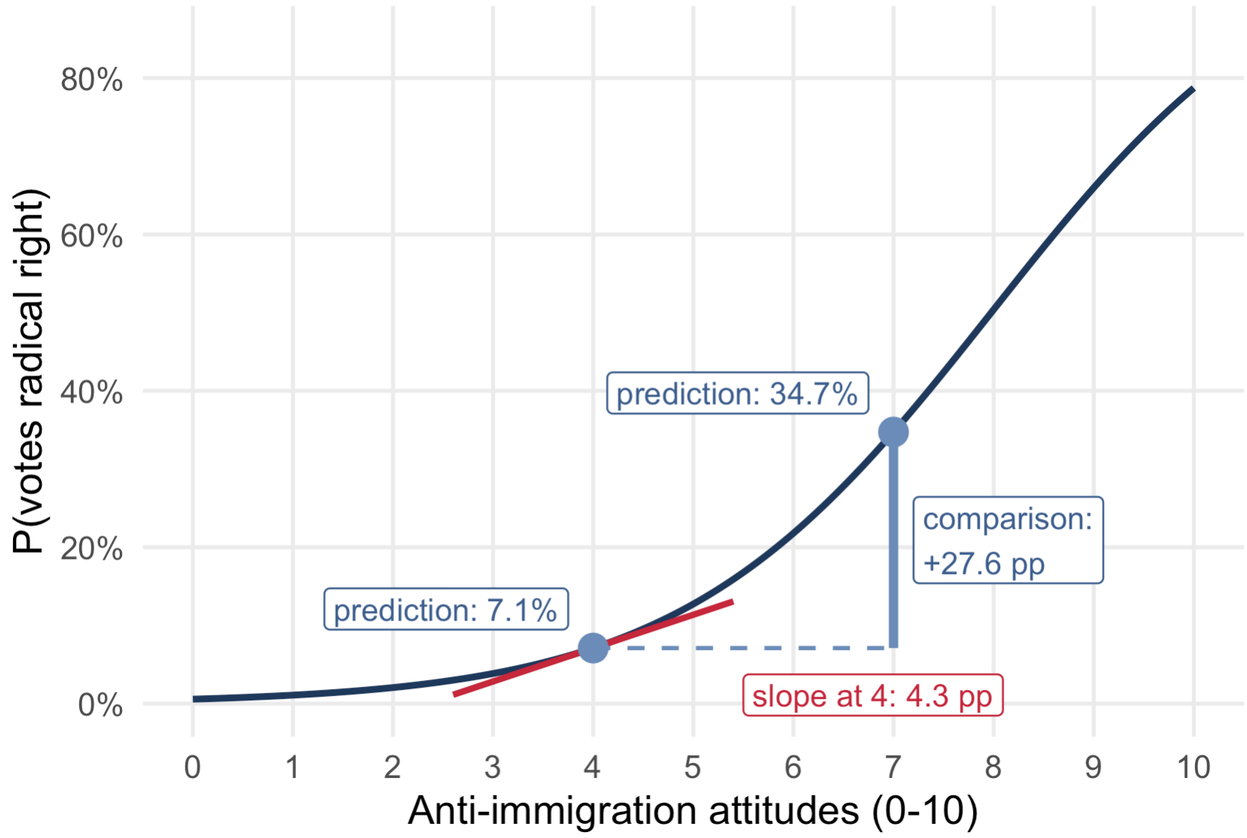
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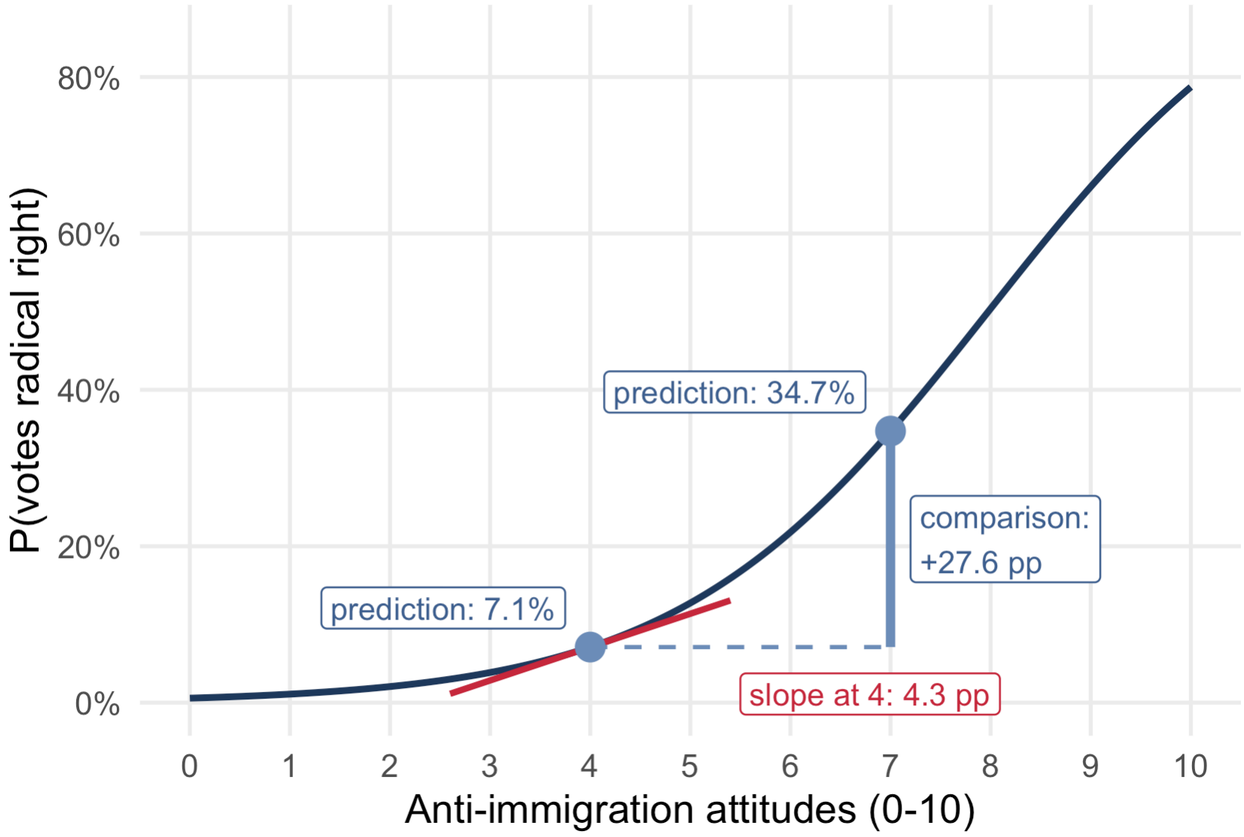
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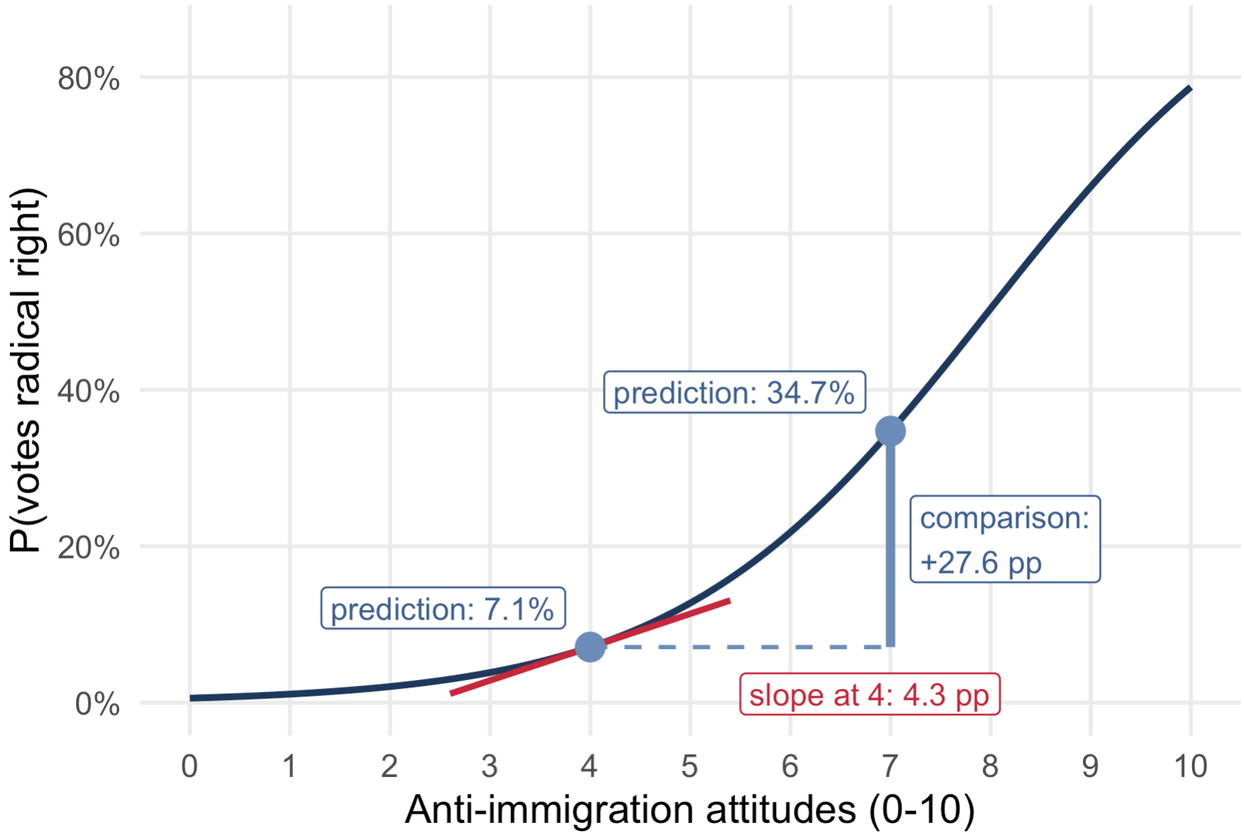
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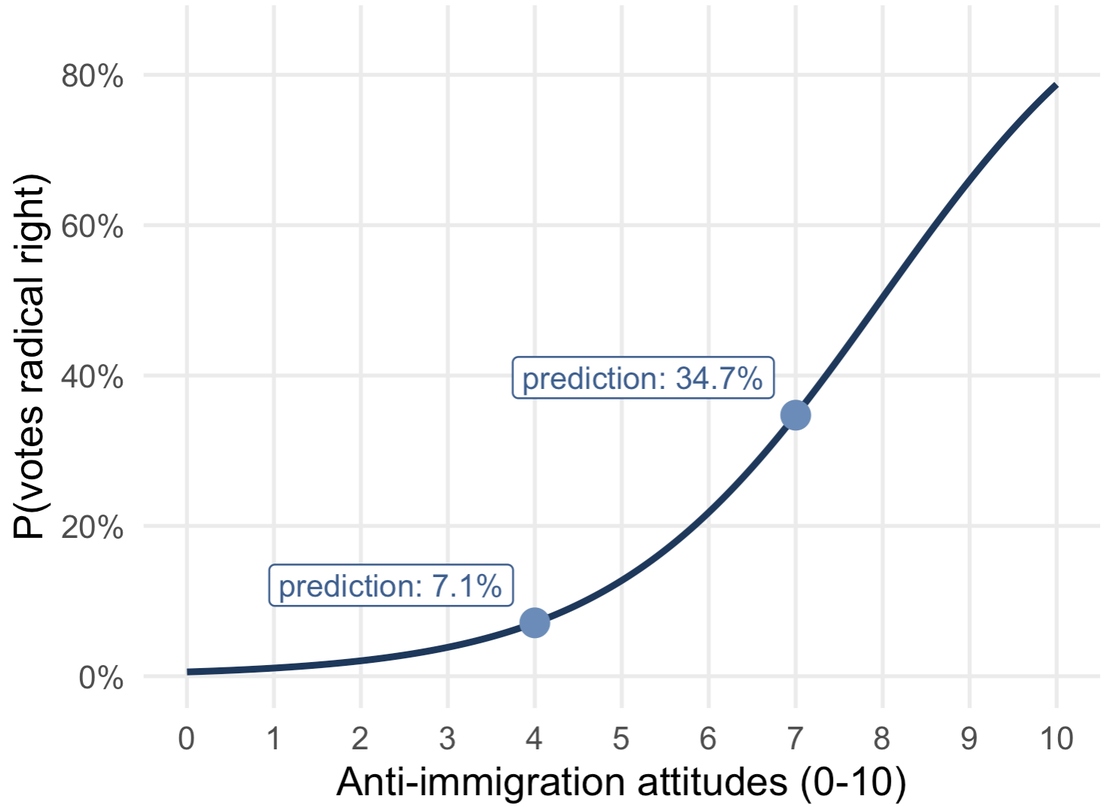
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- Every quantity in the rest of this part is one of these, for some set of people.



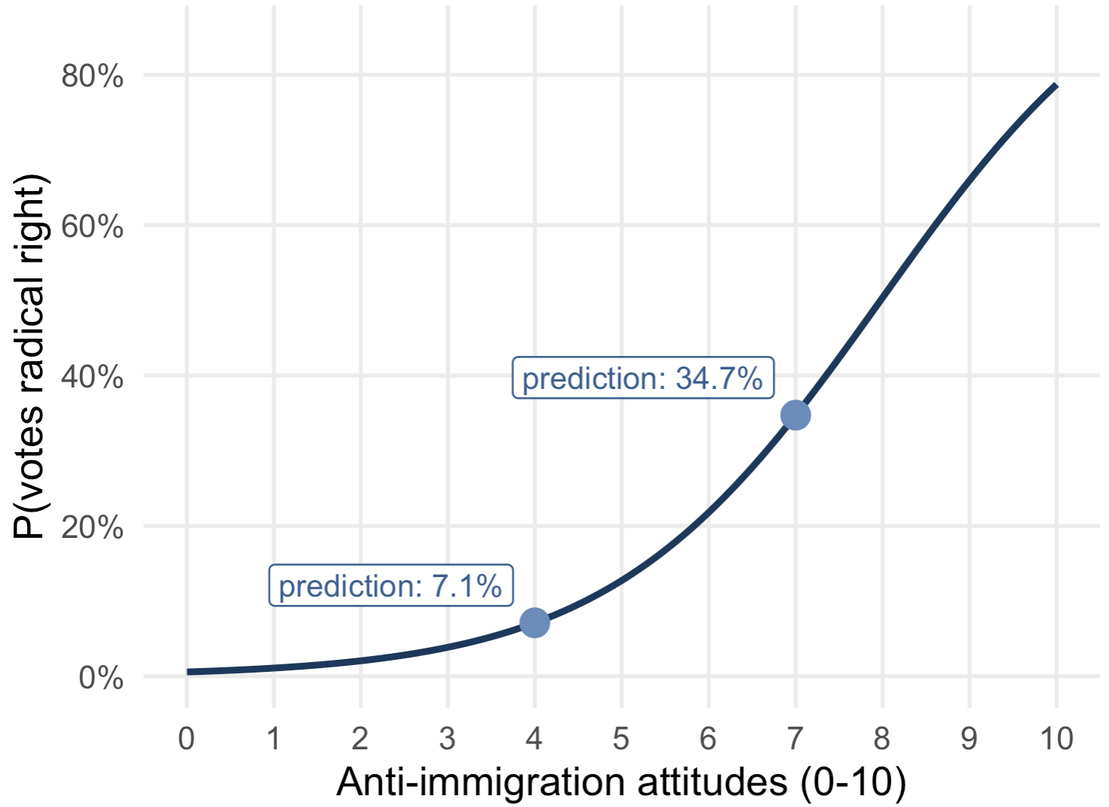
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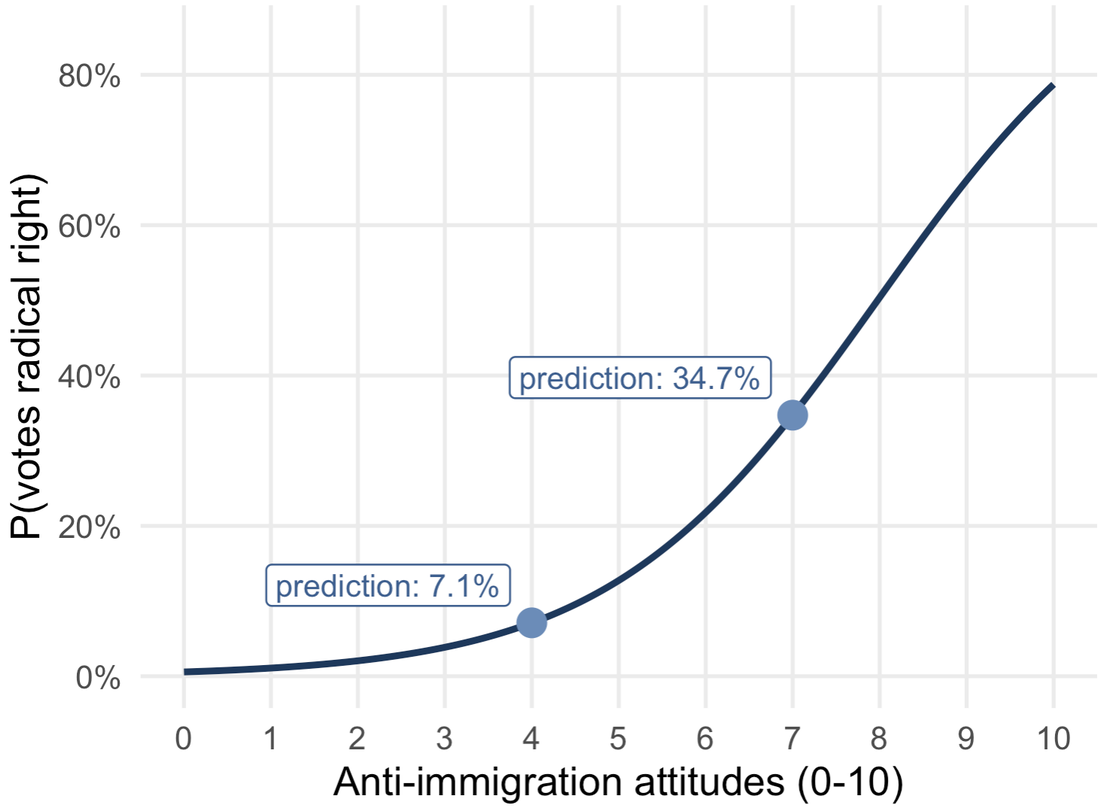
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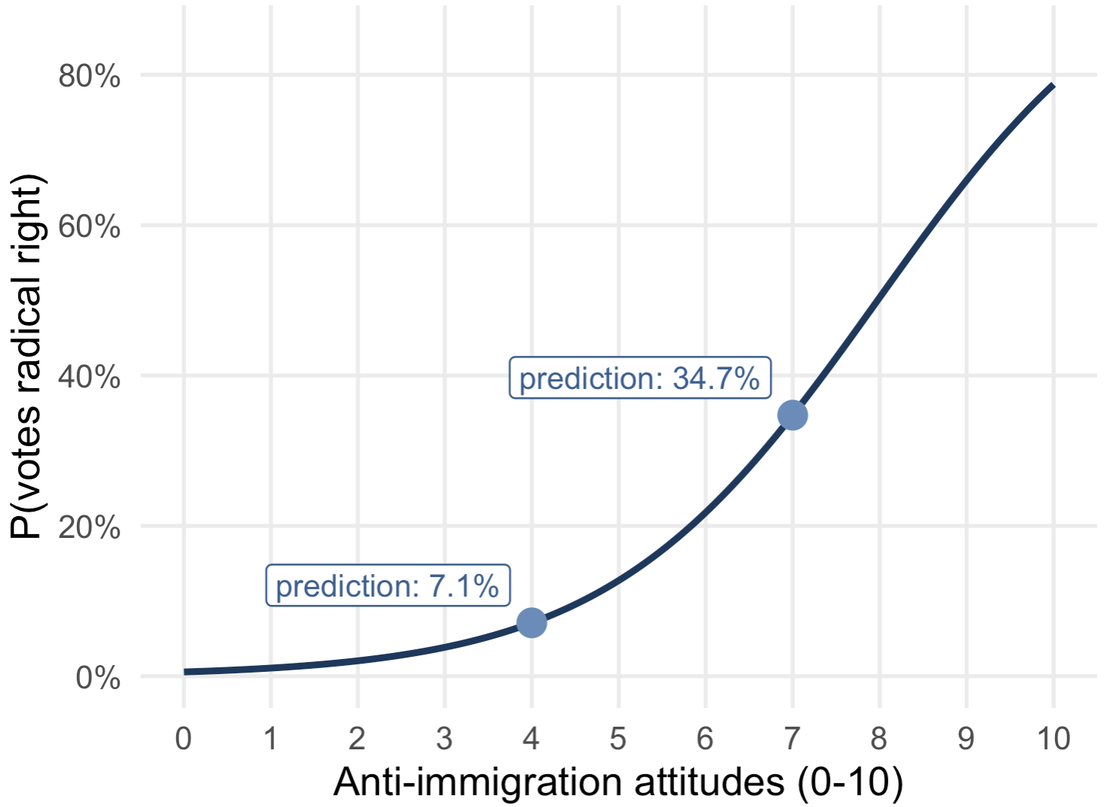
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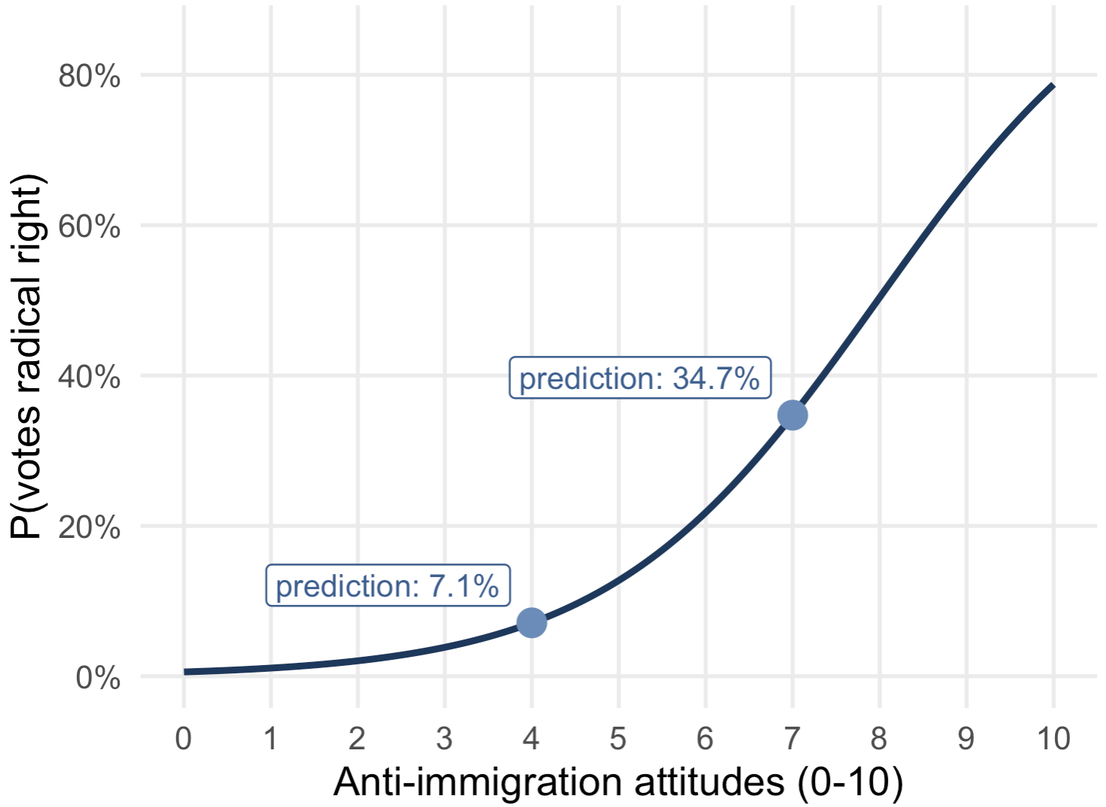
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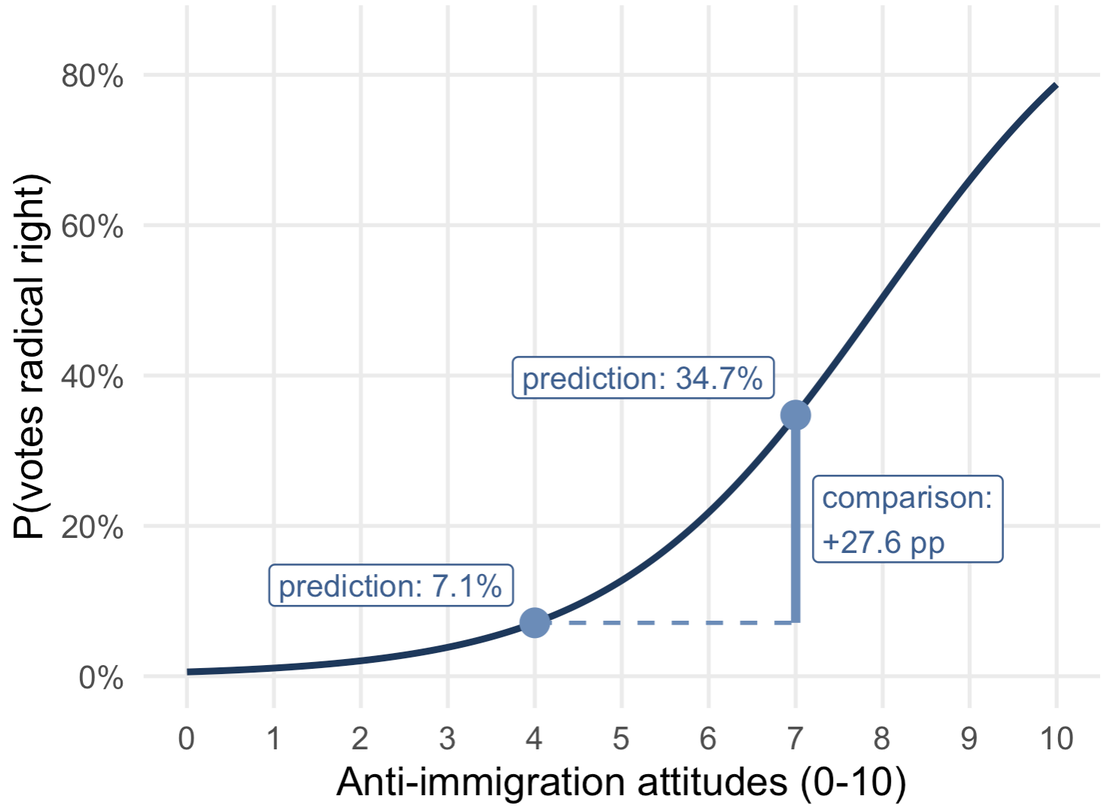
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```
marginaleffects::predictions(fit,
  newdata = marginaleffects::datagrid(
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```

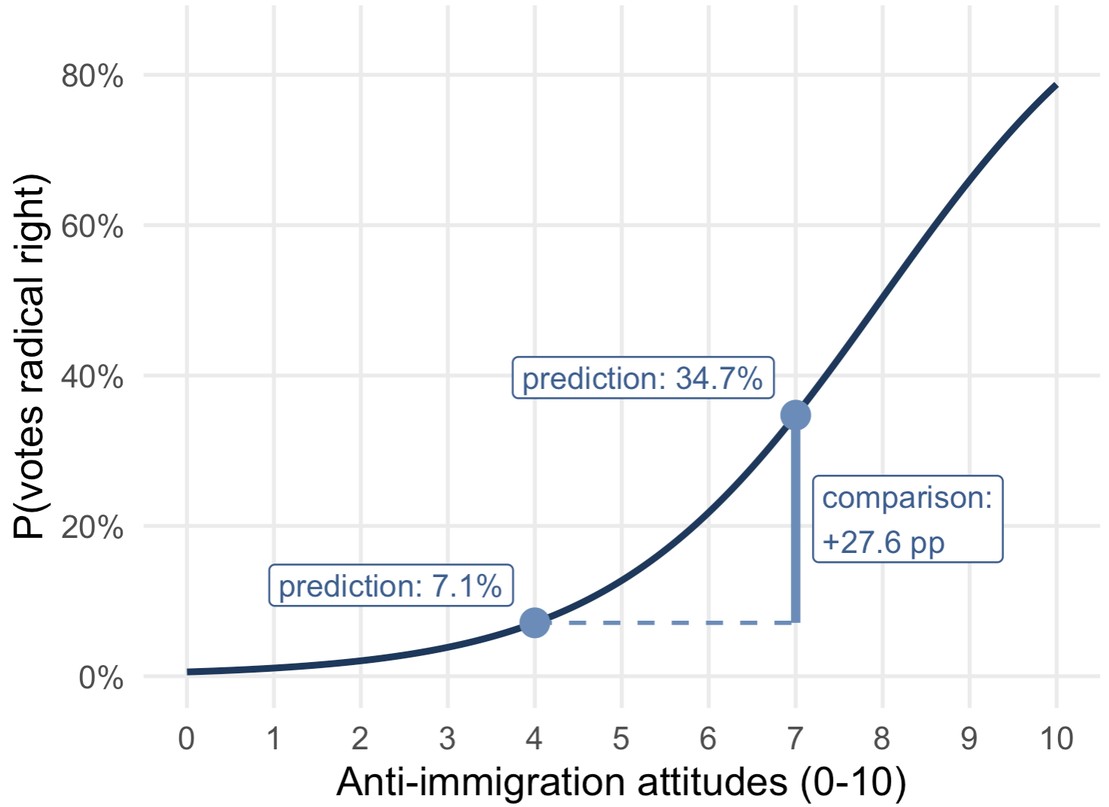


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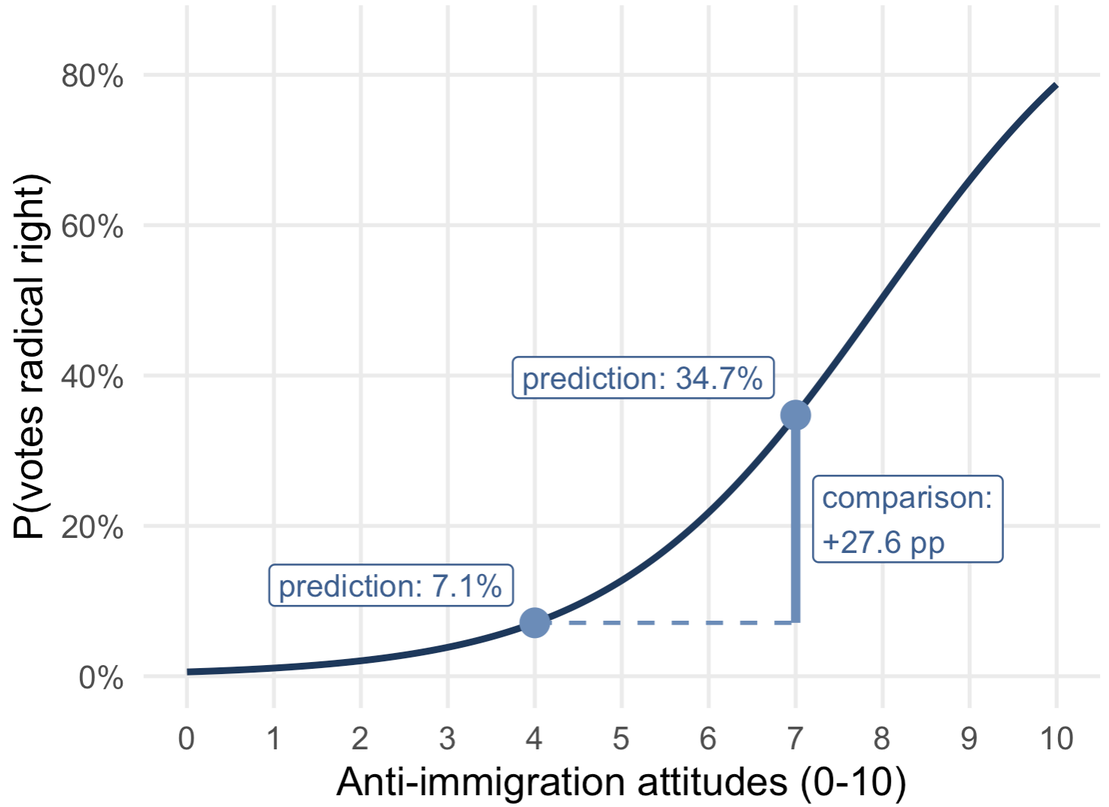
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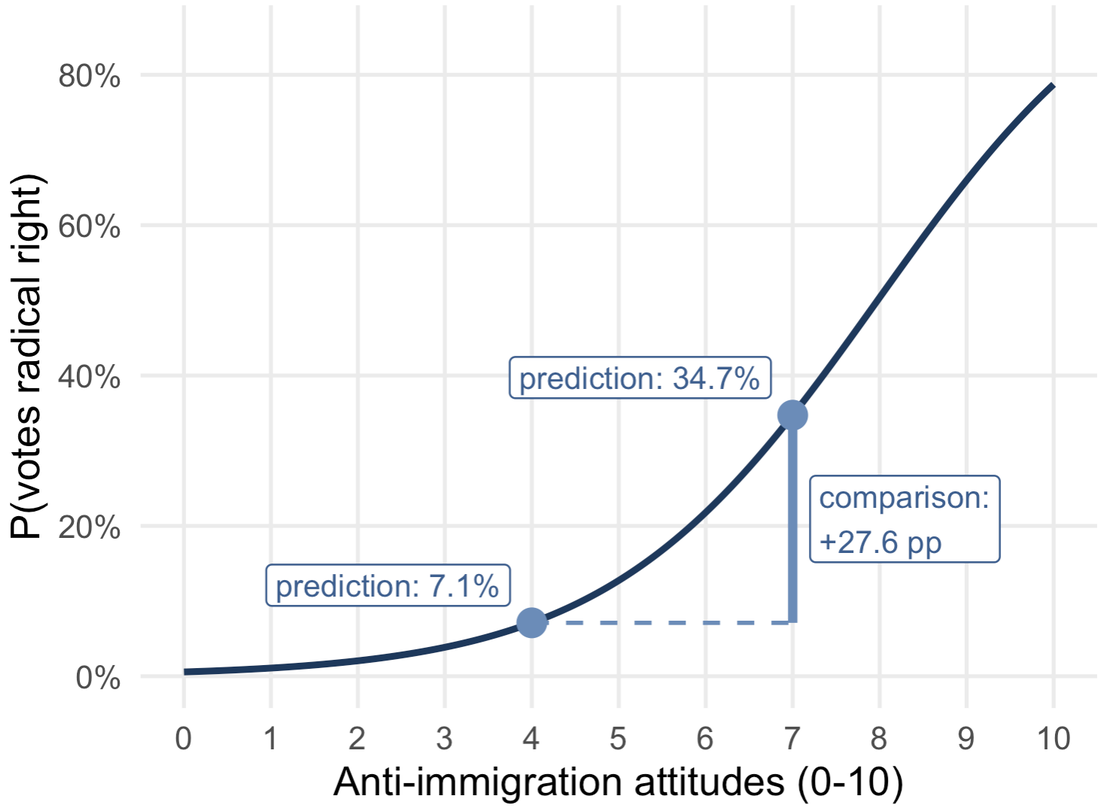
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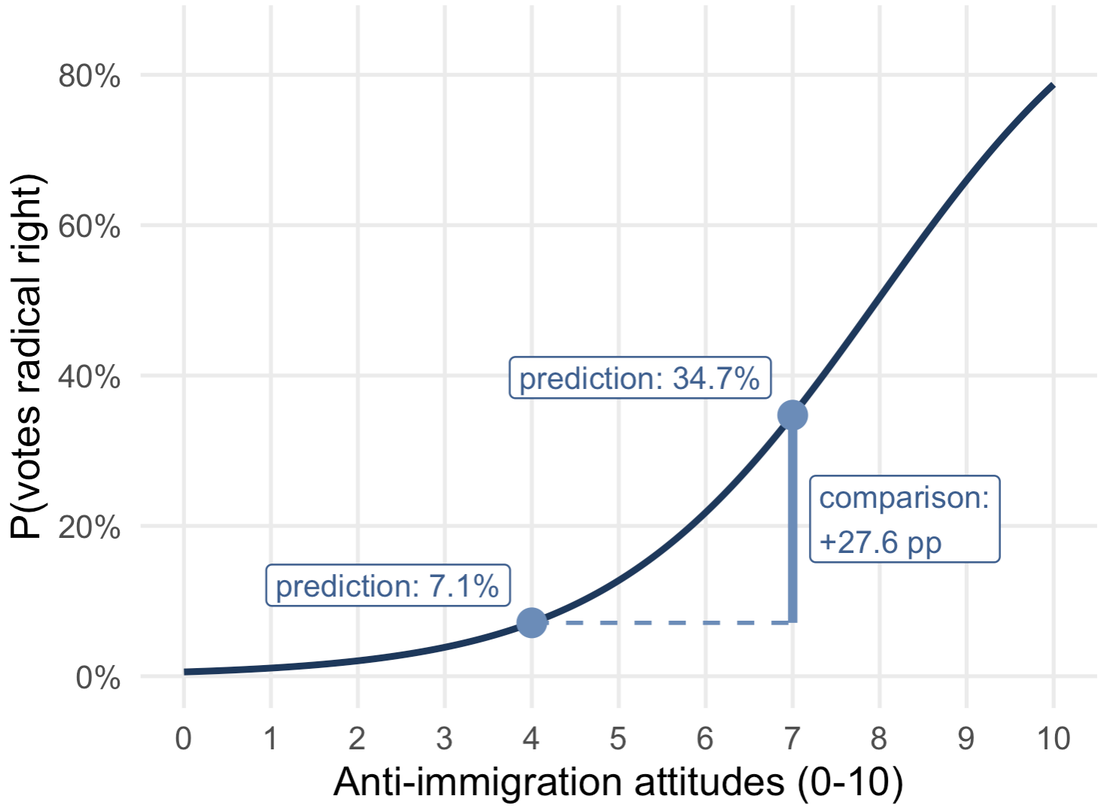
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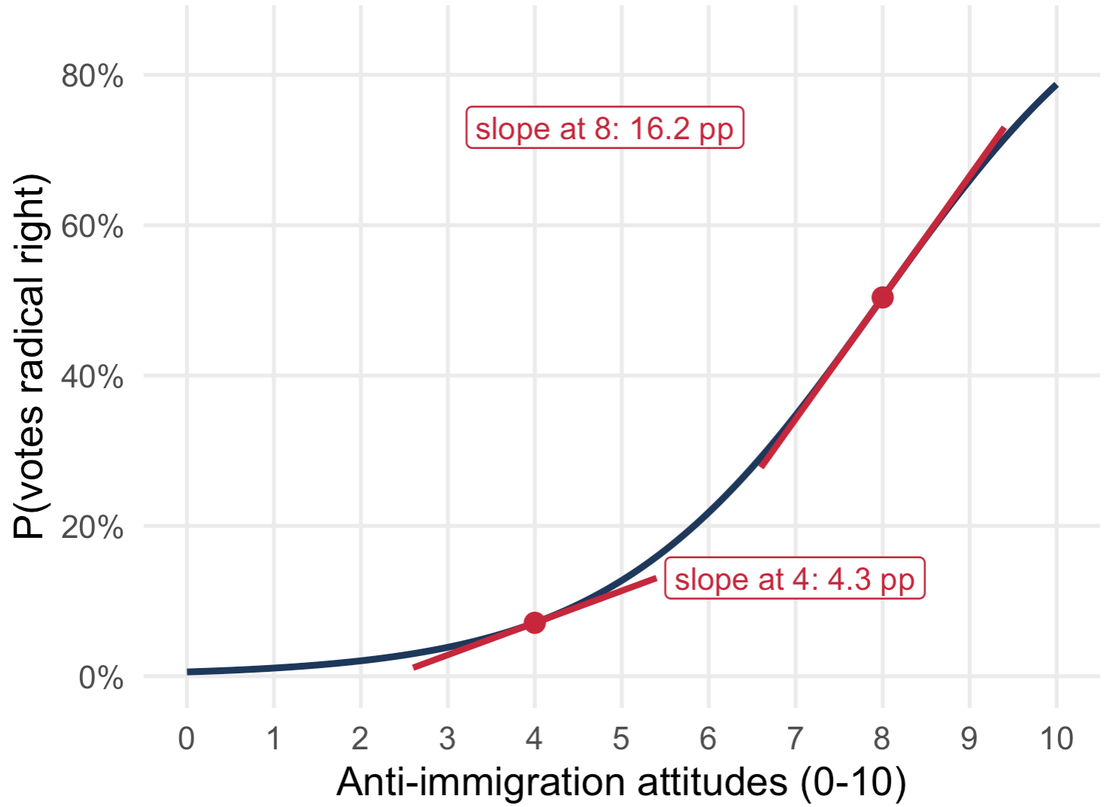
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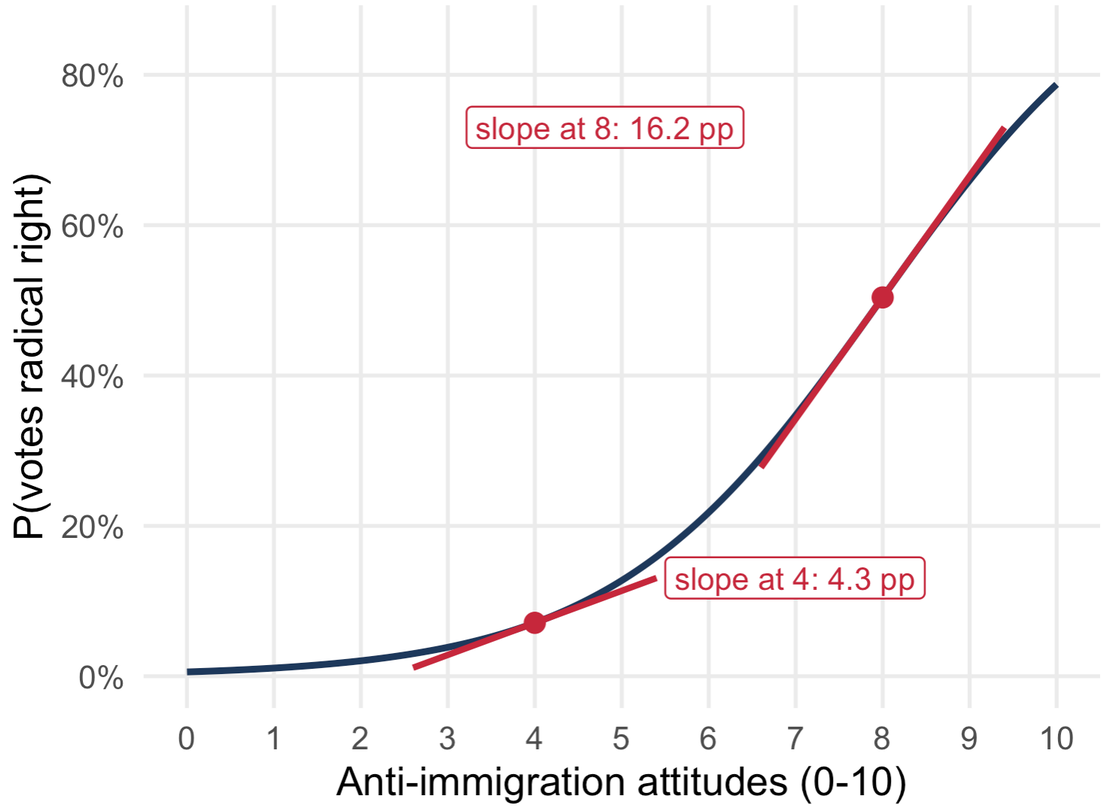


Slopes


80%
60%
40%
20%
0%
P(votes radical right)
slope at 8: 16.2 pp
slope at 4: 4.3 pp
0
1
2
3
4
5
6
7
8
9
10
Anti-immigration attitudes (0-10)

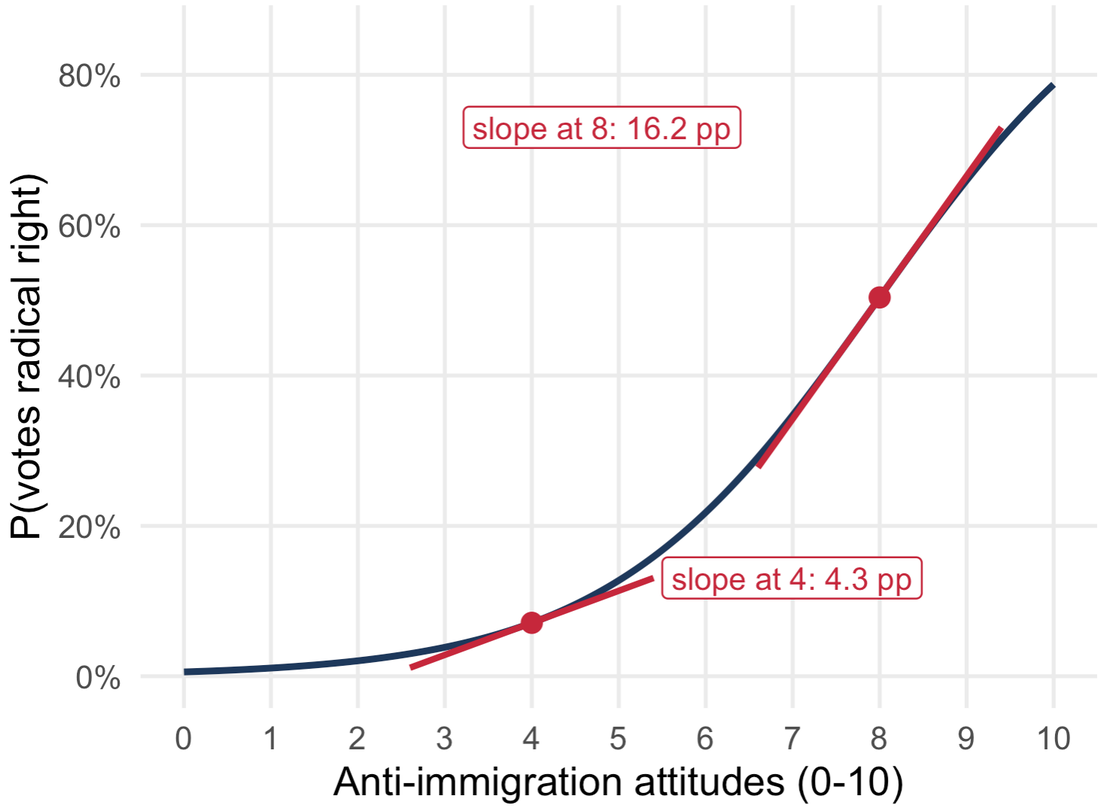
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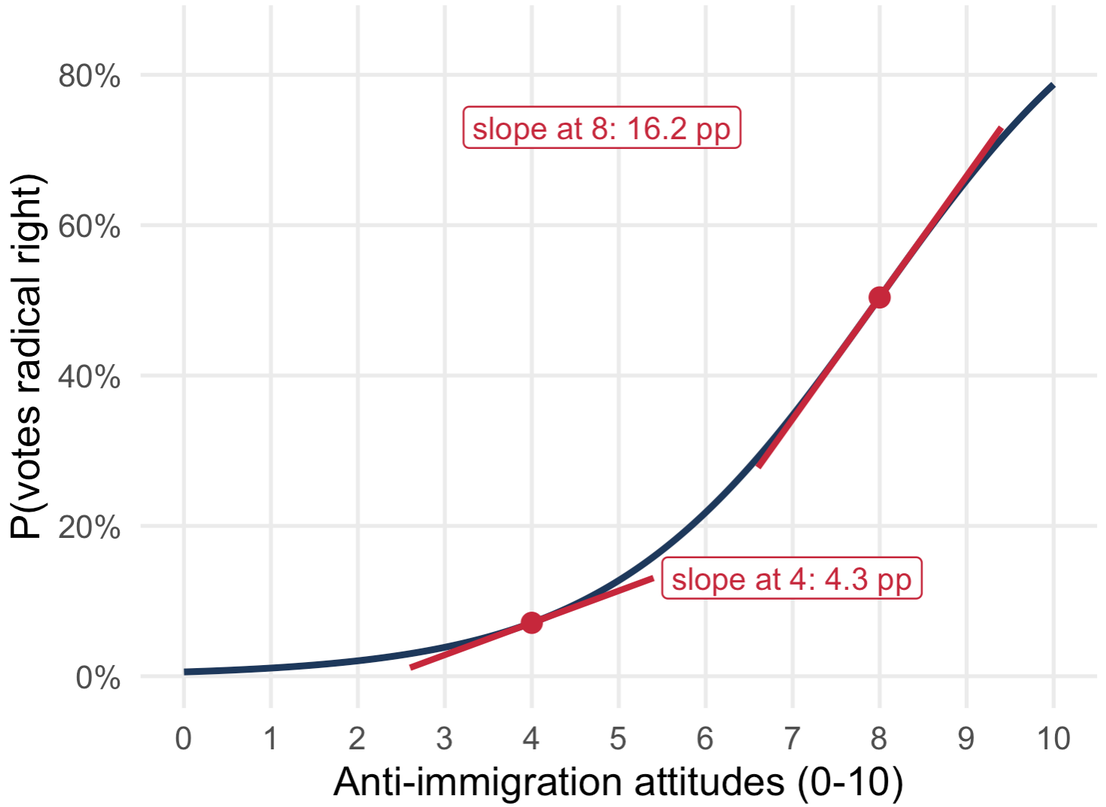
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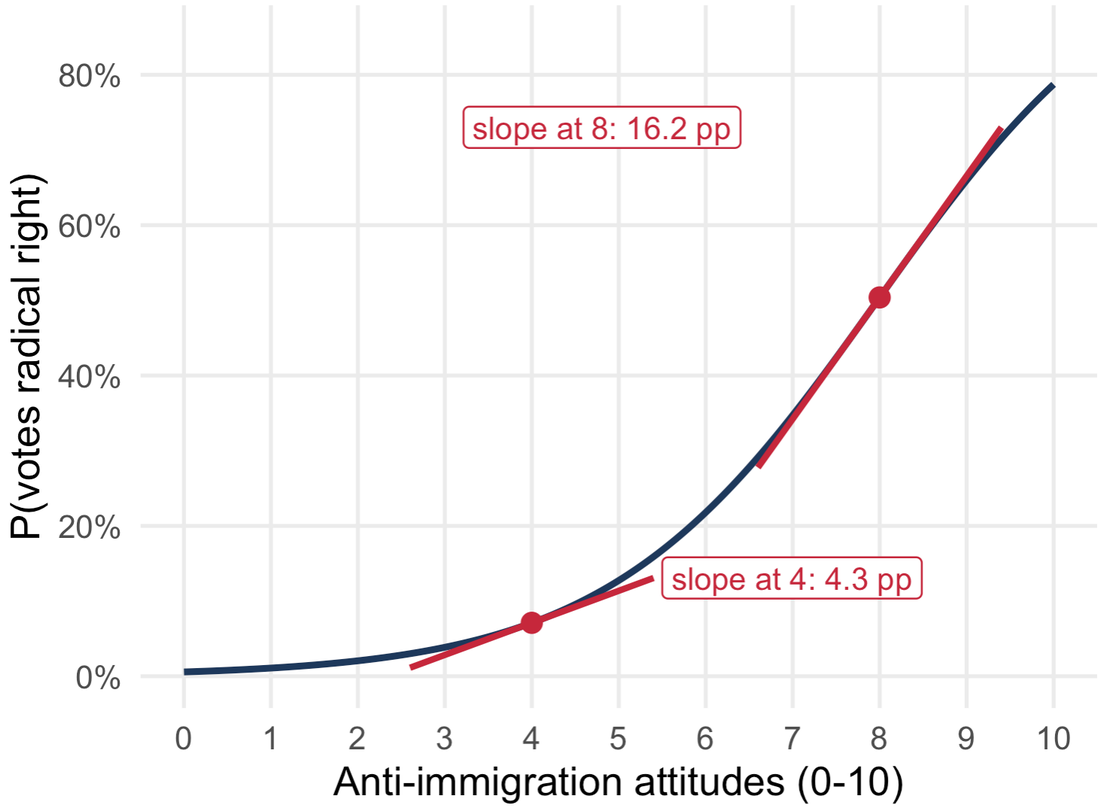
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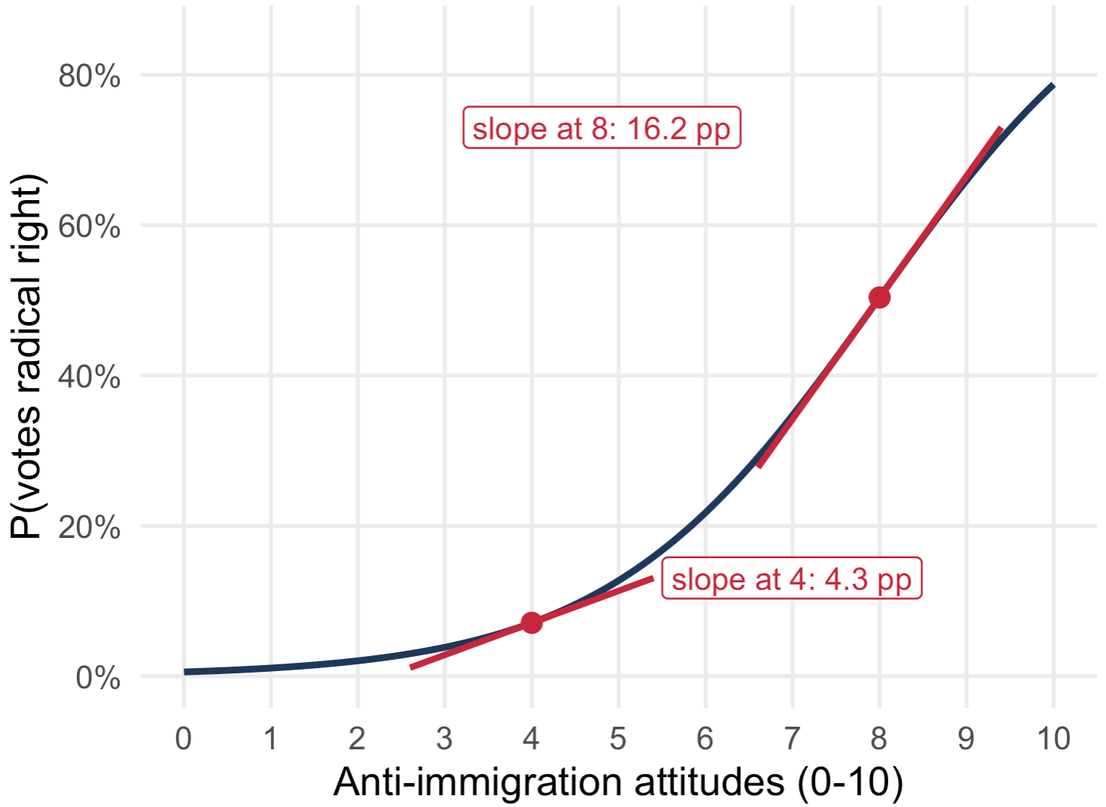
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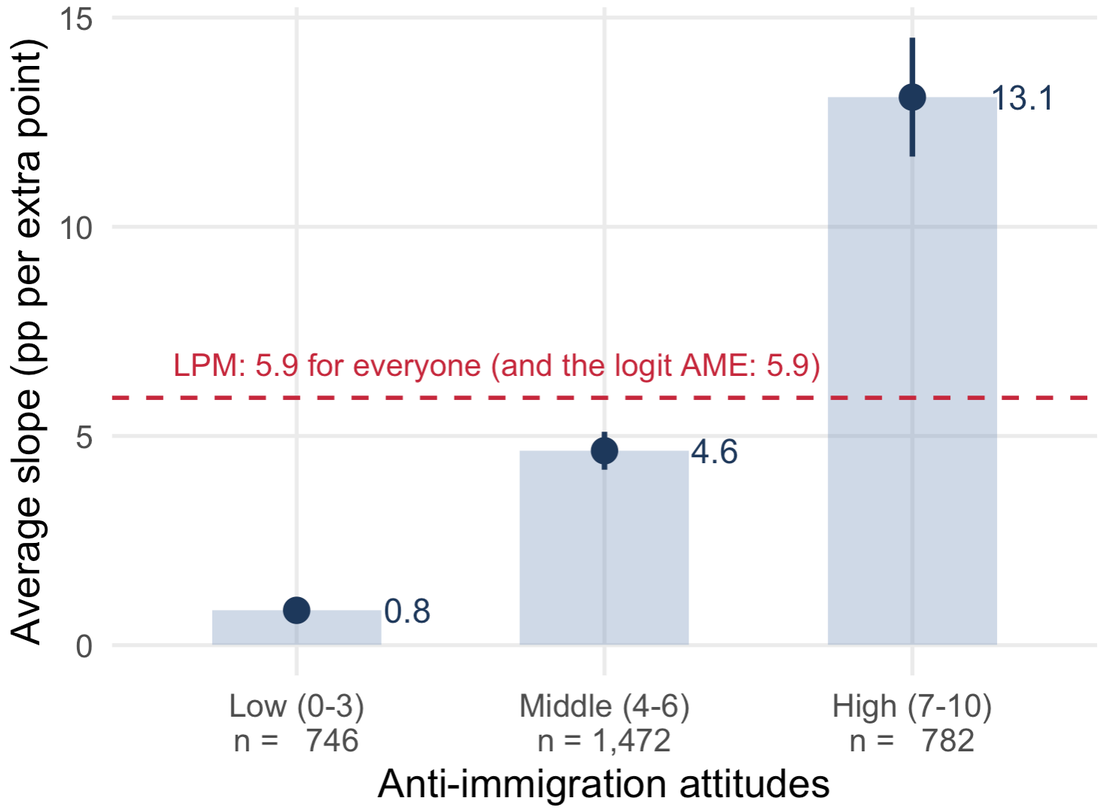
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In a logit, slope $= \beta \cdot P\,(1 - P)$: the coefficient (0.65) times a factor that peaks at $P = 0.5$.

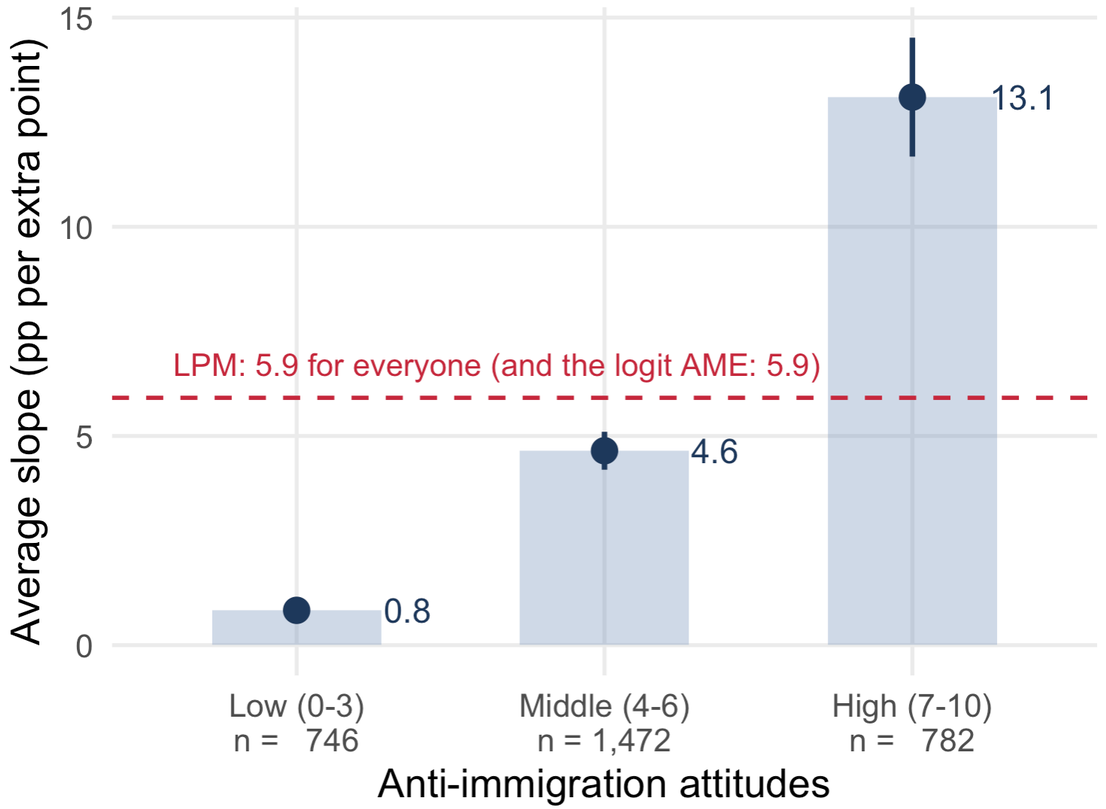
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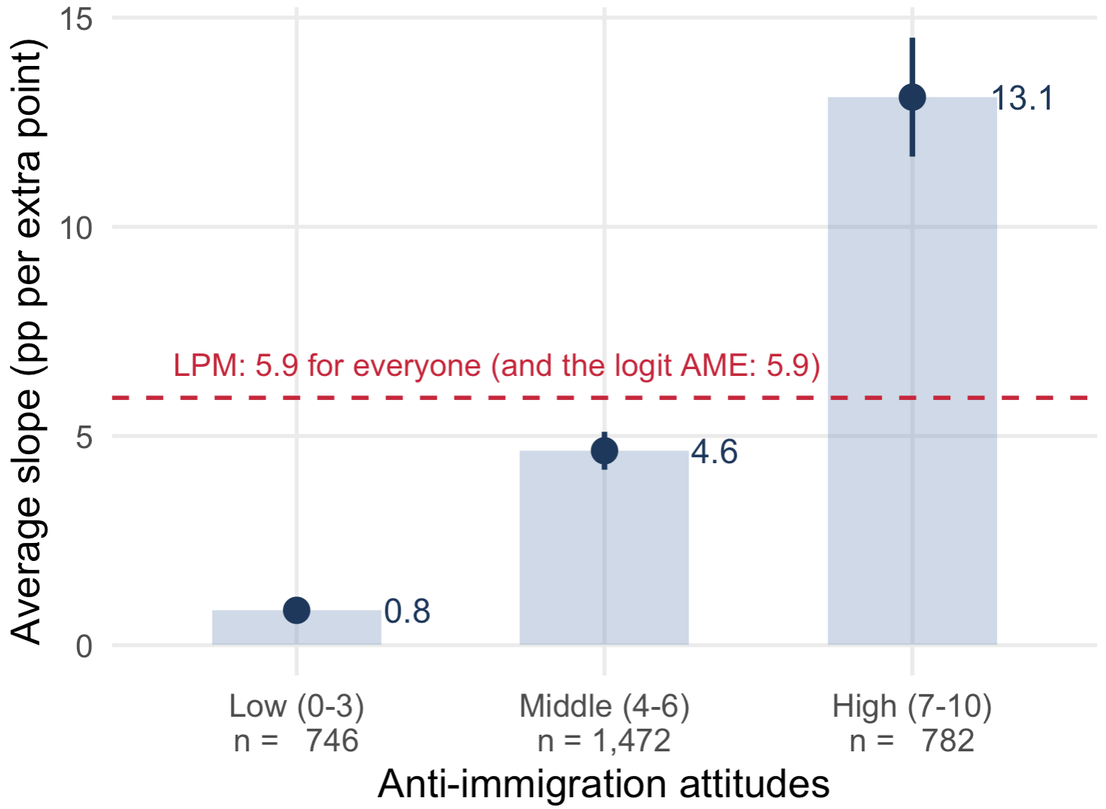
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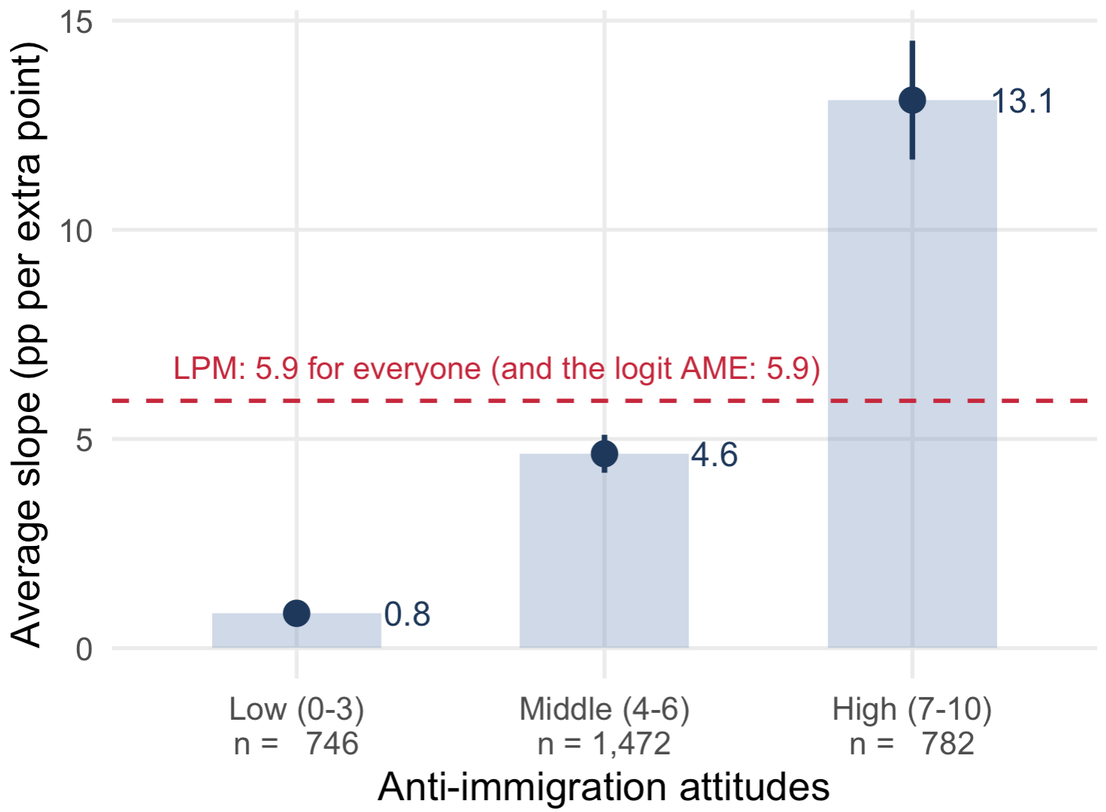
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- Averaged **within groups** of voters: 0.8 points among the least opposed, **13.1** among the most opposed. The LPM gives everyone 5.9.
- So slopes show **whether an effect differs across groups**, and let you test it: the gap is 12.3 points (95% CI 10.7 to 13.8). You cannot read this off a coefficient table, where attitudes get one coefficient for everyone.

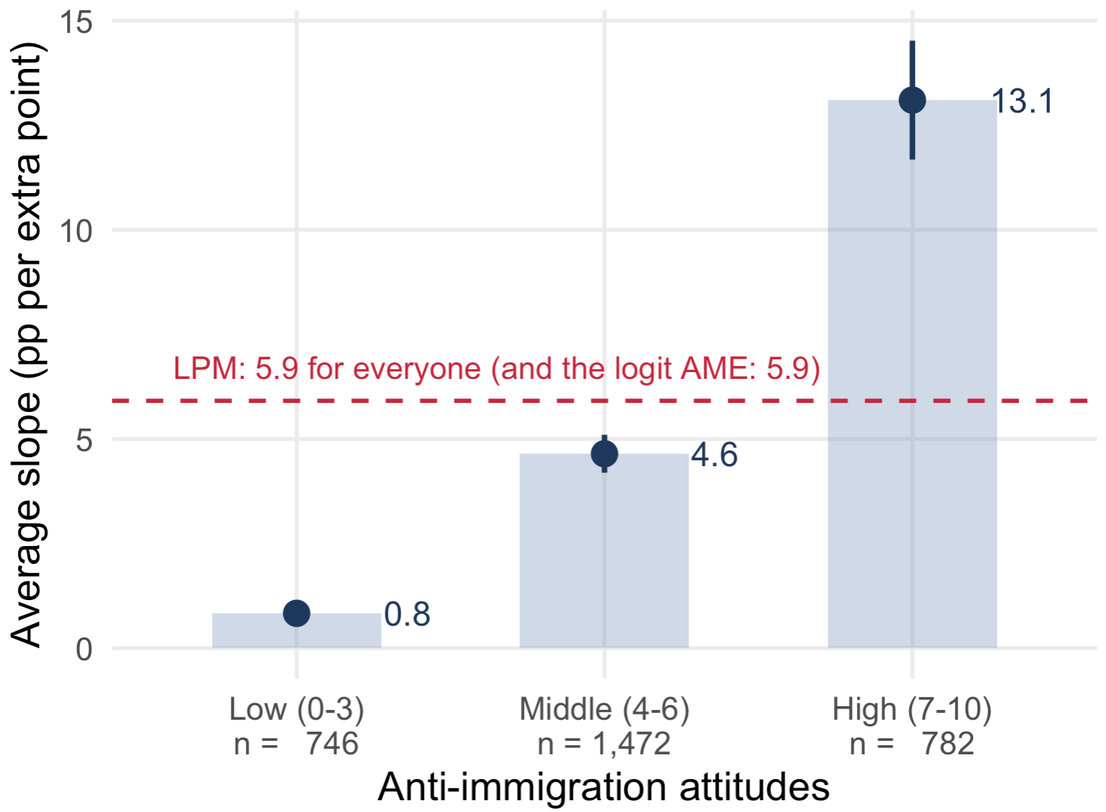


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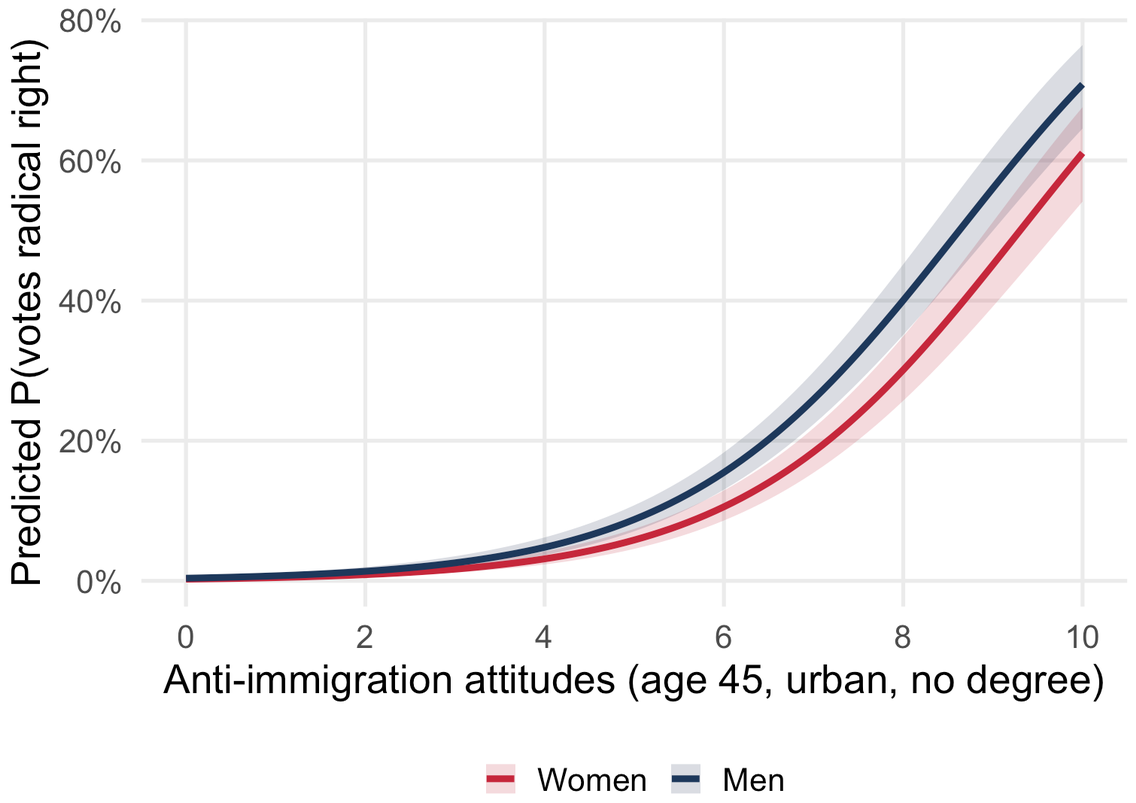
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- Averaged **within groups** of voters: 0.8 points among the least opposed, **13.1** among the most opposed. The LPM gives everyone 5.9.
- So slopes show **whether an effect differs across groups**, and let you test it: the gap is 12.3 points (95% CI 10.7 to 13.8). You cannot read this off a coefficient table, where attitudes get one coefficient for everyone.

```
marginaleffects::avg_slopes(fit, variables = "immig",
  newdata = rr_grp, by = "attitudes",
  hypothesis = "b3 - b1 = 0")
```



Average slopes within each attitude group, with 95% CIs: percentage points of probability per extra point on the 0–10 scale.

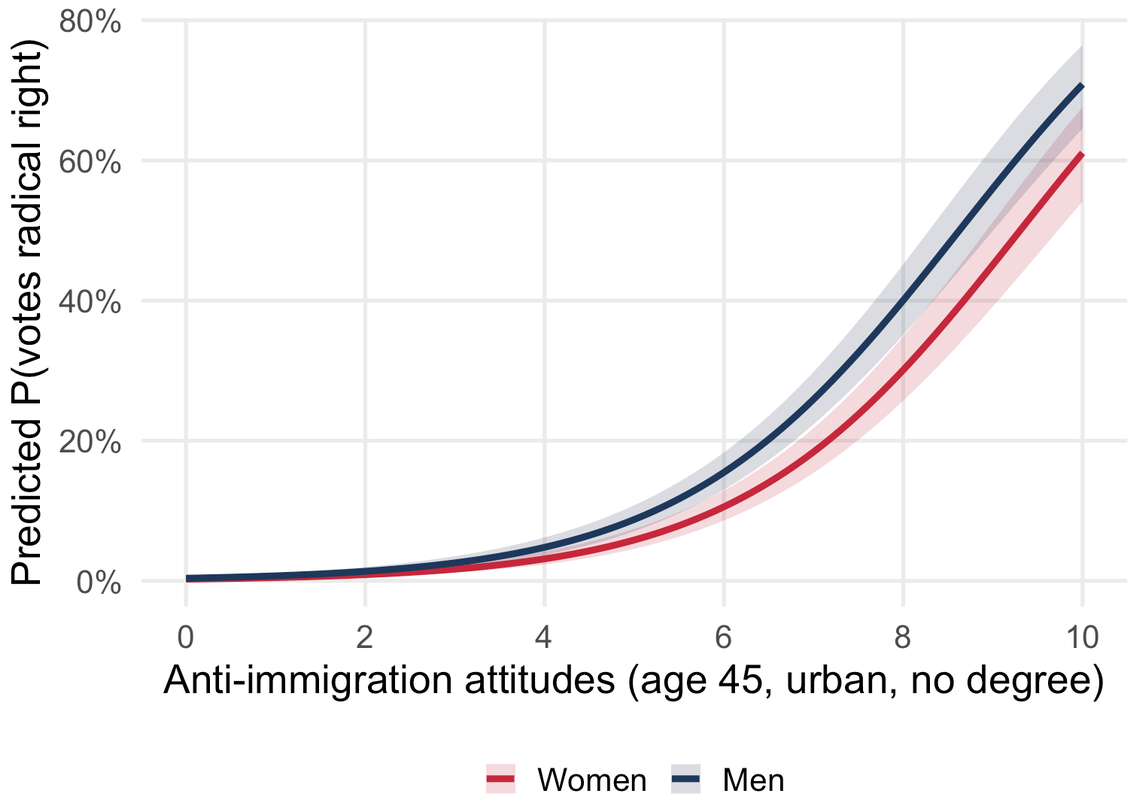
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Predicted probabilities with 95% confidence intervals (`marginaleffects::predictions()`).

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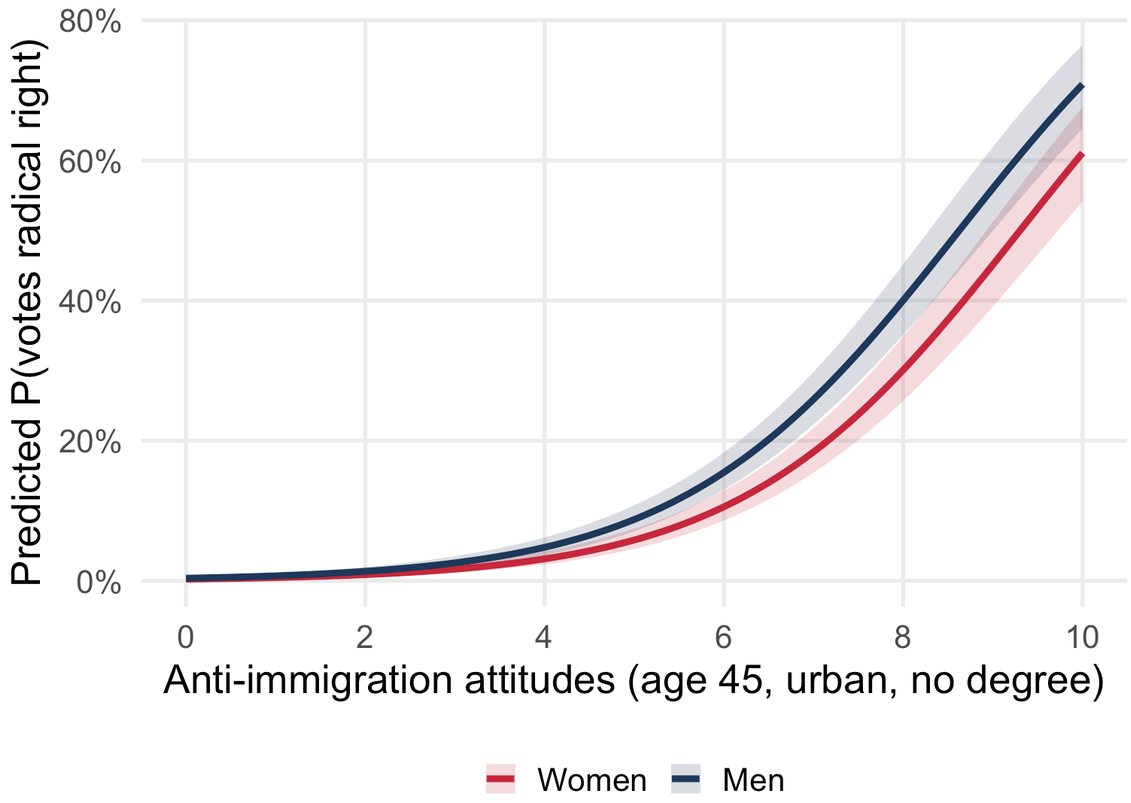
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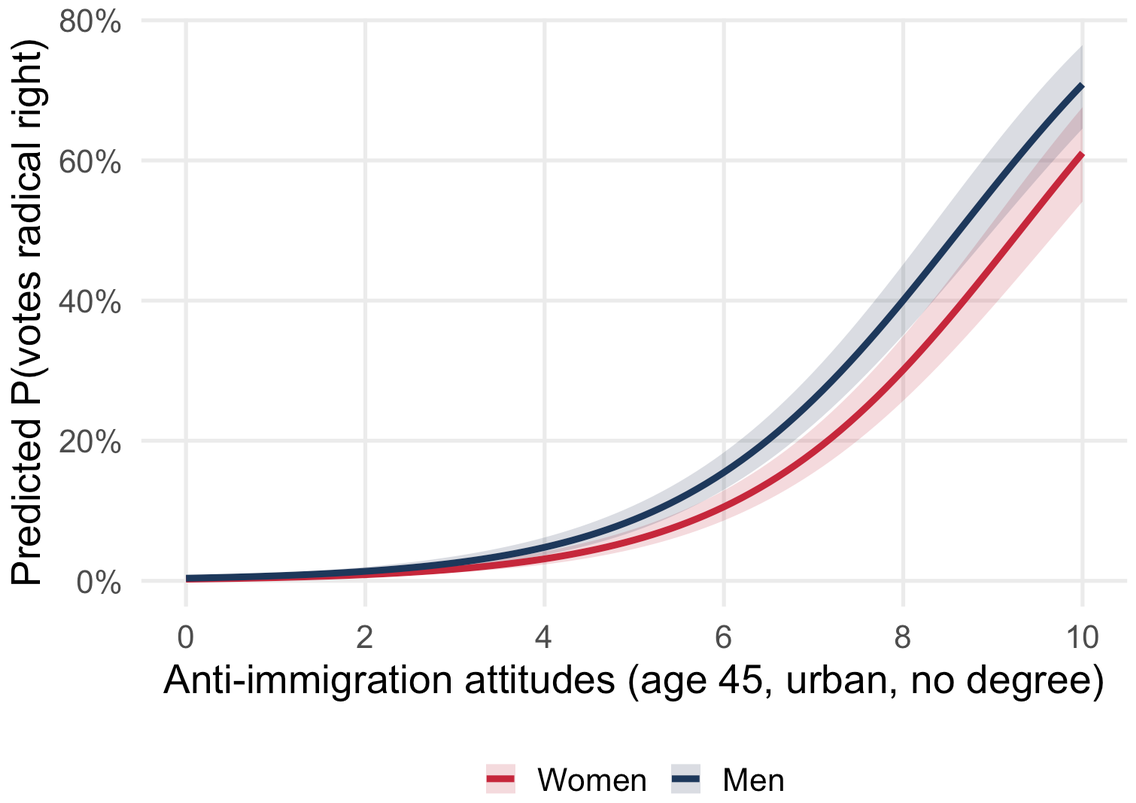
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- Predictions for a whole grid of profiles: men and women, across attitudes.
- The comparison between the curves, the radical-right gender gap, is tiny among voters relaxed about immigration and **largest among the most opposed**.
- So "the effect of being a man" has no single value until we say **for whom**.



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Say whom your number is about. The AME and effects at representative values are good defaults (Hanmer & Kalkan 2013); whichever you pick is your **estimand**.

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You want…	Function or argument
Predictions	`marginaleffects::predictions()`
Comparisons	`marginaleffects::comparisons()`
Slopes	`marginaleffects::slopes()`
Averages	the same with `avg_`, e.g. `marginaleffects::avg_slopes()`
For whom?	`newdata = marginaleffects::datagrid(...)`, `by =`, `wts =`
Tests	`hypothesis =`, `equivalence =`

In code

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fit <- glm(rrv ~ immig + male + degree + rural + age, data = rr, family = binomial)

# Average gap between men and women in the probability of voting radical right (AME)
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# The gap at representative levels of anti-immigration attitudes (MER)
marginaleffects::comparisons(fit, variables = "male",
                             newdata = marginaleffects::datagrid(immig = c(2, 5, 8)))
```

```
 immig Estimate Std. Error    z Pr(>|z|)    S   2.5 %  97.5 %
     2  0.00456    0.00146 3.13  0.00173  9.2 0.00171 0.00741
     5  0.02806    0.00798 3.51  < 0.001 11.1 0.01241 0.04370
     8  0.09783    0.02716 3.60  < 0.001 11.6 0.04459 0.15107

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Published online by Cambridge University Press: **26 October 2026**

MERLIN SCHAEFFER, ALESSIO ROMARRI, DINA ROSENBERG and KRZYSZTOF KRAKOWSKI

Show author details

Abstract

Far-right political shifts have been linked to increased racism, raising concerns about their impact on minority access to public services. This study investigates whether far-right electoral victories encourage discrimination by street-level bureaucrats, who are expected to act impartially. We conducted a nested research design in Italy, combining a field-experimental audit with a regression discontinuity design focused on municipalities near the electoral threshold for far-right mayors. Research assistants with native-Italian and West-African accents contacted municipal healthcare offices to request registration with a general medical practitioner. Compared to native-Italian callers, those with West-African accents faced significantly higher rates of scrutiny and dismissive responses. Leveraging close elections, we find that this disparity was exacerbated in municipalities where far-right candidates narrowly won. These findings demonstrate that far-right electoral success can undermine impartiality and equal access to essential services.

Schaeffer, Romarri, Rosenberg & Krakowski (2026), *American Political Science Review*, First View.

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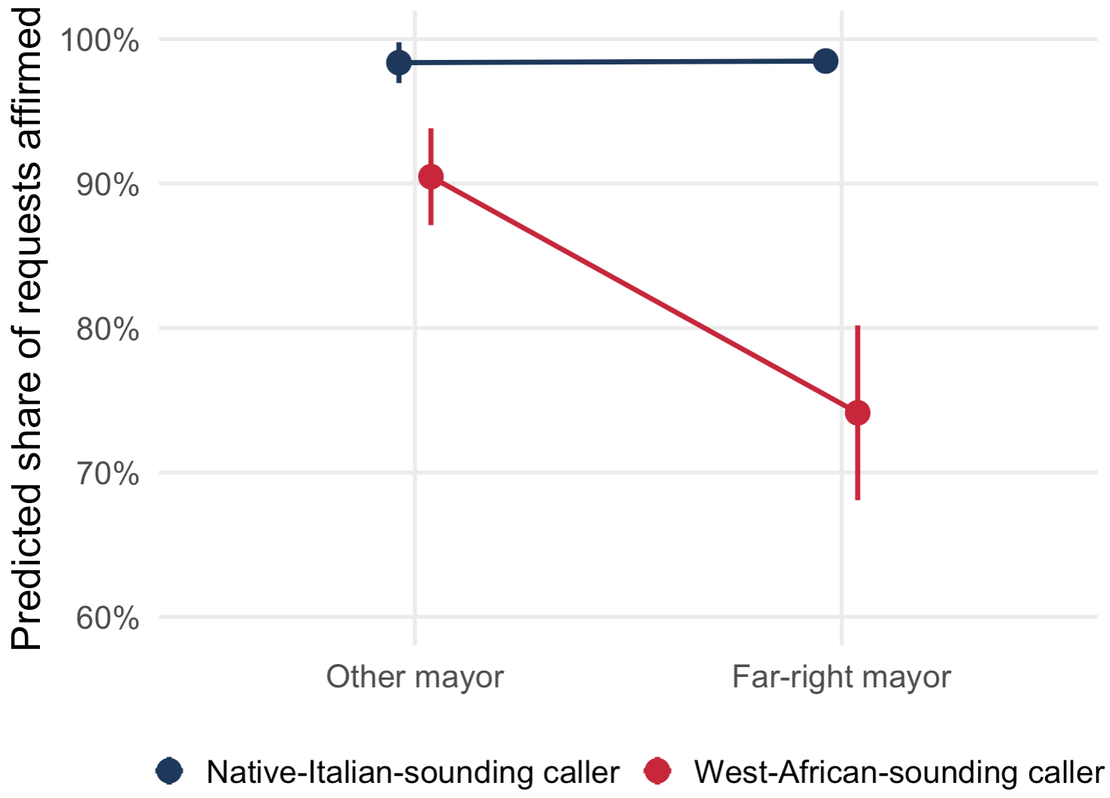
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Near 0 or 1, **ratios explode** while differences stay modest. Report the rate in each group, the difference, and the ratio when it helps. Natural frequencies help most: *about 12 in 100 West-African-sounding callers were not affirmed, against 3 in 1,000 native-Italian-sounding callers.*

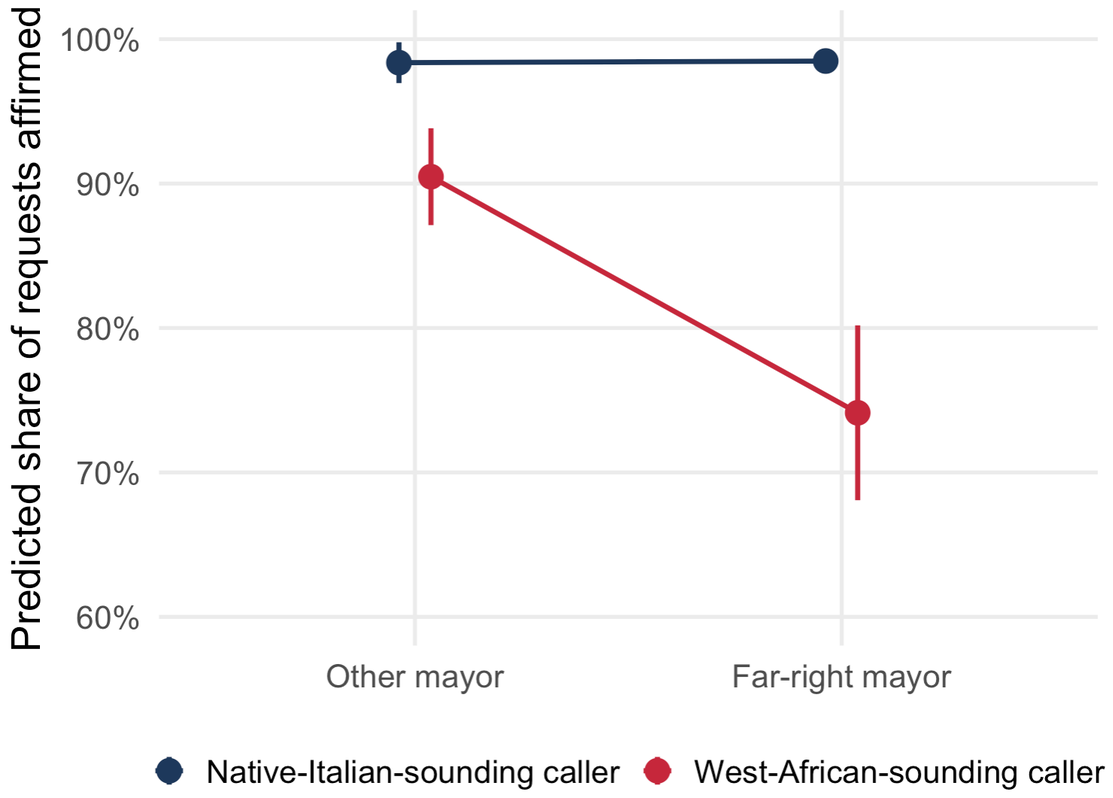
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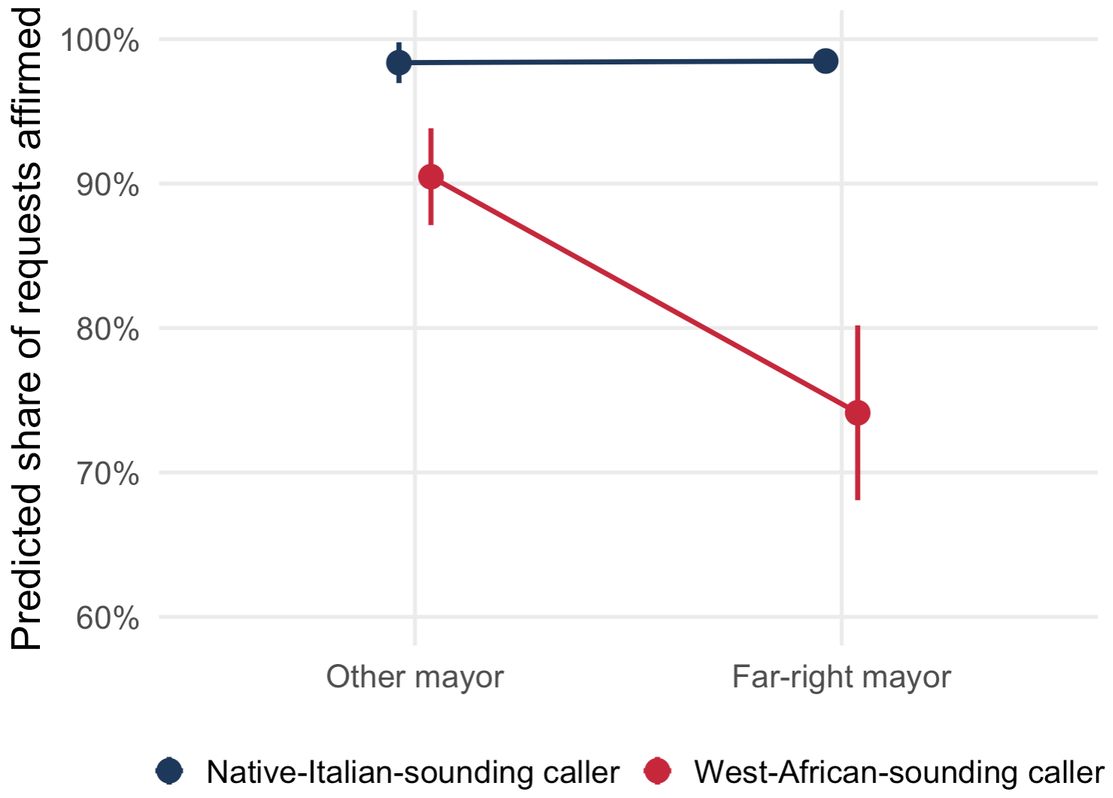
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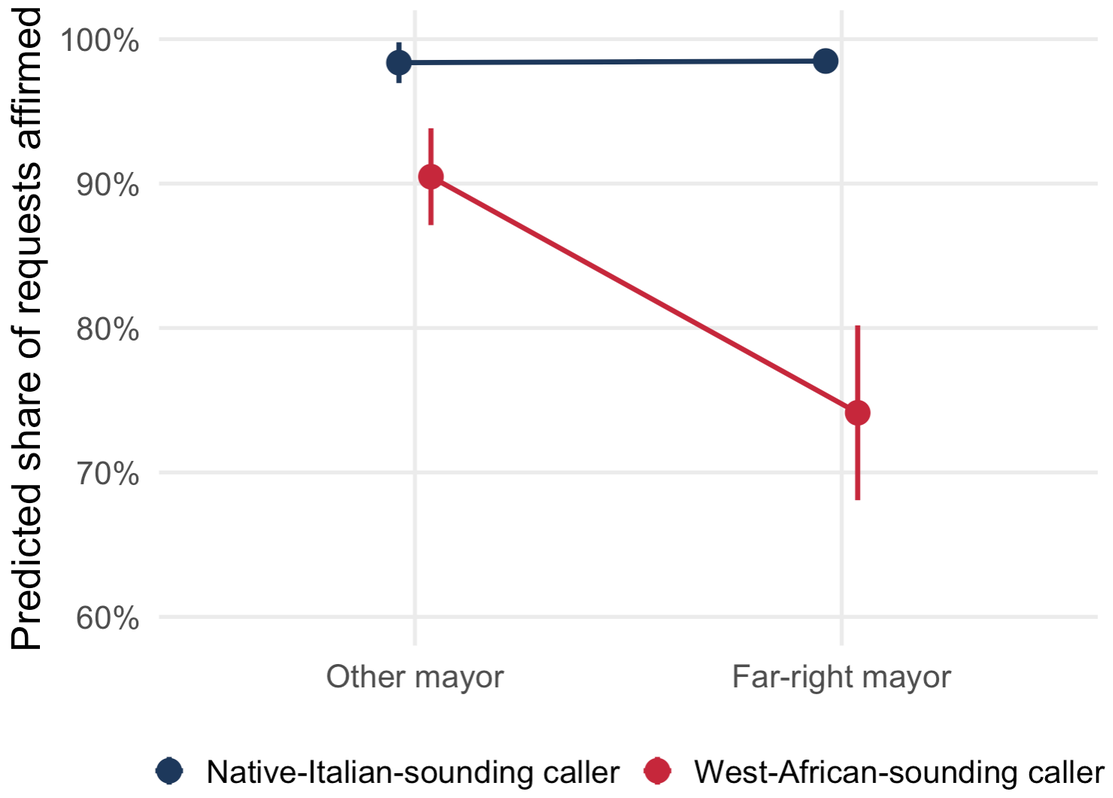
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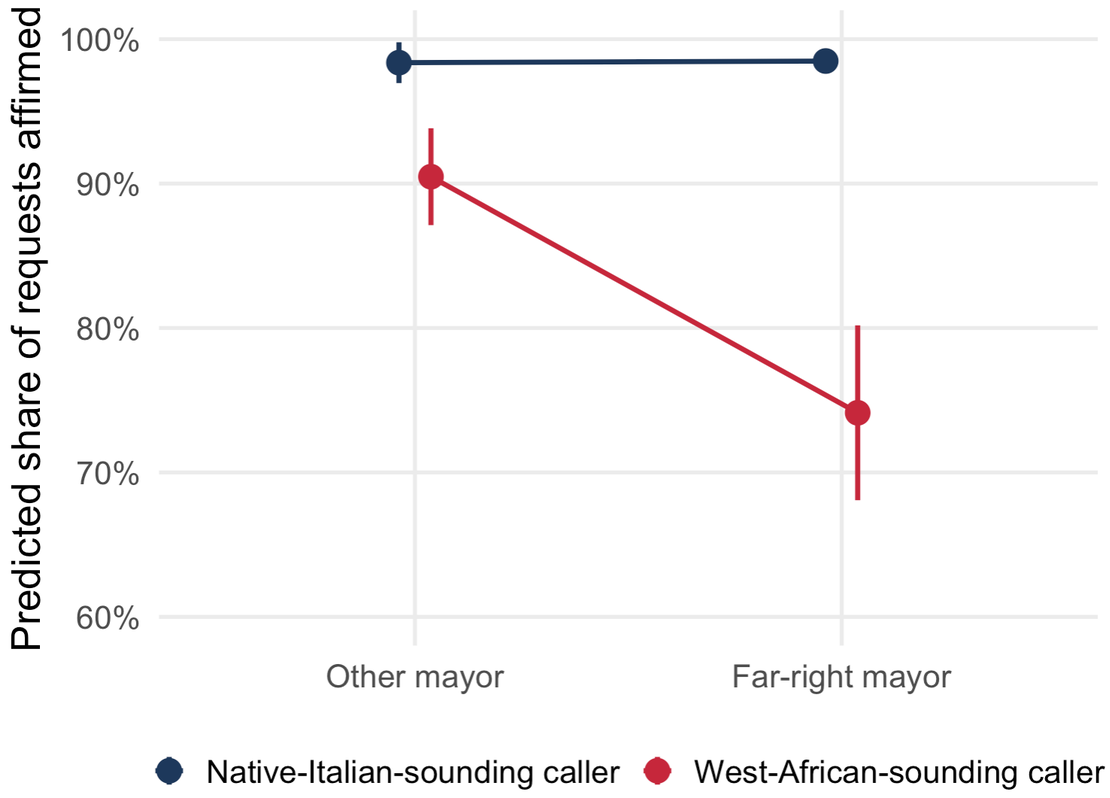
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- The accent gap: 7.9 percentage points in other towns, **24.4 points under far-right mayors**.



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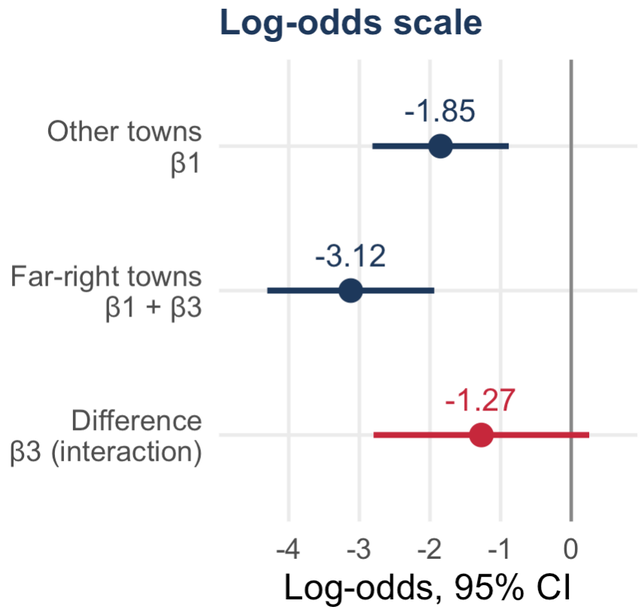
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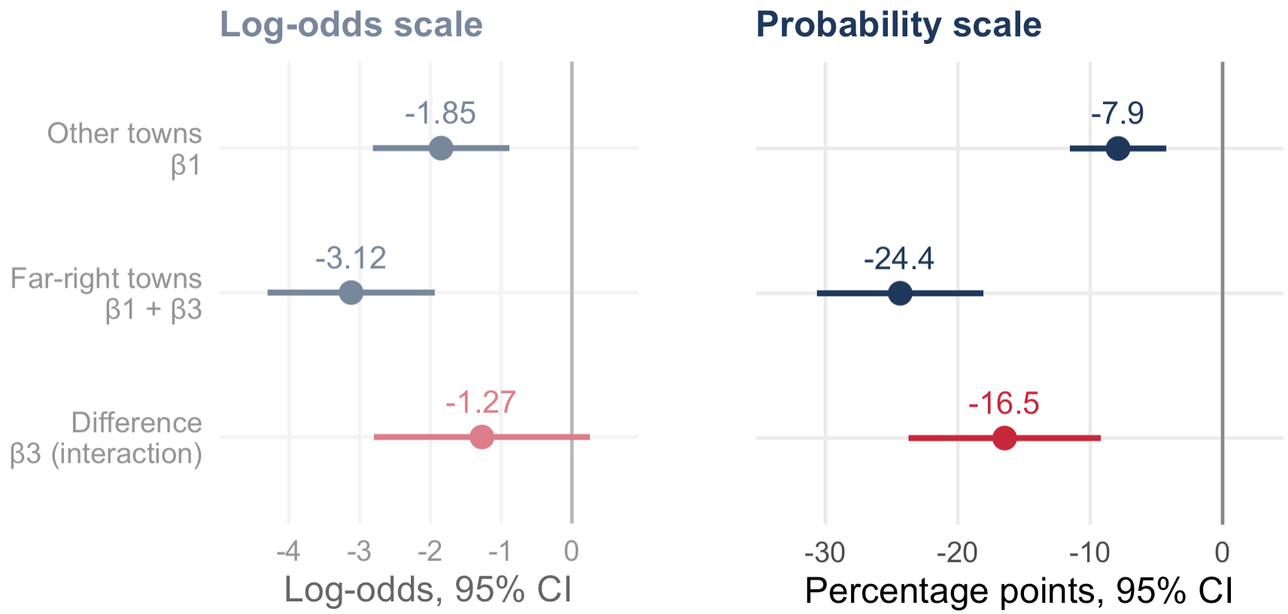
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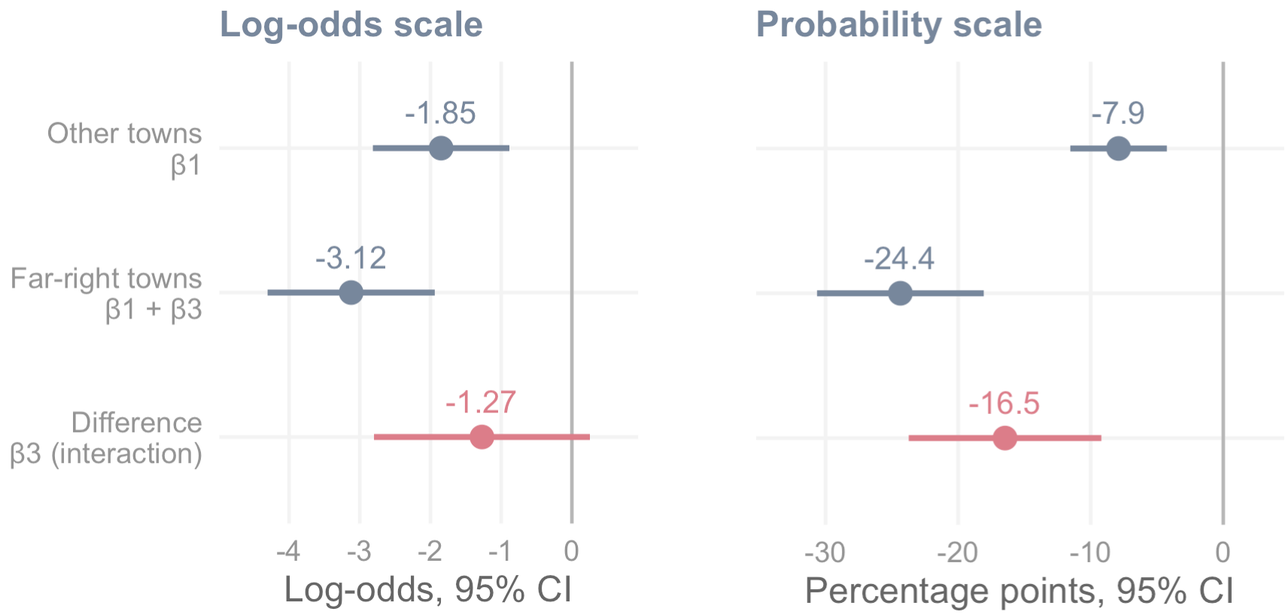
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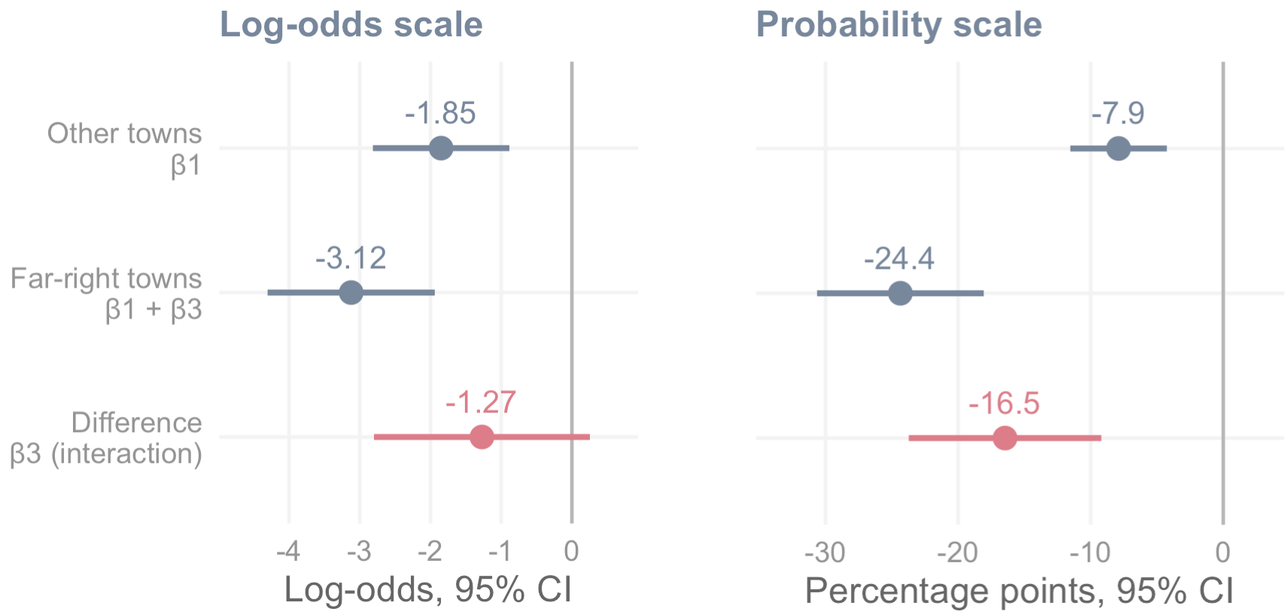
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What to do: choose the scale from your question, before you look at the results. For "how many more callers are turned away", that is the probability scale: report the gap in each kind of town and the difference between the gaps, with its CI. Report the logit interaction only if your question is about odds ratios, and say so.

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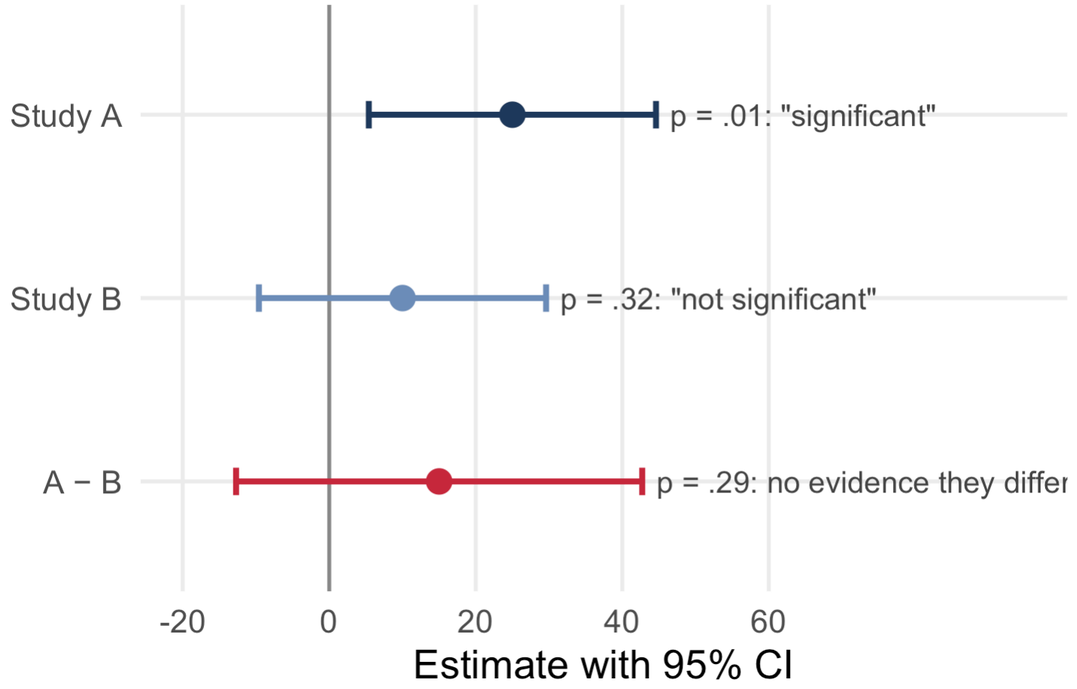
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Gelman & Stern's example: estimates of 25 (SE 10) and 10 (SE 10). Their difference is 15, with SE 14.

Some things you might consider reporting, by design

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An experiment (the turnout letters)

Report: the control-group rate and the average effect in pp, with a CI; a ratio if the outcome is rare.

Show: rates by arm, as dots with intervals.

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Report: the control-group rate and the average effect in pp, with a CI; a ratio if the outcome is rare.

Show: rates by arm, as dots with intervals.

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Whatever you choose, readers need the **baseline rate**, the **unit** (percentage points, not "%"), and the **uncertainty**. Coefficient tables can go to the appendix.

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> One model, one question. Ask the prediction machine about the comparison it was built for, and nothing else.

A (brief) primer: the logic of maximum likelihood

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> The goal is the **logic**, not the math: enough that “likelihood”, “log-likelihood”, and “did not converge” in your output make sense. Six voters are enough.

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> *Suppose the party's vote share is $p = 0.2$. How probable is it that the six votes come out exactly as they did (two for the party, four not)?*
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- Same formula, read in two directions. Maximum likelihood estimation picks the value of p under which **the data we observed are most probable**.

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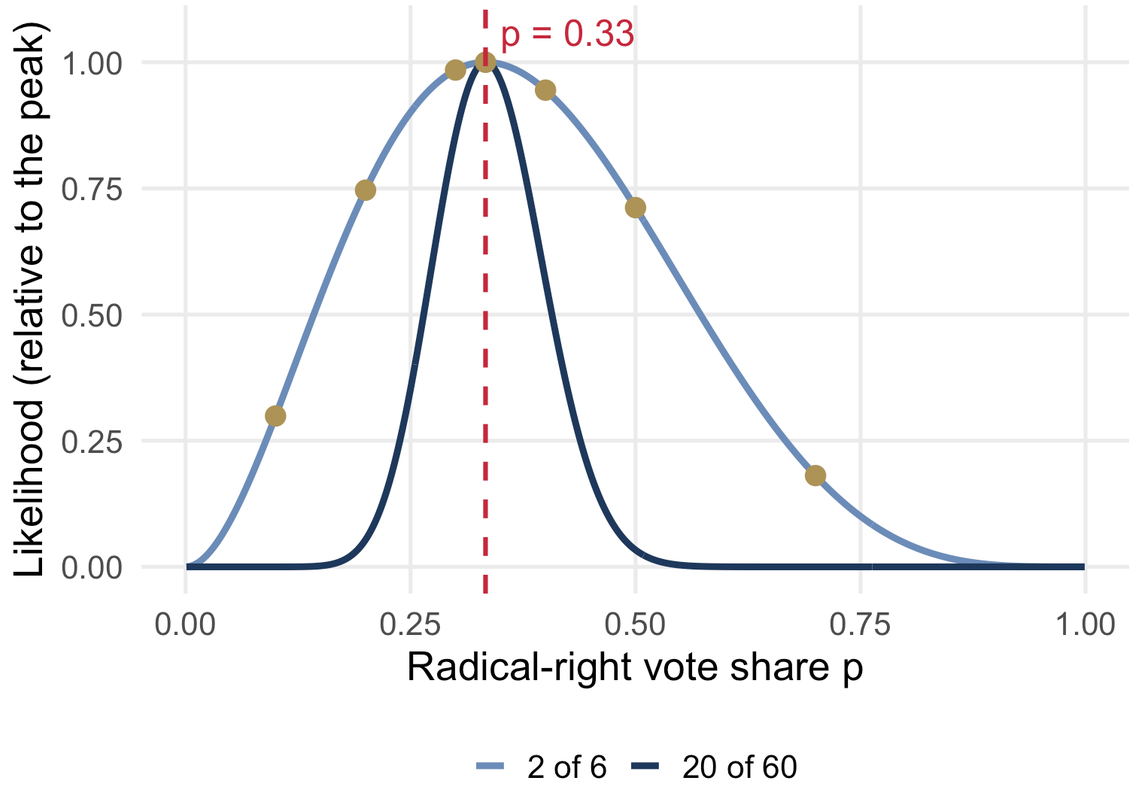
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Voter	Radical right?	Factor	if $p = 0.5$	if $p = 0.3$
1	no	$1 - p$	0.5	0.7
2	yes	p	0.5	0.3
3	no	$1 - p$	0.5	0.7
4	no	$1 - p$	0.5	0.7
5	yes	p	0.5	0.3
6	no	$1 - p$	0.5	0.7
Multiply			**0.016**	**0.022**

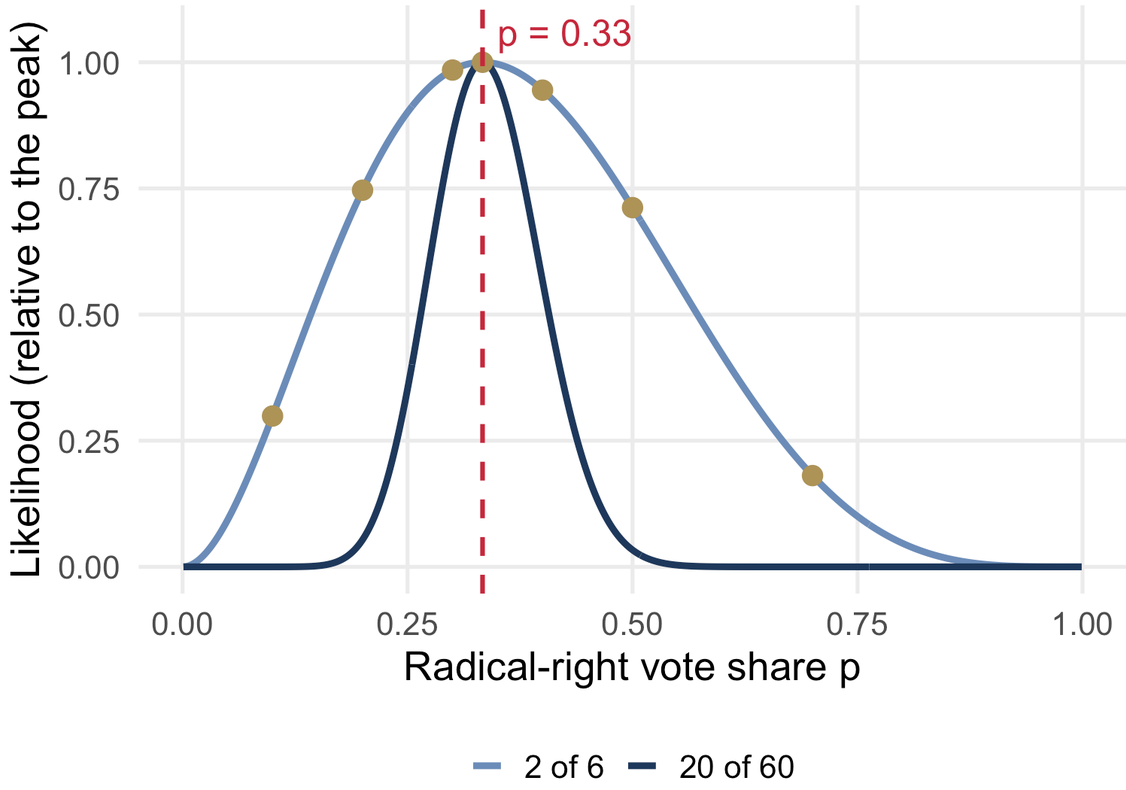
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Gold dots: candidate values tried by hand for 2 of 6.

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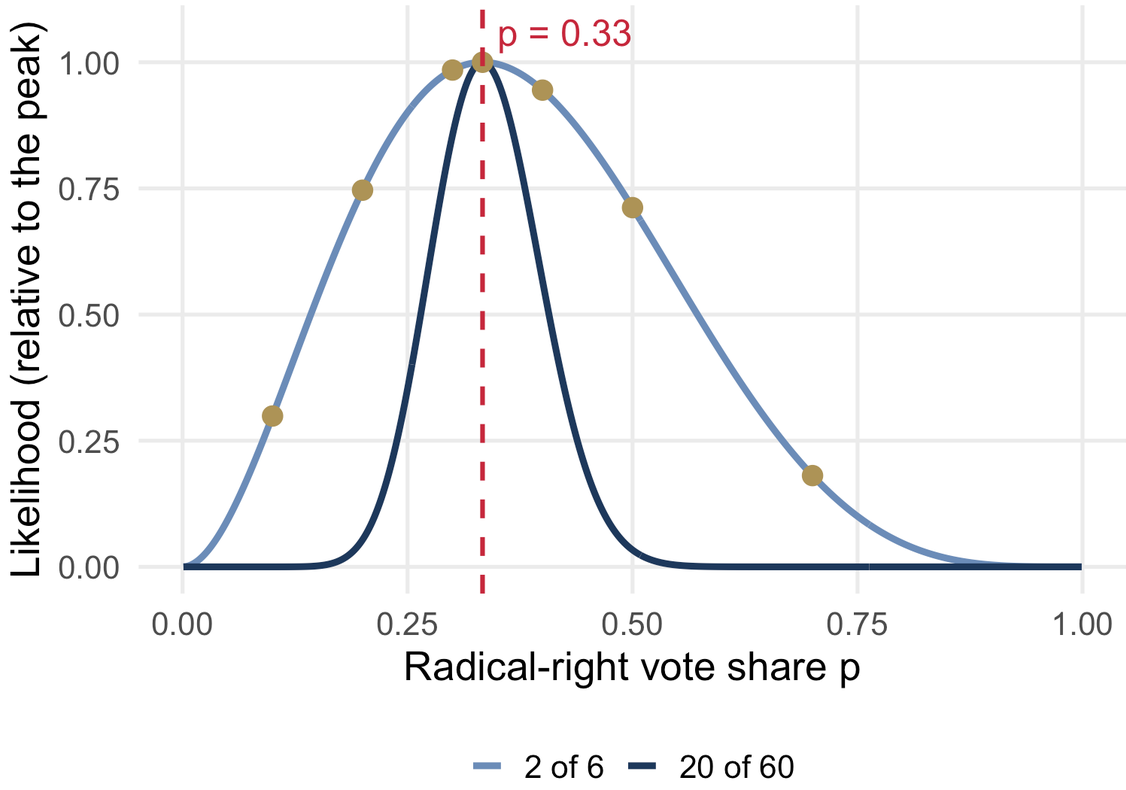
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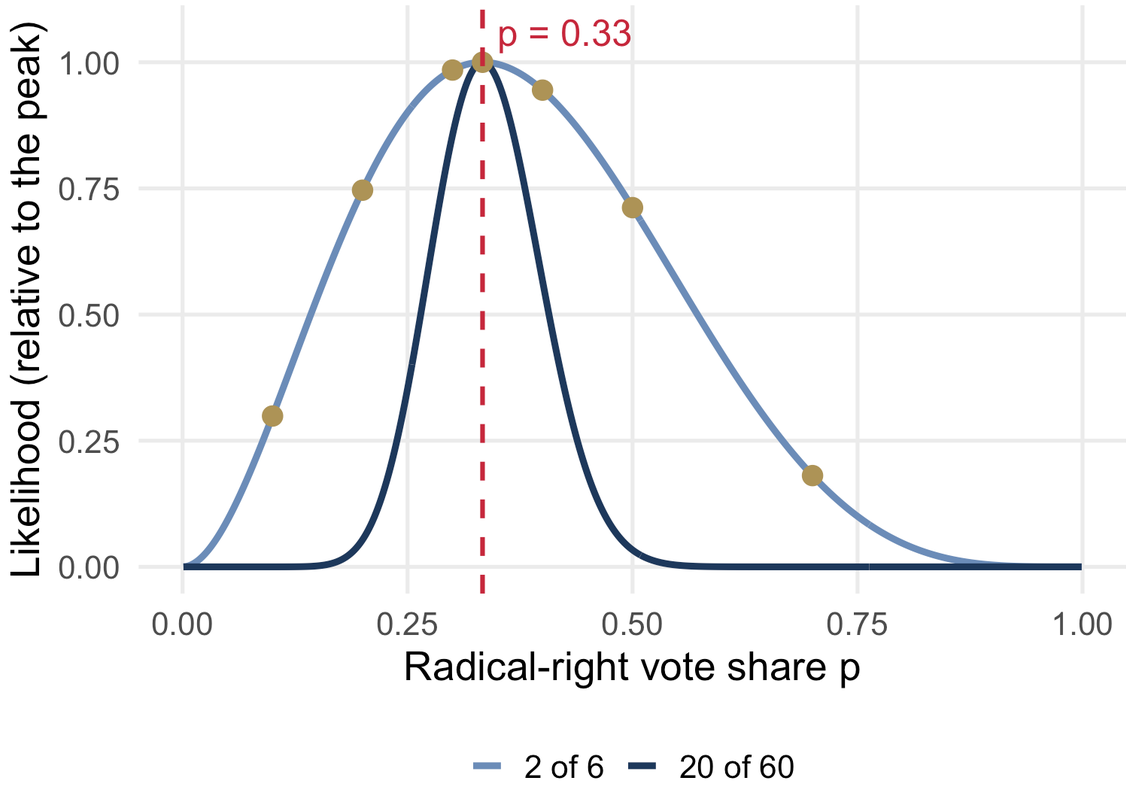
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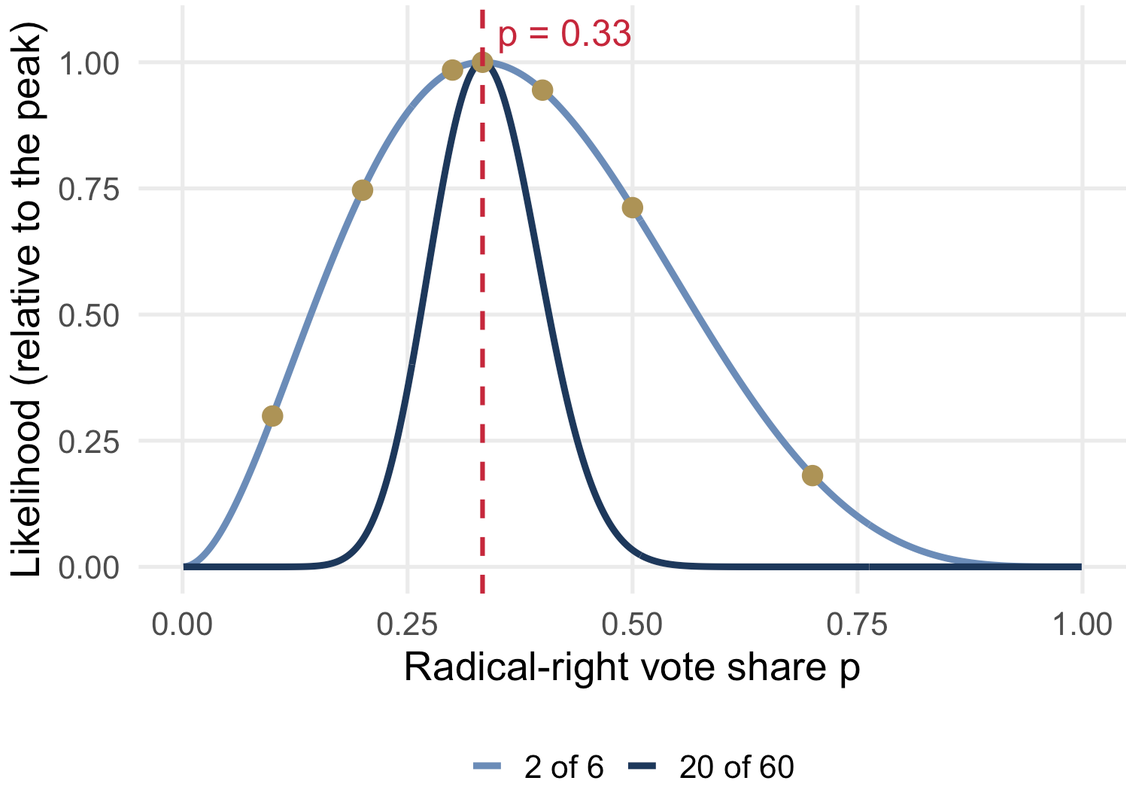
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- With 20 of 60, same peak but **sharper**: we are surer. The standard error comes from that curvature.



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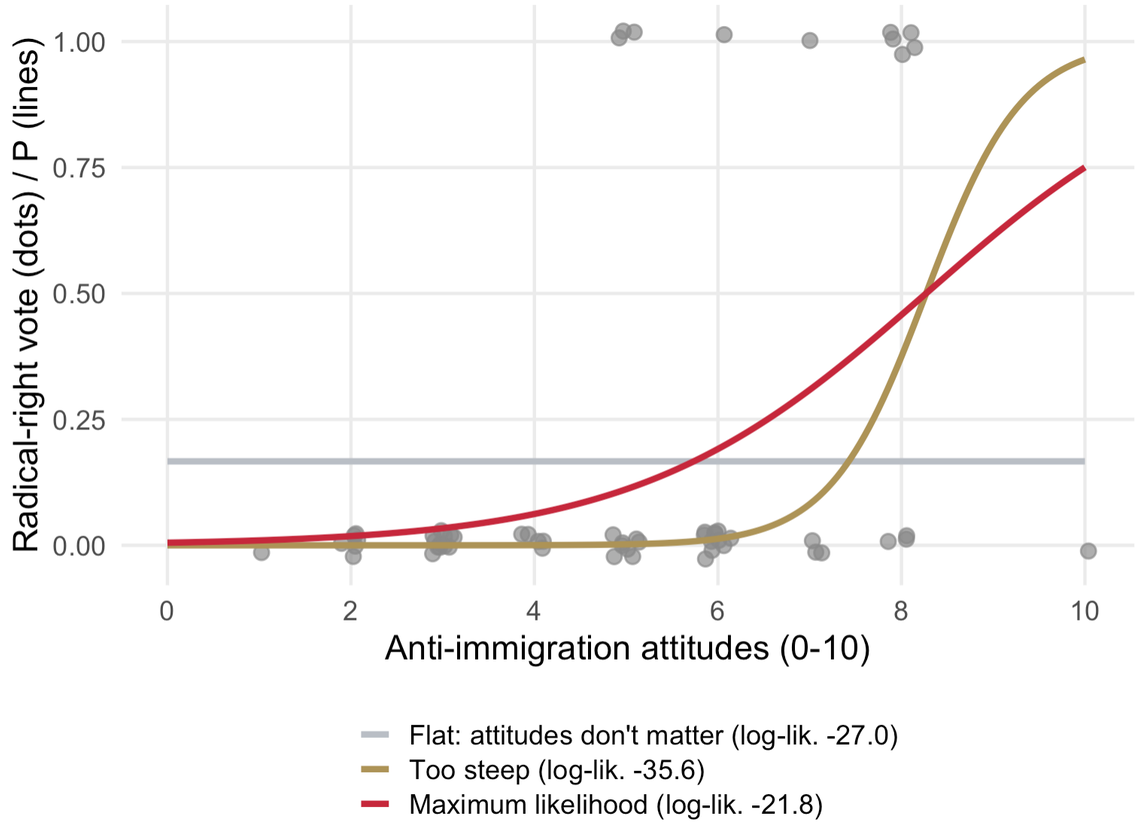
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- Software climbs the **log** of the likelihood: sums instead of products, same peak.



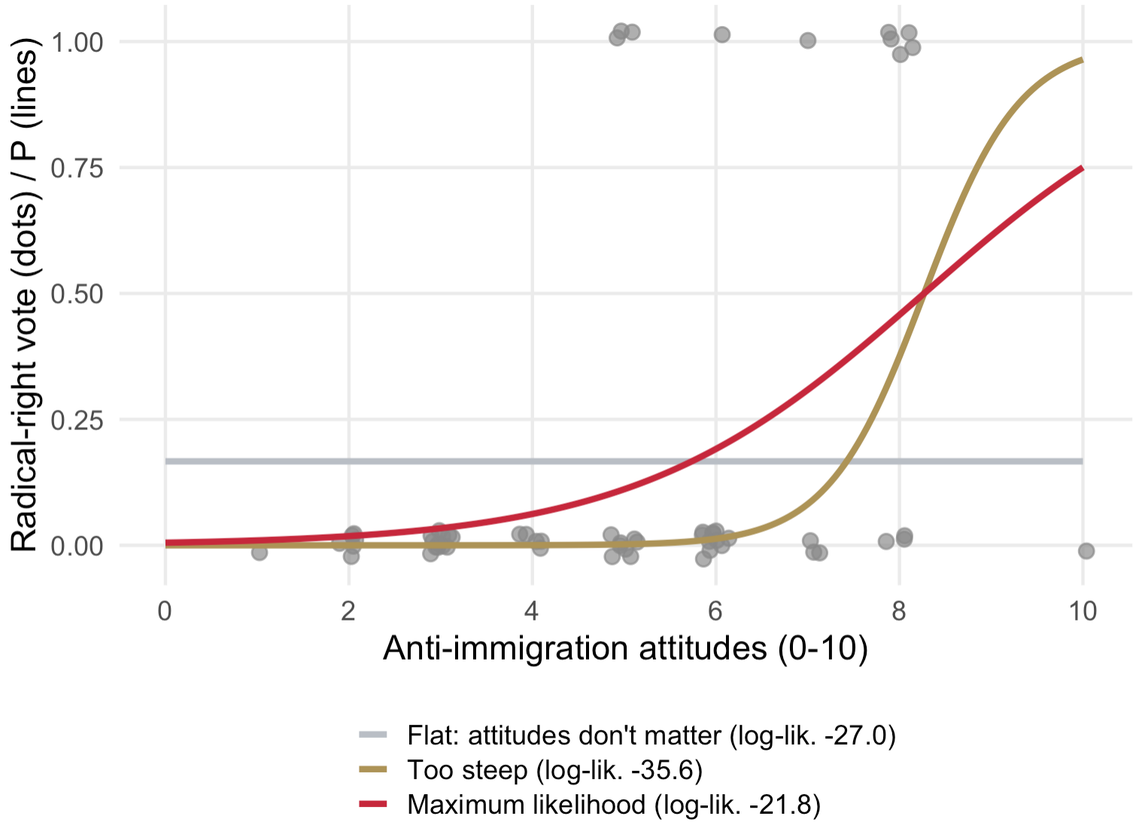
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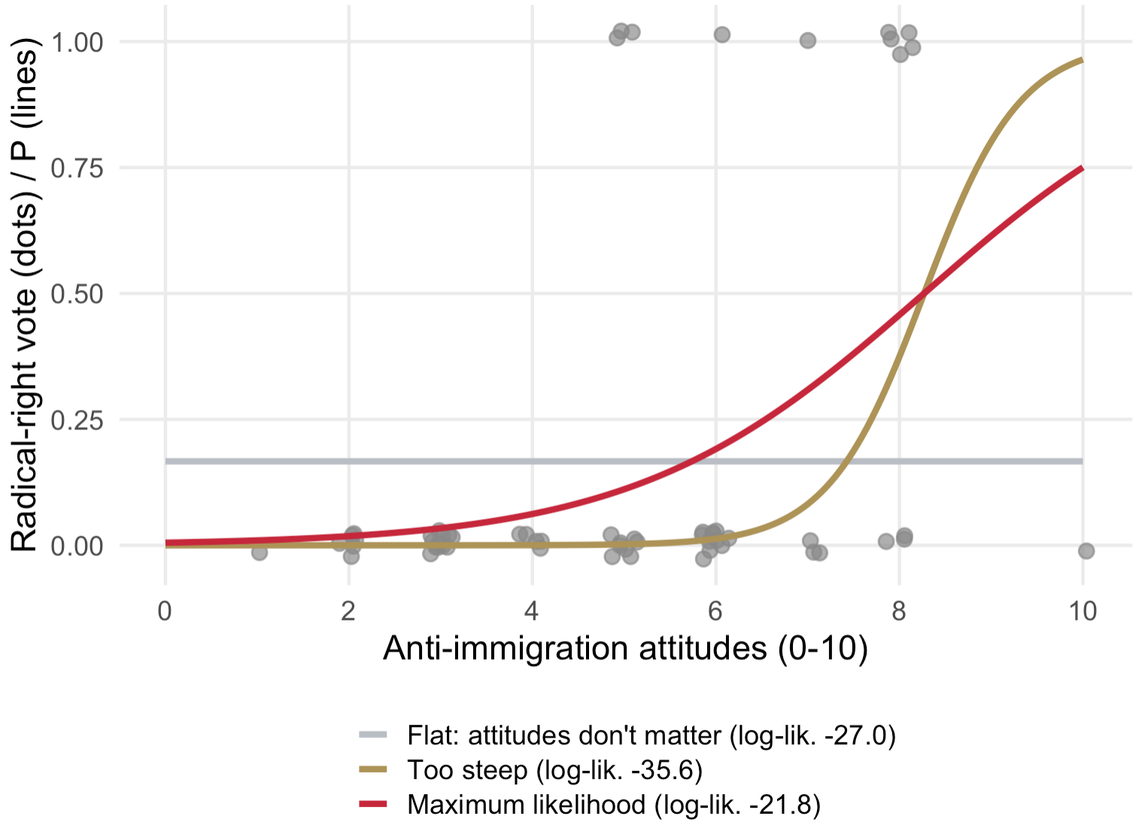
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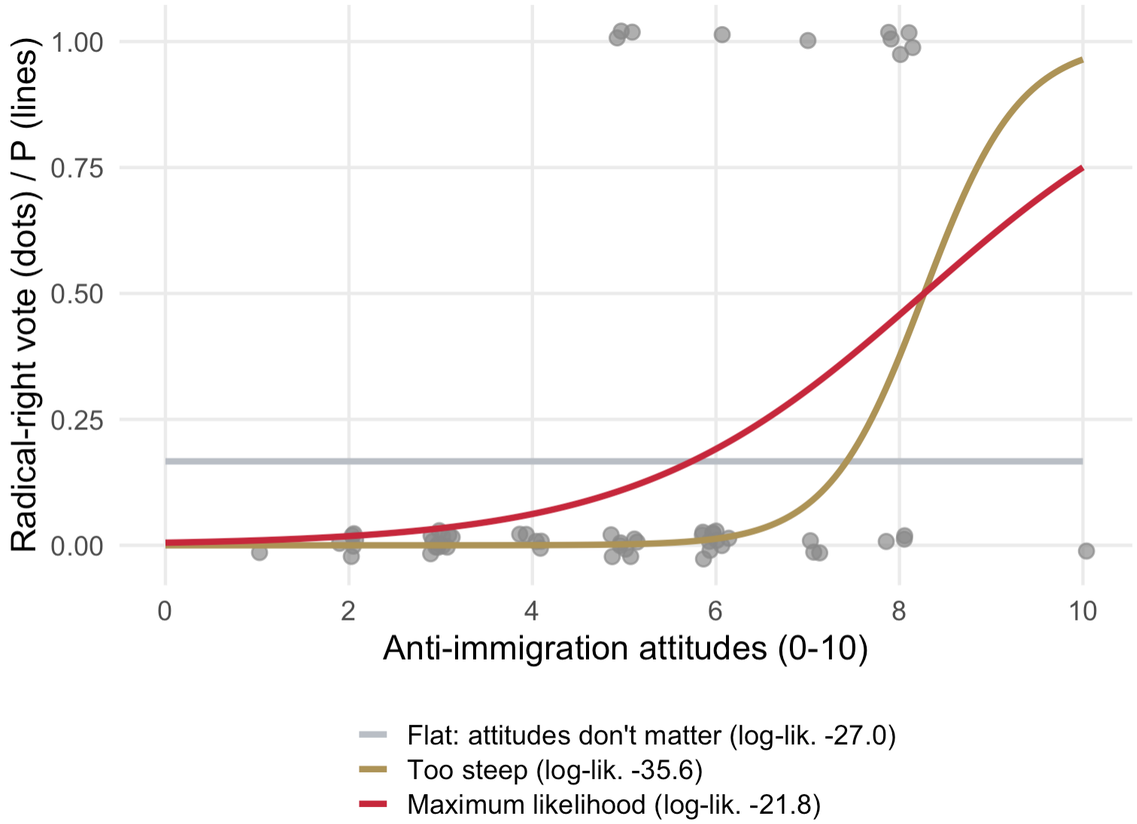
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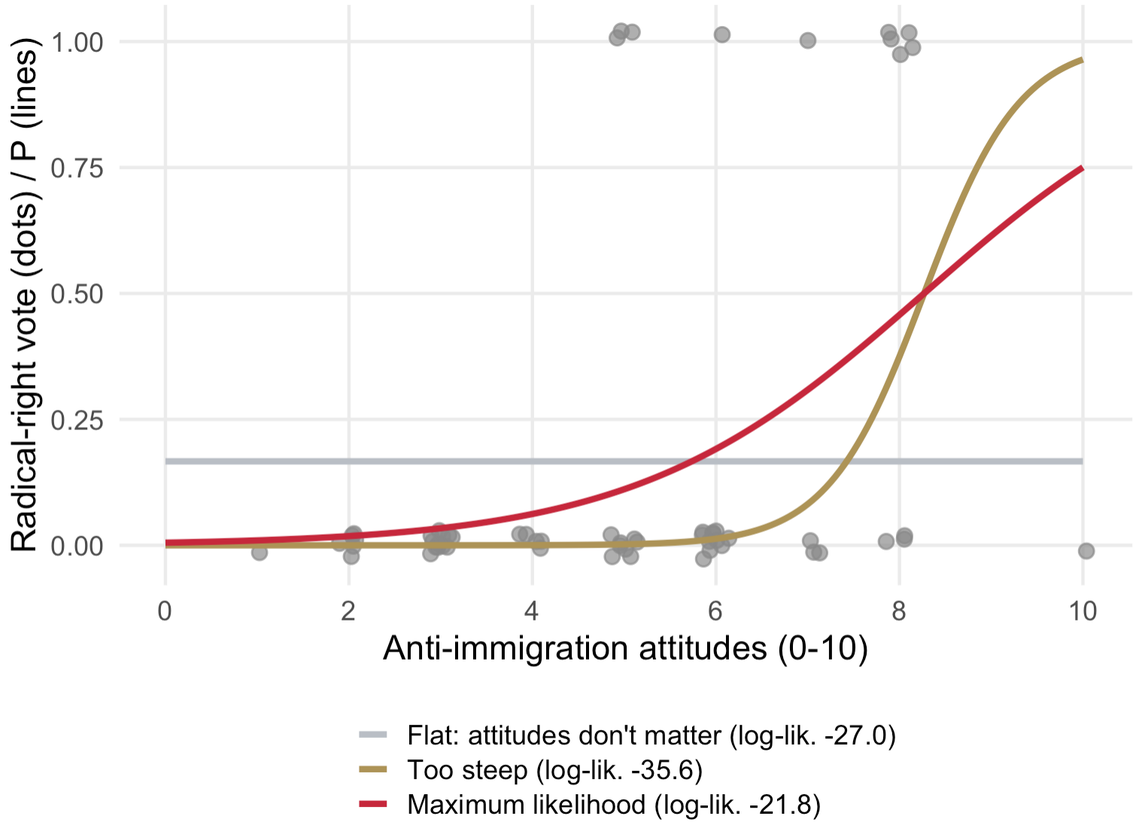
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- The too-steep curve is punished for being **confidently wrong** about a few respondents.



Your turn ✏️ (~5 min, in pairs, if time permits)

Four respondents, and the probability of voting radical right that two models assign to each. **Which model makes the data more likely?**

Respondent	Anti-immigration attitudes	Voted radical right?	Model 1: p_i	Model 2: p_i
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Model 3 says 0.01 for A and B and 0.99 for C and D. Better? Now add E: attitudes 9, *didn't* vote radical right. Which model suffers most?

Wrapping up

Four rules for binary outcomes

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And Week 4's question still comes first: is your 0/1 outcome **measured** well? Self-reported turnout is not turnout.

Your turn ✏️ (on your own time, for reflection)

Find a **yes/no outcome** in your PhD project (or one you could add).

1. **Outcome**: what is the 0/1 variable, and how is it measured? Self-report, records, coding?
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Then check your sentence: is it on the probability scale, and could a journalist misread it?

What we covered today

LIKELIHOOD, BINARY OUTCOMES, AND WHAT TO REPORT

Binary outcomes

OLS on a switch is the LPM: readable and often fine for average effects, but it leaves [0, 1]; logit and probit bend the line; each model speaks its own language

What is a logit coefficient?

Odds ratios are not collapsible: a problem if the outcome is a latent decision, a feature if it is a chance (Mood vs Buis); never read mediation off coefficient changes

Quantities of interest

Treat the model as a prediction machine: predictions, comparisons, and slopes, for the people you care about, on the probability scale; test those quantities, not coefficients

Maximum likelihood

Score every candidate by how likely it makes the data; keep the best; a sharper peak means more precision

Readings

Compulsory

- Stock & Watson. *Introduction to Econometrics*, Ch. 11 (Regression with a binary dependent variable).
- Mood (2010). Logistic regression: Why we cannot do what we think we can do, and what to do about it. *European Sociological Review* 26(1): 67–82.
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Optional

- Rohrer & Arel-Bundock (2026). Models as prediction machines: How to convert confusing coefficients into clear quantities. *Advances in Methods and Practices in Psychological Science* 9(2): 1–21.
- Kuha & Mills (2020). On group comparisons with logistic regression models. *Sociological Methods & Research* 49(2): 498–525.
- Breen, Karlson & Holm (2018). Interpreting and understanding logits, probits, and other nonlinear probability models. *Annual Review of Sociology* 44: 39–54.
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- Arel-Bundock, Greifer & Heiss (2024). How to interpret statistical models using marginaleffects for R and Python. *Journal of Statistical Software* 111(9): 1–32.

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- Allison (1999). *Sociological Methods & Research* 28(2): 186–208.
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- Arel-Bundock (2026). *Model to Meaning: How to Interpret Statistical Models with R and Python*. Chapman and Hall/CRC.
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A logit gives you a coefficient.

Your readers need a probability,

and your estimand tells you which one.

Next week: Nested data · fixed effects, random effects, and multilevel basics

In the lab on Monday

with Martín

In Lab 6 you'll write a likelihood by hand and maximize it, compare LPM, logit, and probit on the radical-right survey, ask the models for predictions, comparisons, and slopes, and test an interaction on two scales. On your own: watch non-collapsibility happen in a simulation.

`optim()` `glm()` `avg_comparisons()` `predictions()` `avg_slopes()`

LAB 6

`06-binary-outcomes-lab.qmd`